

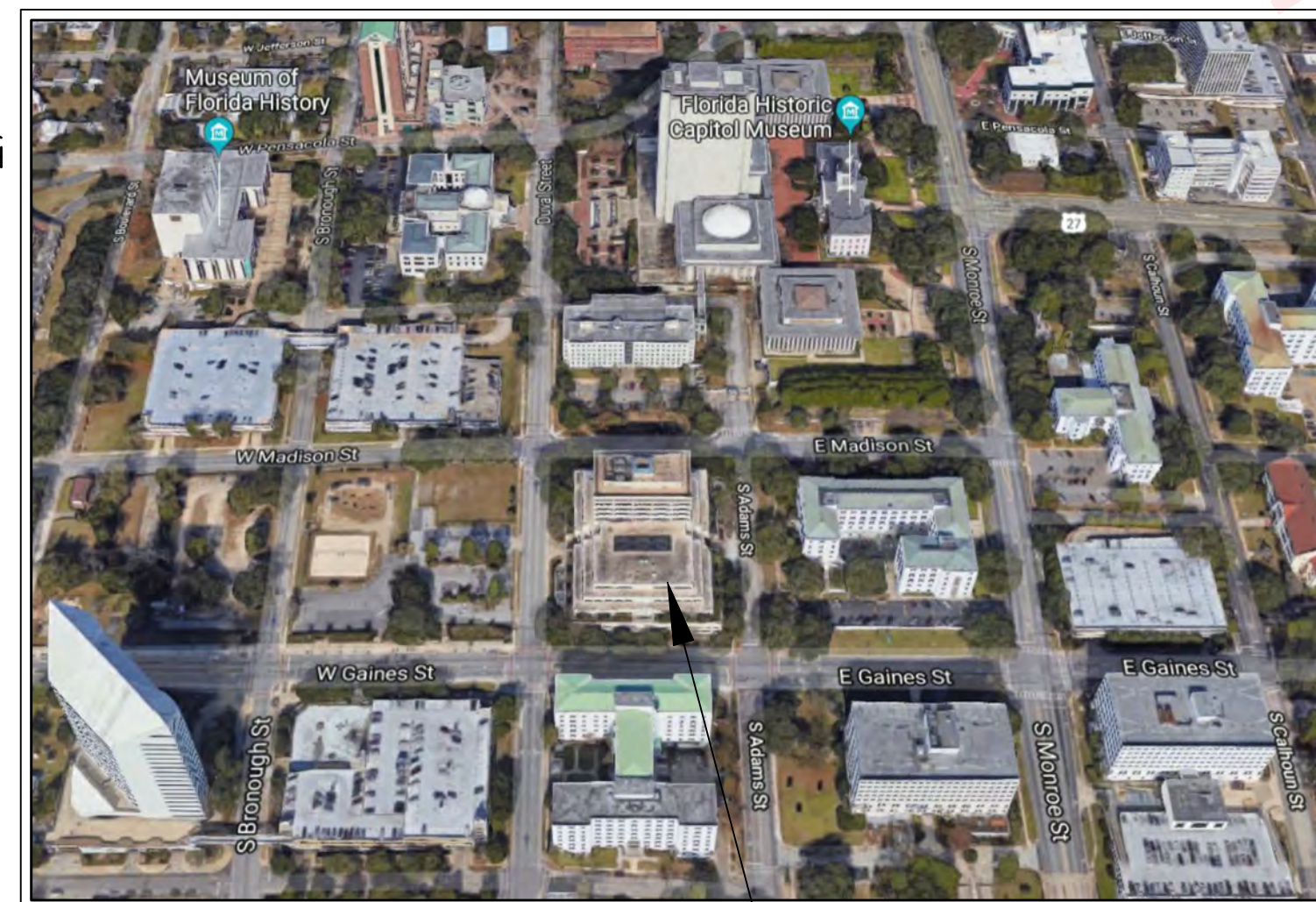


MSFM 02203280

DESIGN REPLACEMENT OF MAJOR ELECTRICAL SYSTEMS - PHASE 1 PEPPER BUILDING

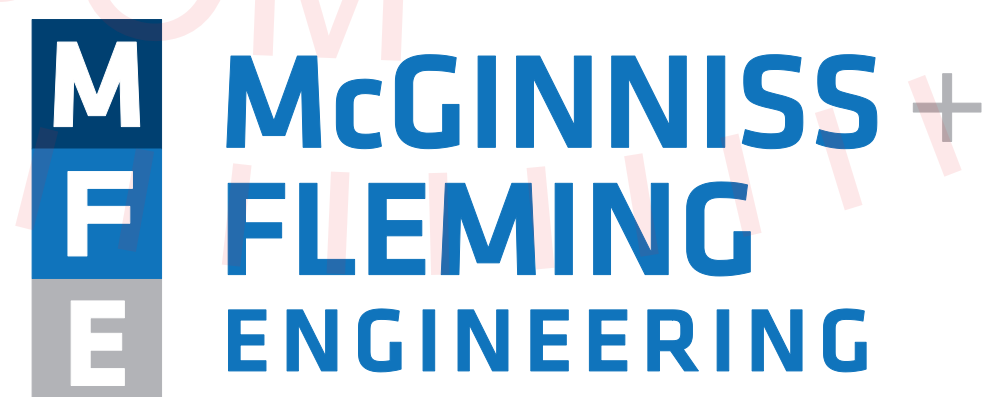
TALLAHASSEE, FLORIDA

Area Map



CLAUDE PEPPER BUILDING

100% CONSTRUCTION DOCUMENTS
March 25, 2024



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MECHANICAL SCOPE OF WORK SUMMARY

FURNISH ALL LABOR, MATERIALS, EQUIPMENT AND INCIDENTALS REQUIRED TO COMPLETE ALL MECHANICAL WORK SHOWN ON THE CONTRACT DRAWINGS. THE WORK REQUIRED CONSISTS BASICALLY OF, BUT IS NOT LIMITED TO, THE FOLLOWING:

- DEMOLITION OF OLD EXHAUST FAN AND DUCTWORK
- PROVIDE AND INSTALL NEW EXHAUST FAN AND DUCTWORK
- PROVIDE AND INSTALL NEW DX DUCTLESS SPLIT AIR CONDITIONERS
- PROVIDE NEW DRAIN FOR CONDENSATE DISPOSAL

GENERAL NOTES:

- FURNISH ALL LABOR, MATERIALS, EQUIPMENT AND INCIDENTALS REQUIRED TO COMPLETE ALL WORK SHOWN ON THE CONTRACT DRAWINGS.

- ALL CONSTRUCTION SHALL CONFORM TO APPLICABLE CODE STANDARDS INCLUDING:

NATIONAL FIRE PROTECTION ASSOCIATION - NFPA
NFPA 70 - 2020; NATIONAL ELECTRICAL CODE
NFPA 72 - 2019; NATIONAL FIRE ALARM AND SIGNALING CODE

FLORIDA BUILDING CODE
FBC-B 2023; THE FLORIDA BUILDING CODE (8th EDITION)
FPC 2023; THE FLORIDA FIRE PREVENTION CODE (8th EDITION)
INCLUDING NFPA 101 - 2021; THE LIFE SAFETY CODE
FBC-M 2023; THE FLORIDA MECHANICAL CODE (8th EDITION)
FBC-A 2023; THE FLORIDA BUILDING CODE, ACCESSIBILITY (8th EDITION)
FBC-EC 2023; THE FLORIDA BUILDING CODE, ENERGY CONSERVATION (7th EDITION)
FBC-FG 2023; THE FLORIDA BUILDING CODE, FUEL GAS (8th EDITION)
FBC-P 2023; THE FLORIDA BUILDING CODE, PLUMBING (8th EDITION)
FBC-EB 2023; THE FLORIDA BUILDING CODE, EXISTING BUILDING (8th EDITION)

STATE REQUIREMENTS FOR EDUCATIONAL FACILITIES - 2014 EDITION

- SHOULD CONFLICT OCCUR BETWEEN PROJECT SPECIFICATIONS & DRAWING NOTES, THE DRAWING NOTES WILL TAKE PRECEDENCE.

- THE CONTRACTOR IS EXPECTED TO PROVIDE PROFESSIONAL WORK PERFORMED IN ACCORDANCE WITH INDUSTRY STANDARDS AND BEST PRACTICES.

- THE WORK SHALL BE COMPLETE, FULLY OPERATIONAL, AND SUITABLE IN EVERY WAY FOR THE SERVICE REQUIRED.

- DRAWINGS INDICATE SCOPE AND DO NOT SHOW ALL DETAILS. DEVICES AND INCIDENTAL MATERIALS NECESSARY TO ACCOMPLISH THE WORK, THEREFORE, IT SHALL BE UNDERSTOOD THAT SUCH DEVICES AND INCIDENTAL MATERIALS REQUIRED SHALL BE FURNISHED AT NO COST TO THE OWNER.

- CONTRACTOR SHALL TAKE INTO ACCOUNT FIELD CONDITIONS AND COORDINATE IN ORDER TO AVOID CONFLICTS WITH EXISTING CONDITIONS AND INTERFERENCE BETWEEN TRADES.

- EQUIPMENT SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURER'S REQUIREMENTS FOR PROPER OPERATION, MAINTENANCE, AND SERVICE. IF CHANGES TO THE CONTRACT DOCUMENTS ARE NECESSARY TO AVOID CONFLICTS, THE CONTRACTOR IS RESPONSIBLE FOR REQUESTING CLARIFICATION IN A TIMELY FASHION.

- THE CONTRACTOR SHALL BE RESPONSIBLE FOR ANY DEFICIENCIES ASSOCIATED WITH WORK PERFORMED BEFORE OBTAINING WRITTEN CLARIFICATION.

- CONTRACTOR SHALL VERIFY SIZE, FLOW DIRECTION, AND LOCATION OF EXISTING DUCTS/PIPING TO REMAIN, RELATED BUILDING INFRASTRUCTURE/SERVICES, PRIOR TO COMMENCING WORK. ADVISE THE ENGINEER IN WRITING IF MATERIALLY DIFFERENT THAN SHOWN.

- THE CONTRACTOR SHALL TAKE DUE CARE DURING ALL PHASES OF WORK TO PROTECT BUILDING FINISHES, FURNISHINGS, EQUIPMENT, ETC. THE CONTRACTOR SHALL BEAR ALL COSTS TO REPAIR ANY DAMAGED ITEMS, FINISHES, ETC. RESULTING FROM THEIR OR THEIR SUBCONTRACTORS' WORK.

- THE CONTRACTOR SHALL PROVIDE DAILY CLEANUP OF THEIR WORK AREAS. UPON COMPLETION OF THE WORK THE CONTRACTOR SHALL THOROUGHLY CLEAN SPACES THAT WERE OCCUPIED BY TEMPORARY WORK AND TEMPORARY FACILITIES. REMOVE ALL DEBRIS, RUBBISH, AND EXCESS MATERIAL FROM THE SITES.

- REPAIR DAMAGE CAUSED BY INSTALLATION OR USE OF TEMPORARY FACILITIES. THIS INCLUDES HARDSCAPING, LANDSCAPING, FINISHES, ETC.

- THE CONTRACTOR SHALL DELIVER TO THE OWNER, UPON SUBSTANTIAL COMPLETION OF THE WORK, TWO COPIES OF DESCRIPTIVE LITERATURE RELATED TO THE EQUIPMENT INSTALLED UNDER THIS CONTRACT, INCLUDING PARTS LISTS, WIRING DIAGRAMS, MAINTENANCE AND OPERATION MANUALS AND WARRANTIES CUSTOMARILY SUPPLIED BY MANUFACTURERS FOR EQUIPMENT INCORPORATED IN THIS WORK.

- THE LITERATURE SHALL BE NEATLY BOUND IN A 3-RING BINDER, IN ADDITION TO ELECTRONIC FORMAT ON A USB THUMB DRIVE, AND DELIVERED PRIOR TO FINAL ACCEPTANCE

- THE CONTRACTOR SHALL LABEL NEW EQUIPMENT AND ANCILLARY SYSTEMS INCLUDED IN THE SCOPE OF THIS PROJECT.

- THE CONTRACTOR SHALL GIVE PHYSICAL DEMONSTRATION AND VERBAL INSTRUCTIONS FOR PROPER OPERATION AND MAINTENANCE OF EQUIPMENT TO THE OWNER OR HIS DESIGNATED REPRESENTATIVE. SCHEDULE THESE DEMONSTRATIONS AND INSTRUCTIONS AT THE OWNER'S CONVENIENCE.

HAZARDOUS MATERIALS

- CONTRACTORS ARE EXPECTED TO NOTIFY THE OWNER WHEN EVER THEY DISCOVER THAT THEIR WORK WILL EXPOSE THEM TO ANY MATERIALS THAT ARE THE LEAST BIT SUSPICIOUS. REMOVAL OF CONTAMINATED MATERIALS WILL BE THE RESPONSIBILITY OF THE OWNER. HOWEVER THE CONTRACTOR IS RESPONSIBLE TO INSPECT FUTURE WORK AREAS IN A TIMELY FASHION SO AS NOT TO BE HELD UP WAITING FOR ABATEMENT.

CONDENSATE & PIPING

- CONDENSATE DRAIN SHALL INCLUDE A P-TRAP, SEE DETAIL.

- PIPING SHALL BE SAME SIZE AS DISCHARGE CONNECTION, D-W-V COPPER AND FITTINGS. MINIMUM SIZE IS 3/4". SUPPORT PIPING AT P-TRAP AND ON 4' CENTERS AND SLOPE 1/4" PER FOOT TOWARD DRAIN.

- PROVIDE CLEANOUTS WITH SCREW CAPS/PLUGS AT TRAPS, ON VERTICAL DROPS, AND IN HORIZONTAL DIRECTION CHANGES.

MISCELLANEOUS METALS:

- INTERIOR EQUIPMENT/PIPING SUPPORTS, HARDWARE, BRACKETS, FRAMING CHANNEL, ETC. SHALL BE GALVANIZED STEEL AND EQUAL TO B-LINE.

- METAL/ELECTRICAL FRAMING/CHANNEL, SUPPORTS, ETC. IN CONTACT WITH CONCRETE OR INSTALLED OUTDOORS SHALL BE HOT-DIPPED GALVANIZED.

- MISCELLANEOUS INTERIOR SUPPORTS SHALL BE 12 GA, 1-5/8" SQ. ELECTRO-GALVANIZED FRAMING CHANNEL. (MINIMUM).

PIPE/EQUIPMENT INSULATION:

CONDENSATE PIPING - INSULATE CONDENSATE PIPING WITH 3/4" CLOSED CELL FOAM INSULATION WITHIN BUILDING. SEAL ALL JOINTS, SEAMS, ETC. AIR TIGHT. PROVIDE ACCESS PLUGS/CAPS TO FITTINGS THAT REQUIRE MAINTENANCE. FOR EXTERIOR CONDENSATE PIPING, ADD ALUMINUM CLADDING.

DUCTWORK:

- ALL WORK SHALL COMPLY WITH SMACNA HVAC DUCT CONSTRUCTION STANDARDS - METAL AND FLEXIBLE. CONSTRUCT SUPPLY DUCTWORK FOR VAV SYSTEMS TO WITHSTAND 3" (MIN) AND RETURN DUCT -2" STATIC PRESSURES.

- USE EITHER ROUND OR RECTANGULAR DUCT WITH EQUAL OR GREATER EQUIVALENT FREE AREA TO ACCOMMODATE EXISTING STRUCTURE.

- CROSS BREAK DUCTS AND OTHER SHEET METAL OVER 24" WIDE.

- INTERIOR - USE GALVANIZED SHEET METAL FOR RECTANGULAR WITH EXTERIOR INSULATION, U.O.S. USE GALVANIZED SPIRAL SEAM ROUND DUCT WITH EXTERIOR INSULATION.

- PROVIDE SHEET METAL CLOSURE ANGLES, ESCUTCHEONS, OR FLASHING ON BOTH SIDES OF WALL PENETRATIONS (NON RATED) AND SEAL AIR TIGHT. MINIMUM WIDTH IS 1" OR AS REQUIRED TO COVER OPENING.

- PROVIDE SMOKE TIGHT SEAL WHEN PENETRATING SMOKE COMPARTMENT WALLS.

- PROVIDE DOUBLE-THICKNESS TURNING VANES IN ALL RECTANGULAR ELBOWS AND OFFSETS.

- DUCT SIZES MAY BE CHANGED TO ACCOMMODATE CONDITION AS LONG AS THE INTERNAL FREE AREA IS NOT DIMINISHED.

- ALL FABRICATED DUCTWORK LONGITUDINAL AND TRANSVERSE JOINTS, TAPS, AND CONNECTIONS SHALL BE SEALED WITH DUCT MASTIC REGARDLESS OF PRESSURE CLASS.

- TAG ALL DAMPER LOCATIONS WITH ORANGE FLAG TAPE.

- PERMANENTLY MARK ALL DAMPER SHAFTS TO INDICATE DAMPER POSITION.

EQUIPMENT INSTALLATION

GENERAL EQUIPMENT INSTALLATION REQUIREMENTS:

INSTALL UNIT IN STRICT ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS. CONTRACTOR IS REQUIRED TO HAVE EQUIPMENT INSTALLATION INSTRUCTIONS ON SITE FOR ALL EQUIPMENT THAT IS ON SITE.

ALL EQUIPMENT SHALL BE SECURED TO PADS OR BUILDING STRUCTURE. INSURE THAT PROPER ACCESS TO THE UNIT IS MAINTAINED. DO NOT RUN PIPING IN FRONT OF ACCESS PANELS.

INSTALL MISCELLANEOUS DEVICES SHIPPED LOOSE.

COORDINATE CONTROLS AND POWER WIRING INSTALLATION.

START-UP ALL EQUIPMENT IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS.

CLEAN FACTORY-FINISHED SURFACES. REPAIR ANY MARRED OR SCRATCHED SURFACES WITH MANUFACTURER'S TOUCH-UP PAINT. TURNOVER ANY SPECIAL TOOLS PROVIDED BY THE EQUIPMENT MANUFACTURER.

CONDENSING UNIT INSTALLATION:

INSTALL CONDENSING UNIT ON MANUFACTURER SUPPLIED WALL BRACKET. SECURE UNIT TO BRACKET PER MANUFACTURER'S INSTRUCTIONS.

INSTALL SITE GLASS AND FILTER/DRYER.

START-UP & CHECK-OUT THE SYSTEM IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATION. THE MECHANICAL CONTRACTOR SHALL TEST AND CERTIFY THE PROPER OPERATION OF THE SYSTEM AND CONTROLS.

DUCTLESS SPLIT FAN COIL INSTALLATION

COORDINATE EXACT LOCATION OF THE FAN COIL UNIT WITH THE OWNER.

INSTALL WIRED CONTROLLER AND CONTROL WIRING TO THE CONDENSING UNIT. CONTROL CABLE MUST BE IN CONDUIT EXCEPT; ACCESSIBLE CONCEALED LOCATIONS PLENUM RATED CABLE MAY BE USED.

NO AIR BALANCE TEST IS REQUIRED. SET FAN AIR FLOW ON HIGH SPEED.

PIPE PUMPED CONDENSATE (PUMP PROVIDED WITH EQUIPMENT) TO CONDENSATE DRAIN. INSTALL REFRIGERANT PIPING TO CONDENSING UNIT; SEE SCHEDULE FOR PIPE SIZES.

REFRIGERANT PIPING INSTALLATION:

REFRIGERANT PIPING SHALL BE ACR TYPE L. PROVIDE HARD DRAWN COPPER TUBING WITH BRAZED LONG RADIUS WROUGHT COPPER FITTINGS AT ALL AIR HANDLING UNITS, CONDENSING UNITS, IN MECHANICAL ROOMS AND OTHER EXPOSED LOCATIONS. WHERE CONCEALED, REFRIGERANT PIPING MAY BE SOFT COPPER.

PROVIDE SIGHT GLASS AND FILTER DRYER FOR EACH REFRIGERANT CIRCUIT. TAKE CARE NOT TO DOUBLE UP WHERE FILTER DRYER IS PROVIDED WITH CONDENSING UNIT.

LEAK TEST ALL REFRIGERANT PIPING. EVACUATE, DEHYDRATE AND CHARGE SYSTEM PER MANUFACTURER'S INSTRUCTIONS.

SUPPORT HORIZONTAL REFRIGERANT SUCTION PIPING 4 FEET ON CENTER. LIQUID LINE MAY BE STRAPPED TO THE INSULATED SUCTION LINE WITH DUCT TAPE.

INSURE THAT EXPOSED METAL PIPES DO NOT CONTACT METAL OR CONCRETE SURFACES. PROVIDE INSULATION MATERIALS OR SLEEVES AT ANY SUCH LOCATIONS.

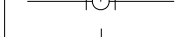
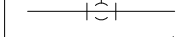
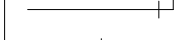
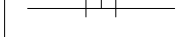
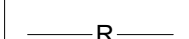
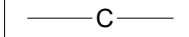
DO NOT CONCEAL ANY REFRIGERANT PIPING INSIDE BUILDING MATERIALS UNTIL IT HAS BEEN INSPECTED BY THE ENGINEER.

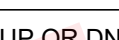
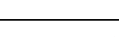
FANS:

- INSTALL FANS WITH REQUISITE LENGTH OF STRAIGHT FULL SIZE DUCTS ON INLET AND DISCHARGE TO MINIMIZE SYSTEM EFFECT. MINIMUM LENGTH IS 3 TIMES THE WHEEL DIAMETER U.O.S. LOCATE TAPS/BRANCHES BEYOND MINIMUM LENGTHS.

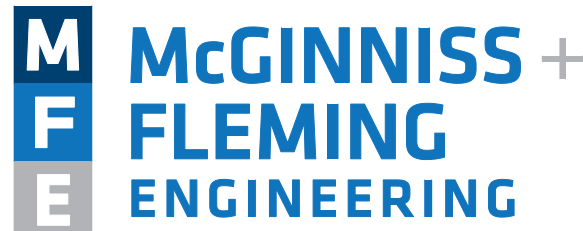
- PROVIDE FLEXIBLE DUCT CONNECTIONS AT FAN.

- SUPPORT FAN FROM STRUCTURE OR WALL.

PIPING SYMBOLS/LEGEND	
DESIGNATION	DESCRIPTION
	PIPE TURNING DOWN
	PIPE TURNING UP
	TEE UP
	TEE DOWN
	DROP & TURN
	DROP & RUN
	TEE OFF TOP
	TEE OFF BOTTOM
	ELBOW
	TEE
	UNDERGROUND PIPING
	REFRIGERANT PIPING
	CONDENSATE PIPING
	CAP OR CLEANOUT

HVAC SYMBOLS/LEGEND	
DESIGNATION	DESCRIPTION
	POINT OF CONNECTION TO EXISTING
	DEMOLITION WORK TERMINATION POINT
	AIR FLOW DIRECTION
	SQUARE-TO-ROUND TRANSITION
	CENTRIFUGAL INLINE CABINET EXHAUST FAN
	DUCT RISE UP OR DOWN
	SERVICE AREA - MAINTAIN CLEAR

ABBREVIATIONS	
AFF	ABOVE FINISHED FLOOR
AHAP	AS HIGH AS POSSIBLE
BAS	BLDG AUTOMATION SYSTEM
BHP	BRAKE HORSE POWER
BJ	BAR JOIST
BTUH	BRITISH THERMAL UNIT/HOUR
BOD	BOTTOM OF DUCT
CFM	CUBIC FEET PER MINUTE
CHWS	CHILLED WATER SUPPLY
CHWR	CHILLED WATER RETURN
C.O.	CLEANOUT
DB	DRY BULB
DIA OR Ø	DIAMETER
DG	DOOR GRILLE
EA	EXHAUST AIR
EF	EXHAUST FAN
ESP	EXTERNAL STATIC PRESSURE
EX OR (E)	EXISTING
EXT	EXTERNAL OR EXTERIOR
FCU	FAN COIL UNIT
FL	FLOOR
FJ	FLEXIBLE JOINT
FFM	FEET PER MINUTE
GPM	GALLONS PER MINUTE
H	HIGH
HDG	HOT-DIP GALVANIZED
HP	HORSE POWER
HHWS/R	HEATING HOT WATER SUPPLY/RETURN
KW	KILOWATT
L	LONG
LAT	LEAVING AIR TEMPERATURE
MU	MAKE UP WATER
MAX	MAXIMUM
MBH	1000 BTU/HOUR
MIN	MINIMUM
NA	NOT APPLICABLE
NC	NORMALLY CLOSED
NIC	NOT IN CONTRACT
NO	NORMALLY OPEN
NPS	NOMINAL PIPE SIZE
OA	OUTSIDE AIR
PD	PRESSURE DROP
PH	PHASE
PNL	PANEL
RA	RETURN AIR
RAG	RETURN AIR GRILLE
RLA	RATED LOAD AMPS
SA	SUPPLY AIR
SAG	SUPPLY AIR GRILLE
SP	STATIC PRESSURE
SQ	SQUARE
SAN SWR	SANITARY SEWER
T	TEMPERATURE
TEFC	TOTALLY ENCL. FAN COOLED
THK	THICK
TOD	TOP OF DUCT
TSP	TOTAL STATIC PRESSURE
TYP	TYPICAL
UG	UNDERGROUND
UOS	UNLESS OTHERWISE SPECIFIED
VAV	VARIABLE AIR VOLUME
VFD	VARIABLE FREQUENCY DRIVE
V	VOLTS
WB	WET BULB
WG	WATER GAUGE
XFA	TRANSFER AIR



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PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 1

MSFM-02203280

Tallahassee, Florida

DATE:

March 25, 2024

REVISED:

DESIGNED BY:

JB

DRAWN BY:

MH

SUBMITTAL:

100% Construction Documents

SHEET TITLE:

GENERAL NOTES & SCHEDULES

SHEET:

M100

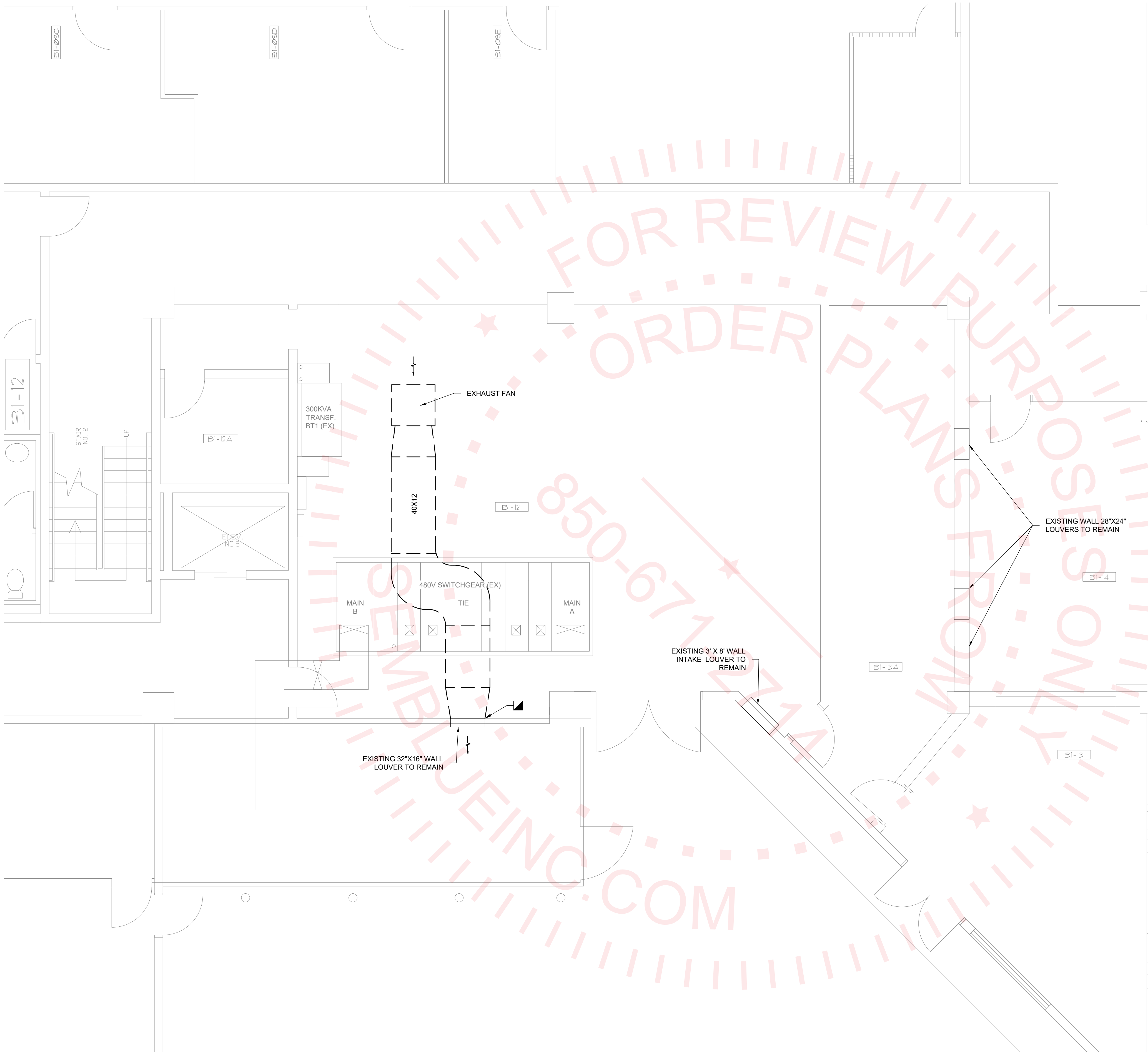
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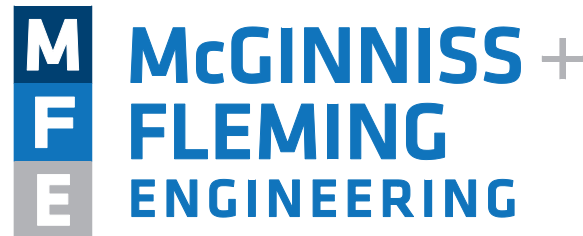
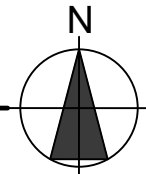
NEW DUCTLESS SPLIT SYSTEM SCHEDULE (COOLING)			
DESIGNATION		AC-1 & FCU-1	AC-2 & FCU-2
TYPE OF UNIT		WALL-MOUNT COOLING ONLY	WALL-MOUNT COOLING ONLY
MANUFACTURER		TRANE	TRANE
FAN COIL UNIT MODEL		TPKA0A030	TPKA0A012
CONDENSING UNIT MODEL		TRUYA030	TRUYA012
SUPPLY AIR	CFM	635, 705, 775	265, 310, 385, 455
VENTILATION AIR	CFM	N/A	N/A
FAN STATIC (EXTERNAL)	IN	N/A	N/A
AHRI RATED COOLING CAPACITY	MBH	12.0	12.0
SEER	BTUH/WATT	21.0	21.0
RATED HEATING CAPACITY	MBH	N/A	N/A
UNIT ELECTRICAL CHARACT.	V/Ø/HZ	208/230 / 1 / 60	208/230 / 1 / 60
MINIMUM CIRCUIT AMPS	AMPS	19	11
BREAKER SIZE	AMPS	25	15
SUCTION/LIQUID LINE SIZE	IN	5/8" / 3/8"	1/2" / 1/4"
MAX PIPING LENGTH	FT	165	165
FAN COIL DIMENSIONS (LXWXDP)	IN	36X12X10	46 X 12 X 15
NOTES		ALL	ALL
NOTES: 1. R-410A REFRIGERANT, INVERTER-DRIVEN COMPRESSOR, COMPRESSOR OVERCURRENT PROTECTION, HIGH & LOW PRESSURE SWITCHES, EXTERNAL SERVICES VALVES, SHORT CYCLE PROTECTION, 5-YR COMPRESSOR WARRANTY. 2. INDOOR UNIT POWERED BY OUTDOOR UNIT. 3. PROVIDE WIRED CONTROLLER 4. INDOOR UNIT TO INCLUDE BACNET INTERFACE FOR MONITORING 5. PROVIDE BLUE DIAMOND MINI CONDENSATE PUMP W/ RESERVOIR & SENSOR 6. INCLUDE MANUFACTURER LINESET OF APPROPRIATE LENGTH. 7. INCLUDE CONDENSER WALL BRACKET.			

FAN SCHEDULE		
DESIGNATION		EF-1
AREA/ROOM SERVED & BUILDING		BATTERY STORAGE
SERVICE		BATTERY EXHAUST
MANUFACTURER		GREENHECK
MODEL		BSQ-70
TYPE		INLINE
FAN CONSTRUCTION		GALV.
DRIVE TYPE		BELT
AIR FLOWRATE DESIGN	CFM	50
DESIGN STATIC PRESSURE	IN	0.25
DESIGN FAN SPEED	RPM	860
RADIATED SOUND POWER	SONES	6.0
ELECTRICAL CHARACTERISTICS	V/Ø/HZ	208 / 1 / 60
MOTOR HORSEPOWER	HP	1/4
MIN CIRCUIT AMPACITY	AMPS	4.0
OPTIONS		3,4,5,6,7
CONTROL NOTES		4
PROJECT QTY.		1
OPTIONS: 1. INLET GUARD. 2. PREWIRED MOTOR DISCONNECT SWITCH, NEMA-1. 3. BACKDRAFT DAMPER		
CONTROL NOTES: 1. FAN WILL OPERATE CONTINUOUSLY		

DEMOLITION NOTES:
1. DEMOLISH EXISTING INLINE EXHAUST FAN AND DUCTWORK BACK TO WALL LOUVER. WALL LOUVER TO REMAIN.



DEMOLITION MECHANICAL PLAN
SCALE: 1/4"=1'-0"



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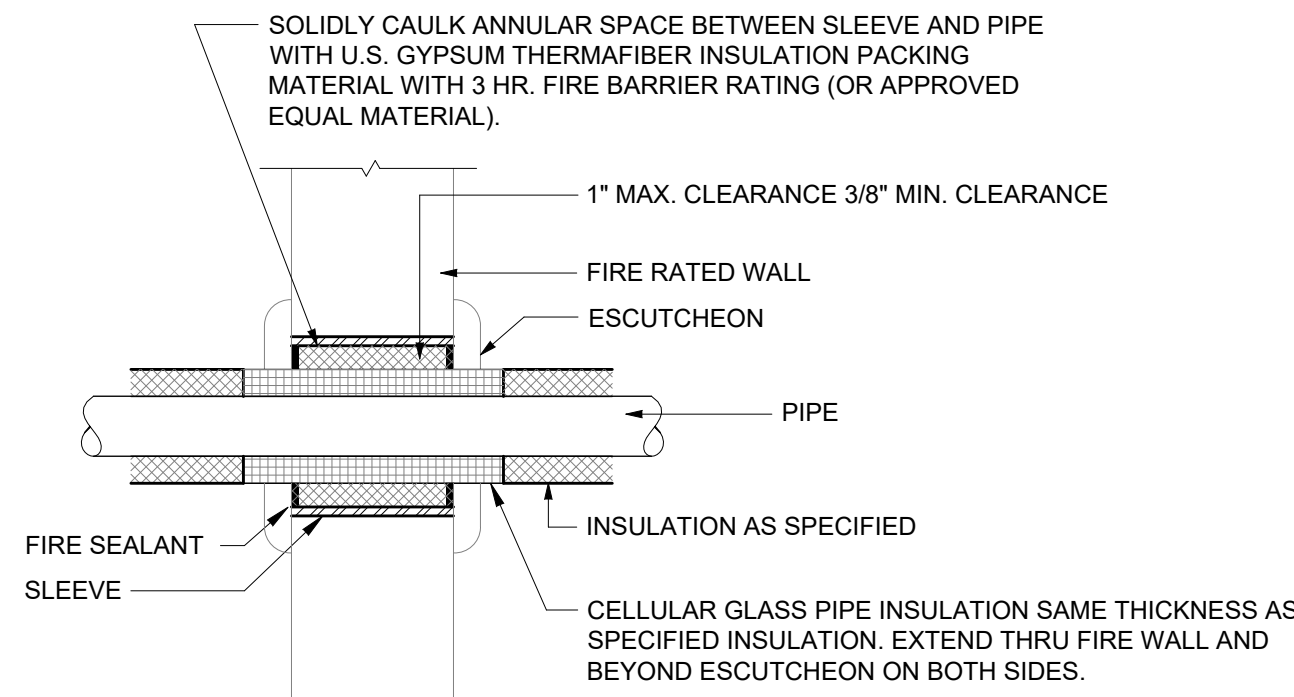
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DEMOLITION
MECHANICAL PLAN

SHEET:

M200

JOB NUMBER:

2311

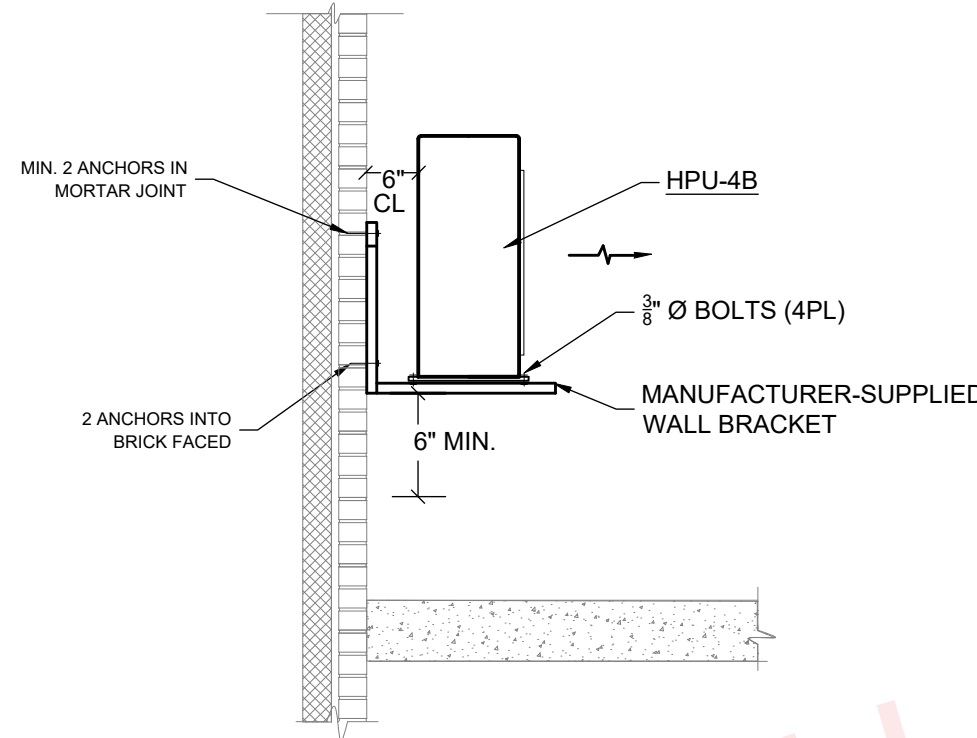


FIRE-RATED WALL PENETRATION

SCALE: NONE

NOTES:

- SUBMIT MANUFACTURER'S UL LISTED APPROVAL FOR WALL SYSTEM AND RATING TO ARCHITECT/ENGINEER FOR REVIEW/APPROVAL.
- SEE PLAN FOR WALL RATINGS.
- INSTALL PRODUCTS IN STRICT ACCORDANCE WITH THE MANUFACTURER'S REQUIREMENTS AND RATING.

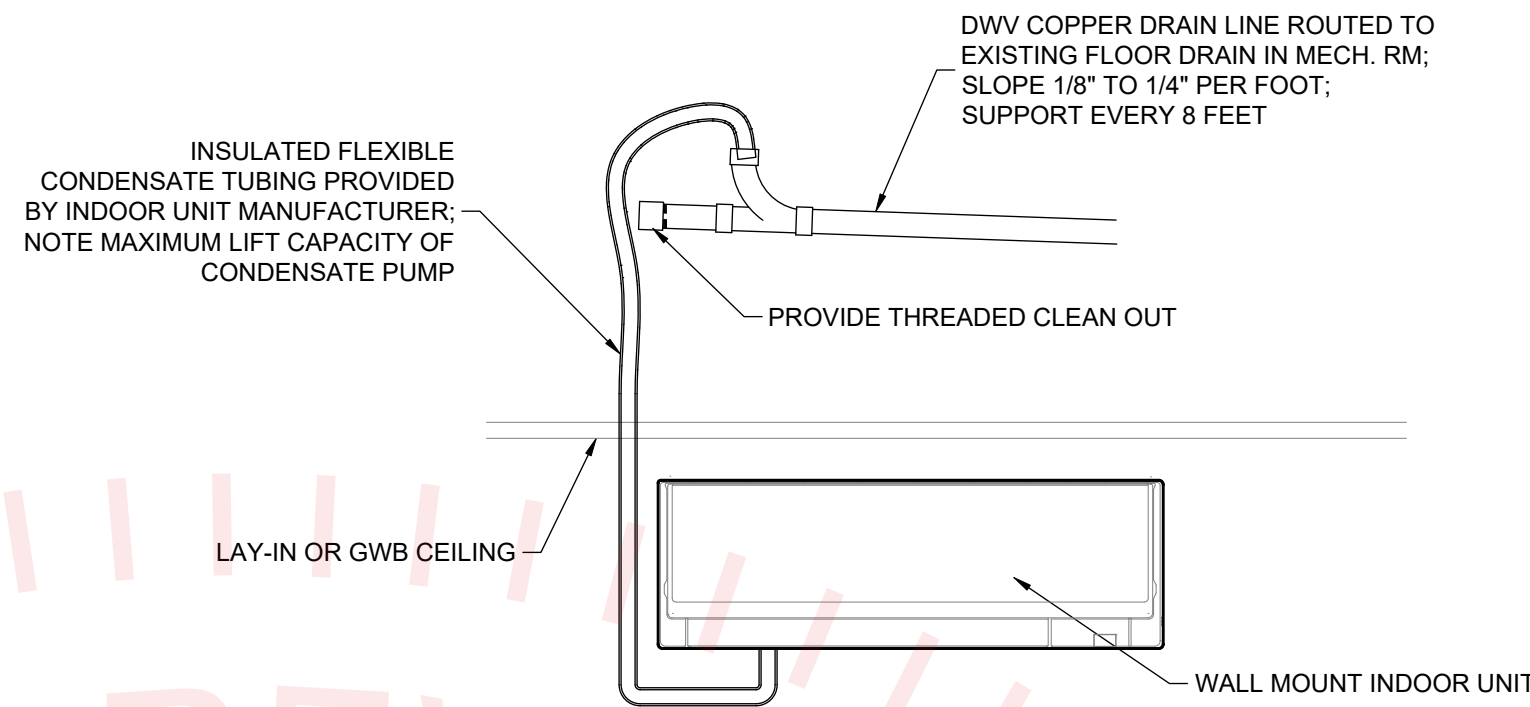


CONDENSING UNIT MOUNTING DETAIL

SCALE: NTS

NOTES:

- UNIT SHALL BE MOUNTED ON FACTORY-SUPPLIED WALL BRACKET. FOLLOW MANUFACTURER'S INSTRUCTIONS. OBSERVE MANUFACTURER'S MINIMUM RECOMMENDED CLEARANCES FOR BOTH SERVICE AND AIRFLOW.
- CONTRACTOR TO INSPECT EXISTING WALL FOR SUITABILITY OF MOUNTING UNIT. LOCATE TOP ANCHORS IN MORTAR JOINT USING MFR SUPPLIED HARDWARE. SEAL BOLT HOLES WATER TIGHT.

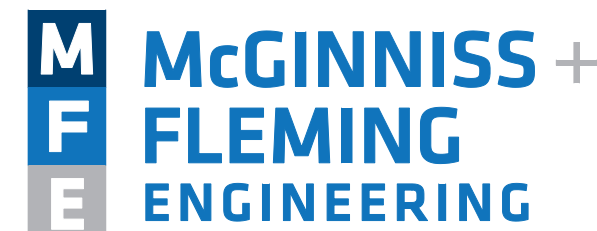


INDOOR UNIT CONDENSATE REMOVAL

SCALE: NTS

NEW WORK NOTES:

- INSTALL NEW EXHAUST FAN AND DUCTWORK AS SHOWN. EXHAUST FAN SHALL OPERATE 24/7 AND BE MONITORED FOR STATUS.
- INSTALL NEW EXHAUST DUCTWORK AS SHOWN. ROUTE DUCTWORK AHAP. DO NOT ROUTE DUCT OVER ANY ELECTRICAL EQUIPMENT. CONNECT TO EXISTING WALL LOUVER.
- INSTALL NEW COOLING ONLY DUCTLESS SPLIT UNITS WHERE SHOWN. SET FAN SPEED TO AUTO. COORDINATE EXACT EQUIPMENT LOCATION WITH ELECTRICAL TO AVOID CONFLICTS.
- INSTALL NEW CONDENSING UNITS ON WALL WHERE SHOWN. USE FACTORY-SUPPLIED WALL MOUNTING HARDWARE. MAINTAIN MANUFACTURER'S RECOMMENDED CLEARANCES. AVOID ROUTING REFRIGERANT PIPING IN FRONT OF EXISTING WALL LOUVERS.
- INSTALL NEW CONDENSATE DRAIN LINE. CONDENSATE PIPING SHALL BE DWV COPPER AND BE FULLY INSULATED ALONG ITS ENTIRE LENGTH. INSTALL DRAIN LINE AS HIGH AS POSSIBLE TO FACILITATE ADEQUATE DRAINAGE. MAINTAIN CONSTANT SLOPE; AVOID TROUGHS OR BELLIES. INSTALL CLEANOUTS WHERE SHOWN.
- DO NOT ROUTE CONDENSATE PIPING OVER ELECTRICAL EQUIPMENT.
- BLANK OFF LOWER HALF OF FILTERED INTAKE LOUVER WITH 2" THICK INSULATED SHEET METAL PANEL.



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PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 1

MSFM-02203280

Tallahassee, Florida

DATE:
March 25, 2024

REVISED:

DESIGNED BY:
JB

DRAWN BY:
MH

SUBMITTAL:
100% Construction Documents

SHEET TITLE:

NEW WORK
MECHANICAL PLAN

SHEET:

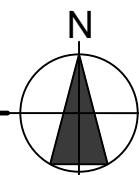
M300

JOB NUMBER:

2311

NEW WORK MECHANICAL PLAN

SCALE: 1/4"=1'-0"



GENERAL NOTES

1. PHASING SHALL BE COORDINATED AND IN COMPLIANCE WITH ALL PHASING DRAWINGS AND NOTES.
2. ALL CONDUCTORS SHALL BE INSTALLED IN METAL CONDUIT OR TUBING. FLEXIBLE CONDUIT INSTALLED OUT-OF-DOORS, IN ANY MECHANICAL EQUIPMENT ROOM, OR IN NORMALLY WET AREAS, SHALL BE LIQUID TIGHT FLEX WITH UL LISTED FITTINGS.
3. CONDUIT SHALL BE INSTALLED TIGHT TO DECK WHERE INSTALLED ABOVE CEILING. RELOCATE INCIDENTAL HANGERS, BRACKETS, ETC. WHERE IN CONFLICT. MAXIMIZE USE OF SPACE.
4. CONDUIT SHALL PASS THROUGH WALLS AT 90 DEGREES AND SHALL BE RUN PARALLEL AND PERPENDICULAR TO WALLS.
5. ALL CONDUCTORS SHALL BE COPPER WITH THHN/THWN INSULATION. CONDUCTORS SIZE # 8 AWG AND LARGER SHALL BE STRANDED.
6. NO MORE THAN 3 PHASE CONDUCTORS SHALL BE INSTALLED IN ONE CONDUIT UNLESS NOTED OTHERWISE.
7. MAINTAIN NEC MINIMUM CLEARANCE IN FRONT OF ALL SAFETY SWITCHES, PANELBOARDS, AND SWITCHGEAR.
8. ALL UNDERGROUND CONDUIT RUNS ENTERING THE BUILDING SHALL BE SEALED TO PREVENT THE ENTRANCE OF MOISTURE AND GASES.
9. ALL ITEMS ON PLANS ARE NEW UNLESS NOTED OTHERWISE.
10. ALL WALL PENETRATIONS SHALL BE SEALED USING AN APPROPRIATE UL ASSEMBLY TO MAINTAIN THE RATING OF THE WALL.
11. ALL SINGLE POLE CIRCUITS SHALL TERMINATE ON A SINGLE POLE CIRCUIT BREAKER ASSEMBLY. DO NOT USE MULTI-POLE CIRCUIT BREAKER OVERCURRENT PROVISIONS FOR 120 VOLT OR 277 VOLT CIRCUITS. PROVIDE INDIVIDUAL GROUNDED CONDUCTORS FOR ALL SINGLE POLE CIRCUITS.
12. EXISTING PANELS TO REMAIN SHALL HAVE THEIR PANEL SCHEDULES UPDATED AND RETYPED AFTER COMPLETION OF NEW WORK. NEW PANEL LOAD DESCRIPTIONS SHALL CLEARLY DEFINE LOAD TYPE AND LOCATION. PANEL SCHEDULES, BOTH FOR NEW AND EXISTING PANELS, SHALL INCLUDE CONTRACTOR COMPANY NAME, PHONE NUMBER, AND DATE SCHEDULE WAS MADE AND APPLIED.
13. WHERE CORE DRILLING OCCURS, CONTRACTOR SHALL USE CAUTION TO AVOID DAMAGING SURROUNDING SURFACES. CONTRACTOR RESPONSIBLE FOR CLEANING UP ALL SLURRY AND ENSURING A CLEAN AND PROFESSIONAL FINISH. CONTRACTOR SHALL REPLACE OR REPAIR BUILDING SURFACES OR EQUIPMENT DAMAGED BY SLURRY OR CORE DRILLING MATERIALS.
14. WHERE INSTALLING ELECTRICAL CONDUIT, MAINTAIN CONDUIT STRAPPING AND FASTENING PER NEC.
15. WHERE INSTALLING ELECTRICAL CONDUIT, MAINTAIN PULL POINTS PER NEC AND DO NOT EXCEED 360 DEGREES IN BENDS IN ANY CONDUIT RUN WITHOUT APPROPRIATE PULL POINTS.
16. ALL JUNCTION BOXES, PULLBOXES, AND DEVICES SHALL BE LABELED WITH CIRCUIT NUMBER, CIRCUIT VOLTAGE, AND FED TO / FROM PANEL LOCATIONS.
17. CONTRACTOR SHALL VERIFY WORKING SPACE AND DEDICATED ELECTRICAL SPACE EXISTS FOR PANELS BEING MODIFIED IN ACCORDANCE WITH NEC. CONTRACTOR SHALL BRING ANY DEFICIENCIES FOUND TO THE ATTENTION OF THE ENGINEER.

SCOPE OF WORK

1. FURNISH ALL LABOR, MATERIALS, EQUIPMENT AND INCIDENTALS REQUIRED TO COMPLETE ALL ELECTRICAL WORK AS SHOWN ON THE CONTRACT DRAWINGS.
2. THIS SHALL INCLUDE THE INSTALLATION OF A COMPLETE AND PROPERLY OPERATING ELECTRICAL SYSTEM. THIS SYSTEM REQUIRED CONSISTS BASICALLY OF, AND IS NOT LIMITED TO, THE FOLLOWING:
 - EXTEND THE DISTRIBUTION SYSTEM FOR POWER INCLUDING THE NECESSARY FEEDERS, BRANCH CIRCUITS, INSTALLATION OF AND CONNECTION TO DEVICES, PANELBOARDS, TRANSFORMERS, SWITCHES, MEDIUM VOLTAGE SWITCHGEAR, AND ALL OTHER EQUIPMENT SHOWN.
 - PERFORM REPLACEMENT OF THE NOTED 15KV SWITCHGEAR AND SUB-SYSTEMS, INCLUDING BUT NOT LIMITED TO, MEDIUM VOLTAGE CIRCUIT BREAKERS, INDICATING LIGHTS, METERING SYSTEM COMPONENTS, RELAY SYSTEM COMPONENTS, CTs, PTs, COMPARTMENT HEATERS, TERMINAL BOARDS, LOW VOLTAGE CONTROL POWER TRANSFORMERS, LOW VOLTAGE LOAD CENTER, SWITCHGEAR ENCLOSURE, AND ALL SWITCHGEAR WIRING AND CONNECTIONS.
 - PERFORM REPLACEMENT OF THE DC BATTERY FOR 15KV SWGR CONTROL POWER.
 - COMPLETE TESTING OF BUSDUCTS AND GROUNDING SYSTEMS AS NOTED IN PLANS AND SPECS.
 - COMPLETE LIGHTING AND LIGHTING CONTROL CHANGES AS NOTED.
 - PERFORM REPLACEMENT OF THE NOTED DRY-TYPE TRANSFORMERS.
 - EXISTING INCOMING AND OUTGOING FEEDERS SHALL BE PROTECTED DURING CONSTRUCTION, UNLESS SHOWN TO BE REPLACED. REFER TO PLANS AND ONE LINES FOR FEEDERS TO BE REPLACED AND / OR REWORKED TO ACCOMMODATE SWITCHGEAR REPLACEMENT. PROVIDE CONDUCTOR LABELING ON NEW AND EXISTING MEDIUM VOLTAGE CONDUCTORS PER DETAIL.
 - COMPLETE INITIAL TESTING OF ALL NEW 15KV CABLE INSTALLED. CABLE TESTING SHALL BE VERY LOW FREQUENCY (VLF) AND BE IN ACCORDANCE WITH THE MAINTENANCE TESTING STANDARDS AND CABLE MANUFACTURER RECOMMENDATIONS. UNLESS MANUFACTURER DATA STATES OTHERWISE, VLF TESTING SHALL BE AT 16KV RMS (22KV PEAK), 0.1Hz, FOR 1 HOUR. RECORD ALL TEST DATA AND SUBMIT TESTING REPORT TO OWNER.
 - PERFORM SWITCHING PROCEDURES AT DIFFERENT BUILDINGS AND SWITCHGEAR LOCATIONS FOR ISOLATION OF SWITCHGEAR TO BE REPLACED. REFER TO SEQUENCING PLANS.
 - PROVIDE NETWORK INFRASTRUCTURE FOR SCADA. PERFORM ALL SCADA WORK AS INDICATED ON SCADA PLAN, INCLUDING PROGRAMMING OF EXISTING PME SYSTEM. REFER TO SCADA PLAN FOR DETAILS.
 - PERFORM THE DEMOLITION WORK INDICATED.
 - THE CONTRACTOR IS RESPONSIBLE FOR ALL TEMPORARY WORK, RIGGING, CRANES AND LIFTS, TRAFFIC CONTROL AND OTHER WORK THAT IS NECESSARY TO ACCOMPLISH THE SWITCHGEAR REPLACEMENT IN A SUITABLE MANNER, INCLUDING BUT NOT LIMITED TO, FENCING AND BOLLARD REMOVAL AND PUTBACK.
 - SURFACE FLOOR WITH PLYWOOD OR SIMILAR SUITABLE MATERIAL ALONG COMPLETE TRANSPORTATION PATHWAY TO PREVENT MARRING DURING REMOVAL AND INSTALLATION OF EQUIPMENT. DO NOT ROLL OR SKID LIFTS, HOISTS, JACKS, ETC. ACROSS FINISH FLOOR. PROTECTION IS TO REMAIN FOR DURATION OF INSTALLATION. ADHERE TO ROUTING INDICATED FOR REMOVAL AND INSTALLATION OF EQUIPMENT.
 - PERFORM SCOPE OF WORK AS INDICATED FOR TEMPORARY POWER PROVISIONS DURING OUTAGES.
 - PERFORM SEALING OF ALL WALLS PER UL DETAILS WHERE EXISTING EQUIPMENT IS REMOVED, LEAVING OPENINGS WITHIN WALL, AS WELL AS, WHEN NEW PENETRATIONS ARE MADE TO RATED WALLS.
 - COMPLETE FLOOR CLEAN-UP AND LEVELING AFTER EXISTING GEAR IS REMOVED PRIOR TO INSTALLING NEW SWITCHGEAR. COMPLETE FLOOR PAINTING AS NOTED AT COMPLETION OF PROJECT SCOPE.
 - PRINT AND APPLY ARC FLASH LABELS TO EQUIPMENT. ARC FLASH LABELS WILL BE PROVIDED TO CONTRACTOR IN PDF FORMAT AND ARE INCLUDED IN THE PLANS FOR REFERENCE.
 - COMPLETE TESTING OF EXISTING EQUIPMENT AS NOTED.
3. THE BIDDER SHALL INSPECT THE PRESENT JOBSITE CONDITIONS BEFORE PREPARING HIS BID. THE SUBMISSION OF A BID WILL BE CONSIDERED EVIDENCE THAT SUCH A VISIT AND INSPECTION WAS PERFORMED BY THE BIDDER AND THAT HE TAKES FULL RESPONSIBILITY FOR ALL FACTORS GOVERNING HIS WORK.
4. THE ELECTRICAL WORK SHALL BE COMPLETE, FULLY OPERATIONAL, AND SUITABLE IN EVERY WAY FOR THE SERVICE REQUIRED. DRAWINGS ARE GENERALLY DIAGRAMMATIC IN NATURE AND DO NOT SHOW ALL DETAIL S, DEVICES AND INCIDENTAL MATERIALS NECESSARY TO ACCOMPLISH THEIR INTENT. THEREFORE, IT SHALL BE UNDERSTOOD THAT SUCH DEVICES AND INCIDENTAL MATERIALS REQUIRED SHALL BE FURNISHED AT NO COST TO THE OWNER.

DEMOLITION NOTES - GENERAL

1. THE LOCATIONS OF ALL ELECTRICAL EQUIPMENT INDICATED (FIXTURES AND DEVICES) MAY VARY FROM DRAWINGS. CONDITIONS AND DEMOLITION WORK WAS DETERMINED BY SITE OBSERVATION AND REVIEW OF EXISTING DOCUMENTS WITHOUT THE BENEFIT OF DESTRUCTIVE INVESTIGATION. VERIFY ACTUAL LOCATIONS, TYPES, CIRCUITING, AND QUANTITIES OF EQUIPMENT AND APPLY DEMOLITION NOTES AS APPROPRIATE FOR THE EQUIPMENT AND ROOM OR AREA.
2. ALL ELECTRICAL ITEMS SHOWN ON DEMOLITION PLANS ARE TO BE REMOVED UNLESS NOTED OTHERWISE. REMOVE ALL ELECTRICAL TRADE MATERIALS ASSOCIATED WITH ITEMS SHOWN TO BE REMOVED. THIS SHALL INCLUDE BUT NOT BE LIMITED TO CONDUIT AND RACEWAYS (BOTH SURFACE MOUNTED AND CONCEALED ACCESSIBLE), WIRE, DEVICES AND CONDUIT SUPPORTS, BACK TO SERVING PANEL.
3. REMOVE ALL CONDUIT STUB-UPS IN FLOOR. CUT OR GRIND CONDUITS BELOW FLOOR LEVEL AND GROUT OPENING. GROUT FINISH SHALL BE ABSOLUTELY FLAT AND READY FOR FLOOR FINISH.
4. ALL REMOVED ELECTRICAL EQUIPMENT, INCLUDING FIXTURES, SWITCHGEAR, BATTERIES, RELAYS, DISCONNECTS, MOTORS, ELECTRICAL CONDUIT, WIRE, DEVICES, AND ALL OTHER MATERIALS SHALL BE DISPOSED OF OFFSITE BY THE CONTRACTOR AT NO ADDITIONAL COST, UNLESS SPECIFICALLY NOTED OTHERWISE.
5. BLANKING PLATES SHALL MATCH NEW PLATES INSTALLED.
7. IF ANY EXISTING ELECTRICAL EQUIPMENT THAT IS TO REMAIN BECOMES ISOLATED BY THE REMOVAL OF WALLS, CEILINGS, ELECTRICAL EQUIPMENT, ETC., IT SHALL BE THE CONTRACTOR'S RESPONSIBILITY TO REROUTE AND RECONNECT THE EQUIPMENT SO THAT THE SYSTEM REMAINS OPERABLE. ANY REQUIRED WORK SHALL BE IN ACCORDANCE WITH SPECIFICATIONS.
8. RELOCATE EXISTING ELECTRICAL EQUIPMENT AS REQUIRED TO AVOID NEW CONSTRUCTION. ALL WORK REQUIRED SHALL BE PERFORMED IN ACCORDANCE WITH THE SPECIFICATIONS. PROJECT MANAGER SHALL BE NOTIFIED BEFORE PERFORMING WORK.
9. WHILE REMOVING ALL ELECTRICAL ITEMS INSIDE THE INDICATED ALTERATION AREA, RECONNECT CIRCUITS TO MAINTAIN INTEGRITY OF EXISTING CIRCUITS AND CONTINUED OPERATION OF LIGHTING, EQUIPMENT AND DEVICES TO REMAIN.
10. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ANY CUTTING OR PATCHING OF WALLS AND CEILINGS REQUIRED. CUTTING AND PATCHING WORK SHALL BE PERFORMED BY SKILLED CRAFTSMEN. RETURN ALL WALLS AND THEIR FINISHES TO ORIGINAL CONDITION.
11. ABANDONED CONDUIT SHALL BE REMOVED WHERE POSSIBLE. ABANDONED CONDUIT IN SLAB OR CONCRETE WALLS WHICH CANNOT BE REMOVED SHALL HAVE WIRES PULLED FROM THEM. CUT OFF CONDUIT FLUSH WITH CONCRETE. PATCH OPENINGS AND PENETRATIONS WITH GROUT AND GRIND OR TROWEL SMOOTH, READY FOR NEW FINISH.
12. ANY AND ALL HOLES OR DAMAGED SURFACES DUE TO DEMOLITION WORK SHALL BE PATCHED BY A SKILLED CRAFTSMAN.
13. WHERE ITEMS ARE DESIGNATED EXISTING TO REMAIN, THEY SHALL BE PROTECTED DURING DEMOLITION.
14. ANY HAZARDOUS MATERIAL OR MATERIAL POSING A THREAT TO HEALTH OR THE ENVIRONMENT SHALL BE CAREFULLY DISPOSED OF IN AN APPROVED MANNER. THIS INCLUDES, BUT IS NOT LIMITED TO, FUEL OIL AND CONTAINERS, LIGHT FIXTURE BALLASTS, BATTERIES, BATTERY ACID, OILS, GREASE OR OTHER PETROLEUM PRODUCTS, OR ANTIFREEZE COOLANT.

15KV SWITCHGEAR AUTOMATIC TRANSFER SYSTEM

AN AUTOMATIC TRANSFER SCHEME SHALL BE PROVIDED WITHIN THE NEW 15KV SWGR FOR THE CONTROL OF THE (2) MAIN CIRCUIT BREAKERS (MAIN 'A' AND MAIN 'B') AND (1) TIE CIRCUIT BREAKER TO TRANSFER THE SECONDARY LOAD SHOULD ONE OR BOTH INCOMING SUPPLIES FAIL. TRANSFER SCHEME SHALL BE SIMILAR TO THAT BELOW.

AUTOMATIC SWITCHING AT NEW 15KV SWITCHGEAR UPON LOSS OF INCOMING POWER SHALL INCLUDE A 10 SECOND DELAY PRIOR TO SWITCHING IN ORDER TO COORDINATE WITH UPSTREAM F-SUBSTATION SWITCHGEAR TO ALLOW F-SUB TO OPERATE FIRST. UPON F-SUB OPERATING, BOTH FEEDERS AT PEPPER SHOULD BECOME RE-ENERGIZED, AND THEREFORE MAY NOT NEED TO OPERATE.

ALL SWITCHING WITHIN 15KV GEAR SHALL BE CLOSED TRANSITION.

THE FOLLOWING PROTECTIVE INTERLOCKS ARE PROVIDED IN THE AUTOMATIC MODE:
AUTOMATIC OPERATION IS DISABLED IF:
1. A BREAKER IS RACKED OUT OF THE CONNECTED POSITION.
2. A BREAKER IS TRIPPED OTHER THAN BY THE AUTO-TRANSFER CONTROL.
3. THE CIRCUIT BREAKER SWITCHGEAR DOOR IS OPEN.
4. THE AUTO-TRANSFER SWITCH (MANUAL/AUTO) IS SET TO MANUAL MODE.

NORMAL MODE OF OPERATION:
MAIN BREAKER 'A' AND MAIN BREAKER 'B' CLOSED.
TIE BREAKER OPEN.

LOSS OF VOLTAGE ON MAIN 'A' ONLY:
MAIN 'A' UNDERVOLTAGE DEVICE 27 DROPS OUT.
AFTER A PROGRAMMABLE TIME DELAY (SET TO 10 SECONDS), MAIN BREAKER 'A' TRIPS OPEN THROUGH AUTOMATIC TRIP CIRCUIT.
AFTER MAIN BREAKER 'A' OPENS, TIE BREAKER TO CLOSE THROUGH AUTOMATIC CLOSE CIRCUIT.
MAIN 'A' SOURCE IS MONITORED FOR POSITIVE SEQUENCE VOLTAGE.

RESTORATION OF VOLTAGE ON MAIN 'A':
MAIN 'A' UNDERVOLTAGE DEVICE 27 PICKS UP.
FOLLOWING PROGRAMMABLE SOURCE-STABILIZATION TIME PERIOD (SET TO 10 MINUTES), MAIN BREAKER 'A' CLOSSES THROUGH THE AUTOMATIC CLOSE CIRCUIT AND TIE BREAKER TRIPS OPEN. (CLOSED TRANSITION)

LOSS OF VOLTAGE ON MAIN 'B' ONLY:
MAIN 'B' UNDERVOLTAGE DEVICE 27 DROPS OUT.
AFTER A PROGRAMMABLE TIME DELAY (SET TO 10 SECONDS), MAIN BREAKER 'B' TRIPS OPEN THROUGH AUTOMATIC TRIP CIRCUIT.
AFTER MAIN BREAKER 'B' OPENS, TIE BREAKER TO CLOSE THROUGH AUTOMATIC CLOSE CIRCUIT.
MAIN 'B' SOURCE IS MONITORED FOR POSITIVE SEQUENCE VOLTAGE.

RESTORATION OF VOLTAGE ON MAIN 'B':
MAIN 'B' UNDERVOLTAGE DEVICE 27 PICKS UP.
FOLLOWING PROGRAMMABLE SOURCE-STABILIZATION TIME PERIOD (SET TO 10 MINUTES), MAIN BREAKER 'B' CLOSSES THROUGH THE AUTOMATIC CLOSE CIRCUIT AND TIE BREAKER TRIPS OPEN. (CLOSED TRANSITION)

SIMULTANEOUS LOSS OF VOLTAGE ON MAIN 'A' AND MAIN 'B':
MAIN 'A' AND 'B' UNDERVOLTAGE DEVICES 27 DROP OUT.
UTILITY SOURCES 'A' AND 'B' ARE MONITORED FOR POSITIVE SEQUENCE VOLTAGE.
MAIN BREAKERS REMAIN CLOSED AND TIE BREAKER REMAINS OPEN.

SIMULTANEOUS RESTORATION OF VOLTAGE ON MAIN 'A' AND MAIN 'B':
MAIN 'A' AND 'B' UNDERVOLTAGE DEVICES 27 PICK UP.
BOTH MAIN BREAKERS 'A' AND 'B' REMAIN IN CLOSED POSITION WITH TIE BREAKER OPEN.

RESTORATION OF VOLTAGE ON SINGLE MAIN AFTER LOSS OF BOTH SOURCES ('A' -OR- 'B'):
ASSOCIATED MAIN UNDERVOLTAGE DEVICE 27 PICKS UP.
ASSOCIATED MAIN BREAKER REMAINS CLOSED.
AFTER PROGRAMMABLE TIME DELAY (SET TO 10 SECONDS), THE OTHER MAIN BREAKER TRIPS OPEN THROUGH AUTOMATIC TRIP CIRCUIT.
AFTER OTHER MAIN BREAKER OPENS, TIE BREAKER TO CLOSE THROUGH AUTOMATIC CLOSE CIRCUIT.
THE OTHER MAIN SOURCE IS MONITORED FOR POSITIVE SEQUENCE VOLTAGE.

RESTORATION OF VOLTAGE ON REMAINING MAIN:
ASSOCIATED MAIN UNDERVOLTAGE DEVICE 27 PICKS UP.
FOLLOWING PROGRAMMABLE SOURCE-STABILIZATION TIME PERIOD (SET TO 10 MINUTES), ASSOCIATED MAIN BREAKER CLOSSES THROUGH THE AUTOMATIC CLOSE CIRCUIT AND TIE BREAKER TRIPS OPEN. (CLOSED TRANSITION)

SCHEDULED OUTAGE NOTES

1. THE CONTRACTOR IS EXPECTED TO PLAN OUT THE EXECUTION OF THE WORK SO AS TO MINIMIZE INTERFERENCE TO THE OWNER'S OPERATIONS. PHASING AND PREPARATION FOR THE WORK SHALL TAKE PLACE SO AS TO ACHIEVE THIS REQUIREMENT.
2. ELECTRICAL POWER OUTAGES TO THE BUILDING IS A REQUIREMENT OF THIS PROJECT AND SHALL BE THOROUGHLY PLANNED AND COORDINATED WITH DMS.
3. PROVIDE COMPLETE PROPOSED CONSTRUCTION SCHEDULE TO ENGINEER AND DMS PROJECT MANAGER FOR REVIEW AND APPROVAL PRIOR TO BEGINNING CONSTRUCTION.
4. ELECTRICAL OUTAGES WILL BE LIMITED TO AFTER HOURS ON NIGHTS AND WEEKENDS. THE CONTRACTOR SHALL PLAN AND STAFF THE PROJECT ACCORDINGLY TO MEET THAT TIMEFRAME.
5. SCHEDULE A MEETING WITH THE ENGINEER, THE OWNER'S PROJECT MANAGER, AND THE CONTRACTOR'S SUPERVISOR WITHIN TWO WEEKS AFTER ACTIVATION TO DISCUSS CONTRACTOR'S STRATEGY FOR OUTAGES AND ACCOMPLISHING THE WORK.
6. COORDINATE ALL POWER OUTAGES WITH DMS PROJECT MANAGER PRIOR TO OUTAGE. PROVIDE 21 DAY MINIMUM NOTICE TO DMS PROJECT MANAGER AS TO WHEN PROPOSED POWER OUTAGES WILL OCCUR TO RECEIVE APPROVAL. ALL POWER OUTAGES AND CHANGEOVERS SHALL BE SCHEDULED WITH THE DMS PROJECT MANAGER AND SCHEDULES STRICTLY ADHERED TO. IN GENERAL, ALL OUTAGES SHALL BE DURING NIGHTS OR WEEKENDS ONLY. OUTAGES DURING NORMAL BUSINESS HOURS WILL NOT BE PERMITTED. OUTAGES THAT ARE NOT SCHEDULED WILL NOT BE ALLOWED.
7. CONTRACTOR SHALL BE RESPONSIBLE FOR TEMPORARY LOW VOLTAGE CONSTRUCTION POWER AND LIGHTING DURING OUTAGES.
8. THE CONTRACTOR SHALL PROVIDE A TRAINED AND KNOWLEDGEABLE STAFF MEMBER REPRESENTATIVE TO CONSULT WITH THE OWNER PRIOR TO ANY SWITCHING, AND TO PERFORM SWITCHING, TEST CIRCUITS, AND DEEM GROUNDED AND SAFE PRIOR TO WORK.
9. COORDINATION WITH DMS MUST OCCUR FOR ENERGIZING AND DE-ENERGIZING FEEDERS.
10. CONTRACTOR IS RESPONSIBLE FOR ALL SWITCHING OF OWNER'S EQUIPMENT AS NEEDED TO COMPLETE SCOPE OF WORK.
11. MULTIPLE SCHEDULED POWER OUTAGES WILL REMOVE NORMAL POWER TO THE BUILDING. BUILDING WILL BE PROVIDED WITH EMERGENCY POWER FROM EXISTING GENERATOR FOR THE DURATION OF THE OUTAGE, WITH EXCEPTION BEING IF ANY WORK TAKES PLACE WITHIN LIFE SAFETY ENERGIZED EQUIPMENT, AT WHICH CASE THE LIFE SAFETY SYSTEM MAY BE DE-ENERGIZED. REFER TO SEQUENCING PLAN FOR QUANTITY AND ESTIMATED DURATION OF OUTAGES.
12. THE OLITS DATA CENTER LOCATED WITHIN PEPPER BUILDING WILL ALSO BE AFFECTED BY BUILDING OUTAGES AND MUST BE COORDINATED WITH PRIOR TO REMOVING POWER TO THE BUILDING. OLITS DATA CENTER WILL UTILIZE THEIR OWN DEDICATED GENERATOR AND CHILLED WATER SYSTEM FOR THE DURATION OF ALL OUTAGES. CONTRACTOR SHALL COORDINATE WITH OLITS PERSONNEL FOR SCHEDULING OUTAGES. OLITS PERSONNEL WILL BE RESPONSIBLE FOR RUNNING THEIR GENERATOR AND CHILLED WATER SYSTEMS DURING OUTAGES.
13. CONTRACTOR SHALL MAINTAIN FUEL DURING ALL OUTAGES TO KEEP BOTH BUILDING LIFE SAFETY AND OLITS GENERATORS RUNNING FOR DURATION OF OUTAGES.
14. CONTRACTOR SHALL PROVIDE GENERATOR FUEL FOLLOWING THE OUTAGE TO TOP OFF THE FUEL TANKS FOR EACH OF THE BUILDING GENERATORS.
15. THE TWO BUILDING GENERATORS AND FUEL TANK SIZES ARE AS FOLLOWS:
 - PEPPER LIFE SAFETY GENERATOR #1 - 500KW, 500 GALLON TANK
 - OLITS DATA CENTER GENERATOR #2 - 1000KW, 2500 GALLON TANK
16. CONTRACTOR SHALL INCLUDE IN THE BID AN ALLOWANCE OF \$8,000 FOR FUEL CHARGES FOR GENERATORS #1 AND #2 NOTED ABOVE AND SHALL COORDINATE TOP OFF OF EACH FUEL STORAGE TANK NOTED ABOVE WITH #2 FUEL OIL AT COMPLETION OF EACH OUTAGE. PORTION OF ALLOWANCE NOT USED SHALL BE RETURNED TO THE OWNER VIA DEDUCTIVE CHANGE ORDER AT COMPLETION OF PROJECT.

15KV RELAY COORDINATION STUDY

MAIN A PHASE CT RATIO: ZERO SEQUENCE GF CT RATIO: 51P1P / 51P1C / 51P1TD 51N1P / 51N1C / 51N1TD 50P1P 50P1D 50N1P 50N1D 50PAPP AF TOL1P	400:5 50:5 2 (160A) / U3 / 1 8 (80A) / U3 / 1 20 (1600A) 0.001 25 (250A) 0.001 4 (320A) 3%	OLITS #1 FEEDER PHASE CT RATIO: ZERO SEQUENCE GF CT RATIO: 51P1P / 51P1C / 51P1TD 51N1P / 51N1C / 51N1TD 50P1P 50P1D 50N1P 50N1D 50PAPP AF TOL1P	200:5 50:5 3 (120A) / U3 / 1 6 (60A) / U3 / 1 24 (960A) 0.001 20 (200A) 0.001 5 (200A) 3%
MAIN B PHASE CT RATIO: ZERO SEQUENCE GF CT RATIO: 51P1P / 51P1C / 51P1TD 51N1P / 51N1C / 51N1TD 50P1P 50P1D 50N1P 50N1D 50PAPP AF TOL1P	400:5 50:5 2 (160A) / U3 / 1 8 (80A) / U3 / 1 20 (1600A) 0.001 25 (250A) 0.001 4 (320A) 3%	OLITS #2 FEEDER PHASE CT RATIO: ZERO SEQUENCE GF CT RATIO: 51P1P / 51P1C / 51P1TD 51N1P / 51N1C / 51N1TD 50P1P 50P1D 50N1P 50N1D 50PAPP AF TOL1P	200:5 50:5 3 (120A) / U3 / 1 6 (60A) / U3 / 1 24 (960A) 0.001 20 (200A) 0.001 5 (200A) 3%
TIE BREAKER PHASE CT RATIO: 51P1P / 51P1C / 51P1TD 51N1P / 51N1C / 51N1TD 50P1P 50P1D 50N1P 50N1D 50PAPP AF TOL1P	400:5 50:5 2 (160A) / U3 / 1 8 (80A) / U3 / 1 20 (1600A) 0.001 25 (250A) 0.001 4 (320A) 3%	TX-A FEEDER PHASE CT RATIO: ZERO SEQUENCE GF CT RATIO: 51P1P / 51P1C / 51P1TD 51N1P / 51N1C / 51N1TD 50P1P 50P1D 50N1P 50N1D 50PAPP AF TOL1P	200:5 50:5 3 (120A) / U3 / 1 6 (60A) / U3 / 1 35 (1400A) 0.001 20 (200A) 0.001 5 (200A) 3%
TX-B FEEDER PHASE CT RATIO: ZERO SEQUENCE GF CT RATIO: 51P1P / 51P1C / 51P1TD 51N1P / 51N1C / 51N1TD 50P1P 50P1D 50N1P 50N1D 50PAPP AF TOL1P	200:5 50:5 3 (120A) / U3 / 1 6 (60A) / U3 / 1 35 (1400A) 0.001 20 (200A) 0.001 5 (200A) 3%		

CODES AND REQUIREMENTS

THE INSTALLATION SHALL COMPLY WITH THE INDICATED EDITION OF THE FOLLOWING CODES AND ORDINANCES. WHERE SPECIFIC EDITION IS NOT INDICATED, COMPLY WITH THE LATEST PUBLISHED EDITION.

NATIONAL FIRE PROTECTION ASSOCIATION - NFPA
NFPA 70 - 2020; NATIONAL ELECTRICAL CODE

FLORIDA BUILDING CODE
FBC-B 2023; THE FLORIDA BUILDING CODE (8th EDITION)
FPC 2023; THE FLORIDA FIRE PREVENTION CODE (8th EDITION)
INCLUDING NFPA 101 - 2021; THE LIFE SAFETY CODE
FBC-M 2023; THE FLORIDA MECHANICAL CODE (8th EDITION)
FBC-A 2023; THE FLORIDA BUILDING CODE, ACCESSIBILITY (8th EDITION)
FBC-EC 2023; THE FLORIDA BUILDING CODE, ENERGY CONSERVATION (8th EDITION)
FBC-EB 2023; THE FLORIDA BUILDING CODE, EXISTING BUILDING (8th EDITION)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

IEEE C2; NATIONAL ELECTRICAL SAFETY CODE

SERVING UTILITY COMPANY POLICIES

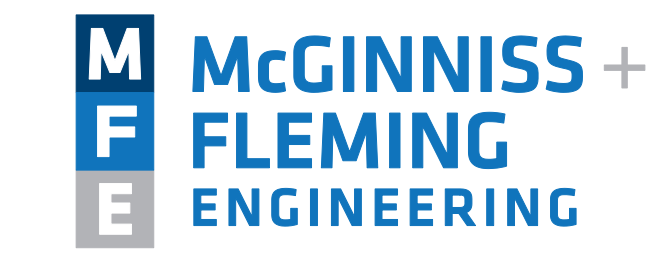
STATE AND MUNICIPAL CODES AND REQUIREMENTS

PM AND TESTING ANNUAL SERVICES REQUEST

CONTRACTOR SHALL INCLUDE WITH THE BID A QUOTE FOR ANNUAL PREVENTATIVE MAINTENANCE ON THE MEDIUM VOLTAGE SWITCHGEAR, LOW VOLTAGE SWITCHGEAR, DRY-TYPE MEDIUM VOLTAGE TRANSFORMERS, AND DC BATTERY SYSTEM.

PREVENTATIVE MAINTENANCE AND TESTING SHALL BE IN ACCORDANCE WITH SPECIFICATION SECTION 01 78 23.

CONTRACTOR'S QUOTE SHALL INCLUDE ANNUAL PM AND TESTING SERVICES FOR TEN (10) YEARS. OWNER WILL REVIEW QUOTE AT TIME OF BID AND DETERMINE IF ACCEPTABLE.



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PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 1
MSFM-02203280
Tallahassee, Florida

DATE:
March 25, 2024

REVISED:

DESIGNED BY: BW	DRAWN BY: TB
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SUBMITTAL:
100% Construction Documents

SHEET TITLE:

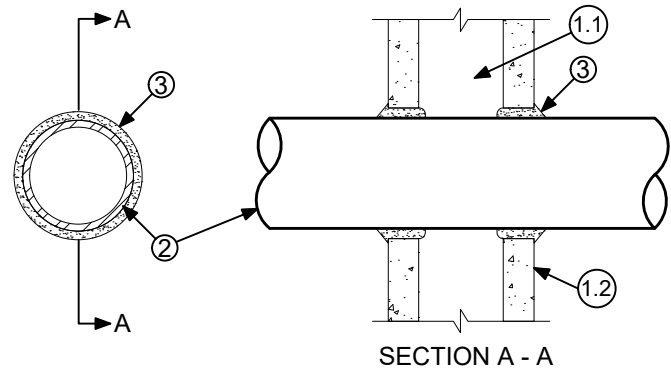
GENERAL NOTES

SHEET:

E101

JOB NUMBER:
2311

SYSTEM NO. W-L-1001
(FORMERLY SYSTEM NO. 147)
F RATINGS - 1, 2, 3 AND 4 HOUR (SEE ITEMS 2 & 3)
T RATING - 0, 1, 2, 3 AND 4 HOUR (SEE ITEM 3)
L RATING AT AMBIENT - LESS THAN 1 CFM/SQ FT
L RATING AT 400°F - LESS THAN 1 CFM/SQ FT

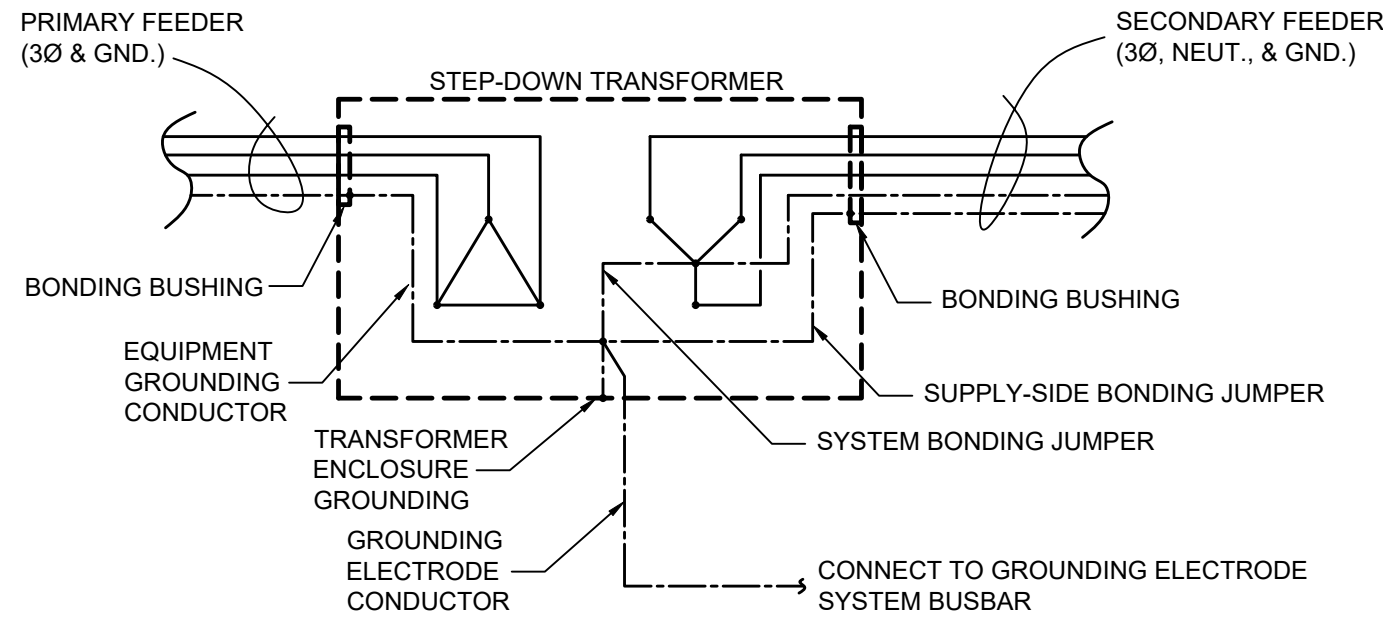


- WALL ASSEMBLY - THE 1, 2, 3 OR 4 HOUR FIRE-RATED GYPSUM WALLBOARD / STUD WALL ASSEMBLY SHALL BE CONSTRUCTED OF THE MATERIALS AND IN THE MANNER DESCRIBED IN THE INDIVIDUAL U300 OR U400 SERIES WALL OR PARTITION DESIGNS IN THE UL FIRE RESISTANCE DIRECTORY AND SHALL INCLUDE THE FOLLOWING CONSTRUCTION FEATURES:
 - STUDS - WALL FRAMING MAY CONSIST OF EITHER WOOD STUDS (MAXIMUM 2 HOUR FIRE RATED ASSEMBLIES) OR STEEL CHANNEL STUDS. WOOD STUDS TO CONSIST OF NOMINAL 2"x4" LUMBER SPACED 16" O.C. WITH NOMINAL 2"x4" LUMBER END PLATES AND CROSS BRACES. STEEL STUDS TO BE MINIMUM 3-5/8" WIDE BY 1-3/8" DEEP CHANNELS SPACED MAXIMUM OF 24" O.C.
 - WALLBOARD, GYPSUM - NOMINAL 1/2" OR 5/8" THICK, 4' WIDE WITH SQUARE OR TAPERED EDGES. THE GYPSUM WALLBOARD TYPE, THICKNESS, NUMBER OF LAYERS, FASTENER TYPE AND SHEET ORIENTATION SHALL BE AS SPECIFIED IN THE INDIVIDUAL U300 OR U400 SERIES DESIGN IN THE UL FIRE RESISTANCE DIRECTORY. MAXIMUM DIAMETER OF OPENING IS 13-1/2".
- PIPE OR CONDUIT - NOMINAL 12" DIAMETER (OR SMALLER) SCHEDULE 10 (OR HEAVIER) STEEL PIPE, NOMINAL 12" DIAMETER (OR SMALLER) SERVICE WEIGHT (OR HEAVIER) CAST IRON SOIL PIPE, NOMINAL 12" DIAMETER (OR SMALLER) CLASS 50 (OR HEAVIER) DUCTILE IRON PRESSURE PIPE, NOMINAL 6" DIAMETER (OR SMALLER) STEEL CONDUIT, NOMINAL 4" DIAMETER (OR SMALLER) STEEL ELECTRICAL METALLIC TUBING, NOMINAL 6" DIAMETER (OR SMALLER) TYPE L (OR HEAVIER) COPPER TUBING OR NOMINAL 1" DIAMETER (OR SMALLER) FLEXIBLE STEEL CONDUIT. WHEN COPPER PIPE IS USED, MAXIMUM F RATING OF FIRESTOP SYSTEM (ITEM 3) IS 2 HOUR. STEEL PIPES OR CONDUITS LARGER THAN NOMINAL 4" DIAMETER MAY ONLY BE USED IN WALLS CONSTRUCTED USING STEEL CHANNEL STUDS. A MAXIMUM OF ONE PIPE OR CONDUIT IS PERMITTED IN THE FIRESTOP SYSTEM. PIPE OR CONDUIT TO BE INSTALLED NEAR CENTER OF STUD CAVITY WIDTH AND TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF WALL ASSEMBLY.
- FILL, VOID OR CAVITY MATERIAL - CAULK - CAULK FILL MATERIAL INSTALLED TO COMPLETELY FILL ANNULAR SPACE BETWEEN PIPE OR CONDUIT AND GYPSUM WALLBOARD AND WITH A MINIMUM 1/4" DIAMETER BEAD OF CAULK APPLIED TO PERIMETER OF PIPE OR CONDUIT AT ITS EGRESS FROM THE WALL. CAULK INSTALLED SYMMETRICALLY ON BOTH SIDES OF WALL ASSEMBLY. THE HOURLY F RATING OF THE FIRESTOP SYSTEM IS DEPENDENT UPON THE HOURLY FIRE RATING OF THE WALL ASSEMBLY IN WHICH IT IS INSTALLED, AS SHOWN IN THE FOLLOWING TABLE. THE HOURLY T RATING OF THE FIRESTOP SYSTEM IS DEPENDENT UPON THE TYPE OR SIZE OF THE PIPE OR CONDUIT AND THE HOURLY FIRE RATING OF THE WALL ASSEMBLY IN WHICH IT IS INSTALLED, AS TABULATED BELOW:

MAXIMUM PIPE DIAMETER (INCHES)	ANNULAR SPACE (INCHES)	F RATING (HOURS)	T RATING (HOURS)
4	0 - 1-1/2	1 OR 2	0

PENETRATION DETAIL — A

NOT TO SCALE

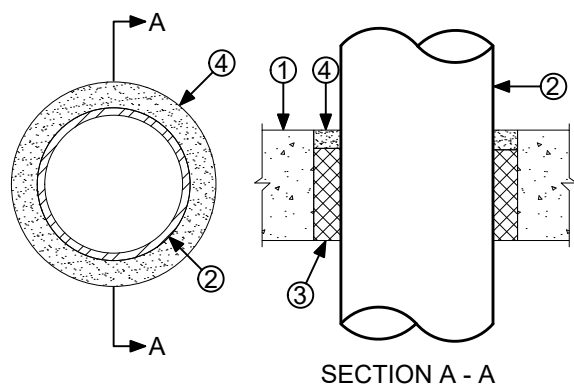


* REFER TO POWER ONE LINE FOR CONDUCTOR SIZING

TRANSFORMER GROUNDING DETAIL

NOT TO SCALE

SYSTEM NO. C-AJ-1014
(FORMERLY SYSTEM NO. 133)
F RATING - 2 HOUR
T RATING - 0 HOUR
L RATING AT AMBIENT - LESS THAN 1 CFM/SQ FT (SEE ITEM 4)
L RATING AT 400°F - LESS THAN 1 CFM/SQ FT (SEE ITEM 4)

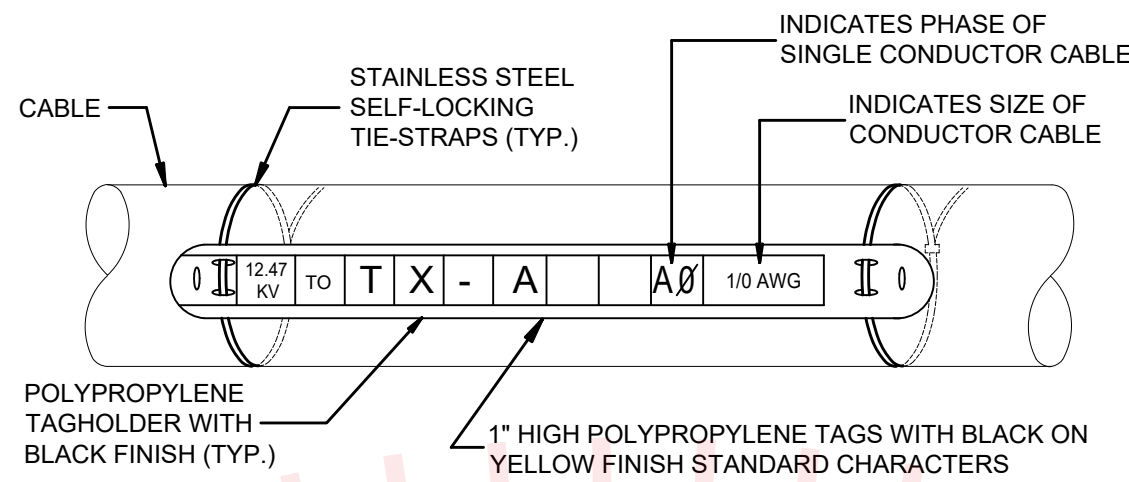


- FLOOR OR WALL ASSEMBLY - MIN 3-1/4" THICK REINFORCED LIGHTWEIGHT OR NORMAL WEIGHT (100-150 PCF) CONCRETE. WALL MAY ALSO BE CONSTRUCTED OF ANY UL CLASSIFIED CONCRETE BLOCKS*. MAX DIAMETER OF CIRCULAR OPENING IS 6". SEE CONCRETE BLOCK (CAZT) CATEGORY IN THE FIRE RESISTANCE DIRECTORY FOR NAMES OF MANUFACTURERS.
- THROUGH PENETRANTS - ONE METALLIC PIPE OR CONDUIT TO BE CENTERED WITHIN THE FIRESTOP SYSTEM. A NOMINAL ANNULAR SPACE OF 3/4" IS REQUIRED WITHIN THE FIRESTOP SYSTEM. PIPE OR CONDUIT TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF FLOOR OR WALL ASSEMBLY. THE FOLLOWING TYPES AND SIZES OF METALLIC PIPES OR CONDUITS MAY BE USED:
 - STEEL PIPE - NOMINAL 4" DIAMETER (OR SMALLER) SCHEDULE 10 (OR HEAVIER) STEEL PIPE.
 - CONDUIT - NOMINAL 4" DIAMETER (OR SMALLER) STEEL ELECTRICAL METALLIC TUBING OR STEEL CONDUIT.
- PACKING MATERIAL - MINIMUM 4 PCF MINERAL WOOL BATT INSULATION FIRMLY PACKED INTO OPENING AS A PERMANENT FORM. PACKING MATERIAL TO BE RECESSED FROM TOP SURFACE OF FLOOR OR FROM BOTH SURFACES OF WALL AS REQUIRED TO ACCOMMODATE THE REQUIRED THICKNESS OF FILL MATERIAL. MINIMUM THICKNESS OF PACKING MATERIAL IN FLOORS AND WALLS TO BE 2-3/4" AND 2-1/4", RESPECTIVELY.
- FILL, VOID OR CAVITY MATERIAL* - SEALANT - MINIMUM 1/2" THICKNESS OF FILL MATERIAL APPLIED WITHIN ANNULUS, FLUSH WITH TOP SURFACE OF FLOOR OR WITH BOTH SURFACES OF WALL. AS AN ALTERNATE, THE PERMANENT FORMING MATERIAL (ITEM 3) MAY BE OMITTED IF THE FILL MATERIAL THICKNESS IS INCREASED TO A MINIMUM OF 1-1/2". MINNESOTA MINING AND MFG CO. - (TYPES FB-2000 OR FB-2000+). (NOTE: L RATINGS APPLY ONLY WHEN FB-2000+ IS USED.)

*BEARING THE UL CLASSIFICATION MARKING.

PENETRATION DETAIL — B

NOT TO SCALE



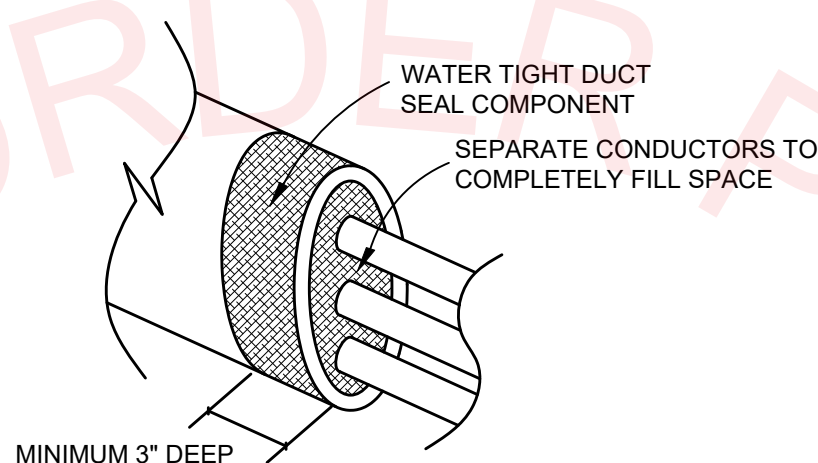
CIRCUITS WITHIN GEAR

PROVIDE THE FOLLOWING FOR EACH PHASE:

- TO TX-A
- TO TX-B
- TO OLITS 1
- TO OLITS 2
- FROM KNOTT SW A
- FROM KNOTT SW B

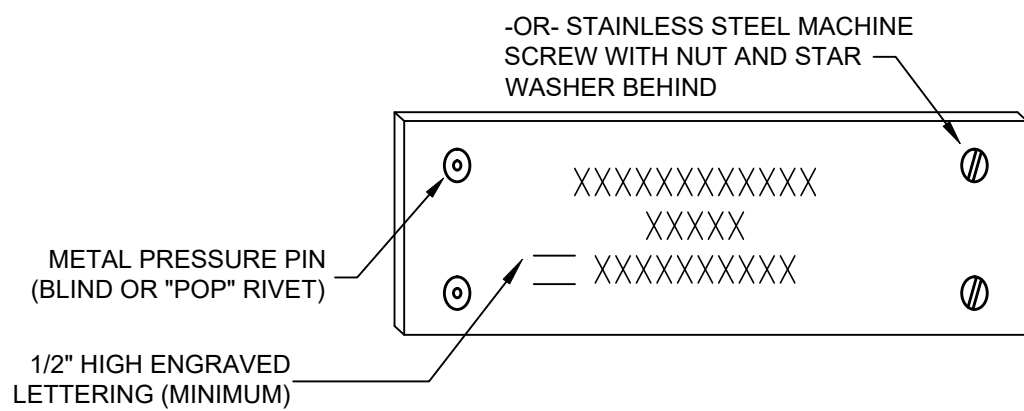
CONDUCTOR LABELS DETAIL

NOT TO SCALE



CONDUIT DUCT SEAL

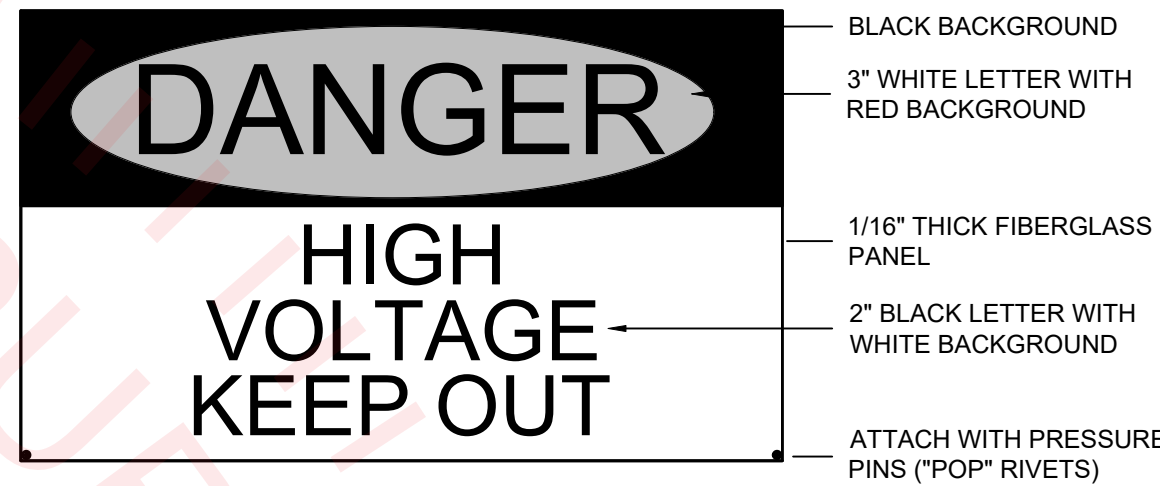
NOT TO SCALE



EQUIPMENT IDENTIFICATION SHALL BE MADE USING ENGRAVED LAMINATED PLASTIC PLATES (INDENTED TAPE LABELS WILL NOT BE PERMITTED). CHARACTERS SHALL BE WHITE ON A BLACK BACKGROUND AND 1/2" HIGH MINIMUM. PLATES SHALL BE SECURED TO THE PANELS BY MEANS OF SCREWS OR METAL PRESSURE PINS. ADHESIVES FASTENING ALONE WILL NOT BE ACCEPTABLE. ALL NAMEPLATES SHALL BE MOUNTED ON THE OUTSIDE SURFACE OF THE PIECE OF EQUIPMENT.

NAMEPLATE DETAIL

NOT TO SCALE

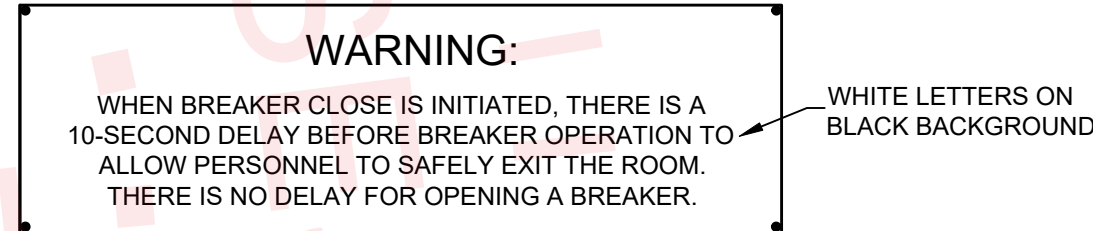


PROVIDE NEW WARNING LABELS FOR EACH DOOR OF SWITCHGEAR

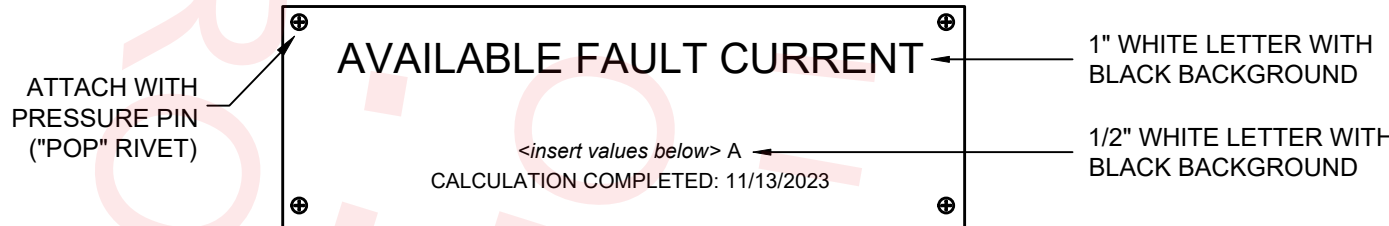
WARNING PLATE DETAIL

NOT TO SCALE

INSTALL ON 15KV SWITCHGEAR ADJACENT TO BREAKER CONTROLS AND TOUCHSCREEN

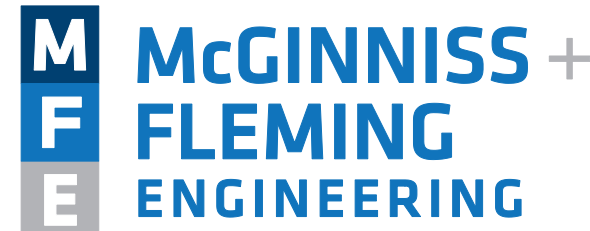


INSTALL ON MAIN BREAKER SECTIONS AT 15KV GEAR



BREAKER	VALUE
15KV MAIN A	25.76 kA
15KV MAIN B	25.76 kA
OLITS 15KV MAIN A	25.62 kA
OLITS 15KV MAIN B	25.68 kA

VALUES SHOWN HERE ARE PRELIMINARY AND WILL BE FINALIZED BY ENGINEER PRIOR TO COMPLETION OF CONSTRUCTION. SIGNAGE SHALL INCLUDE FINAL VALUES.



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PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 1

MSFM-02203280

Tallahassee, Florida

DATE:
March 25, 2024

REVISED:

DESIGNED BY:
BW

DRAWN BY:
TB

SUBMITTAL:
100% Construction Documents

SHEET TITLE:
ELECTRICAL DETAILS

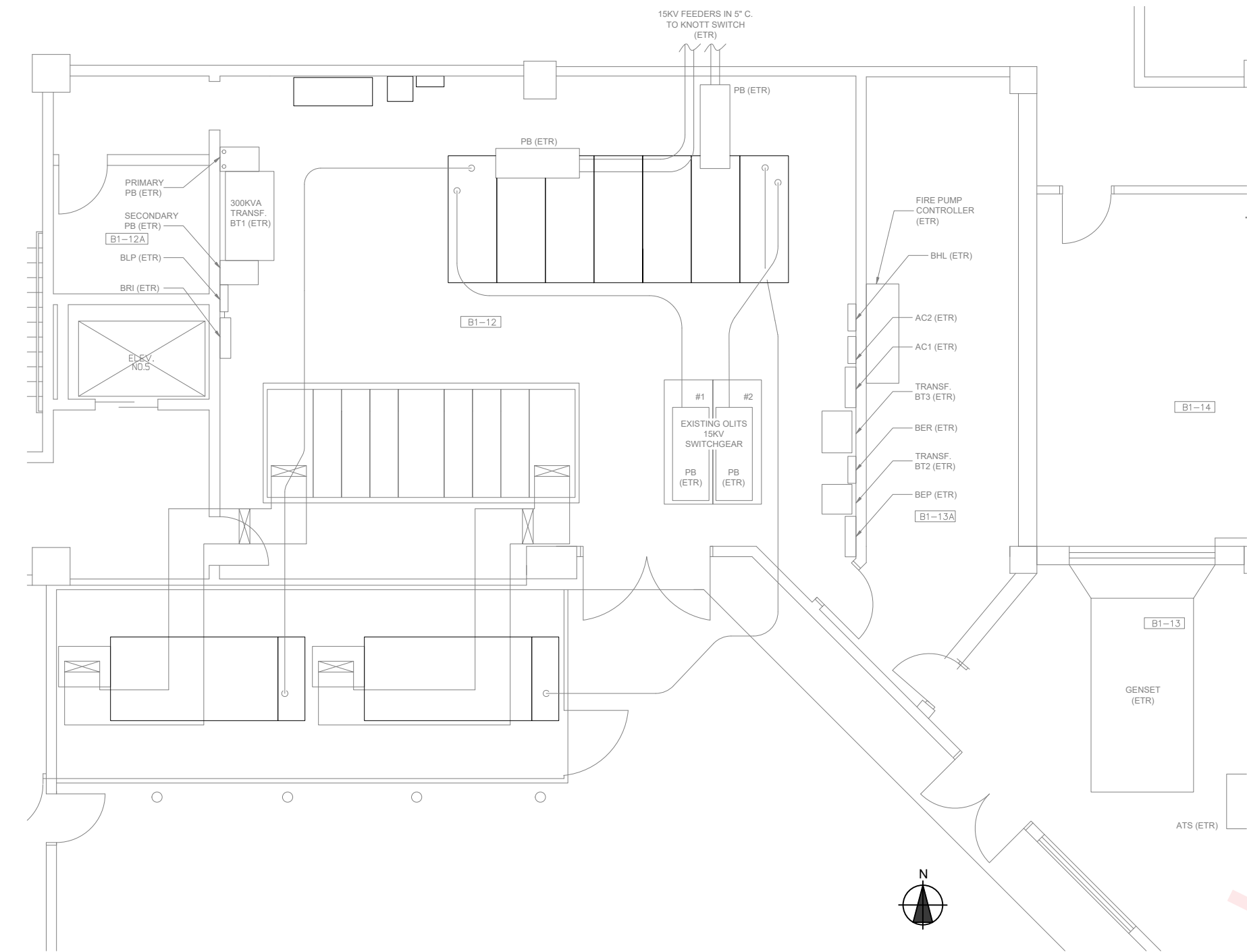
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E102

JOB NUMBER:

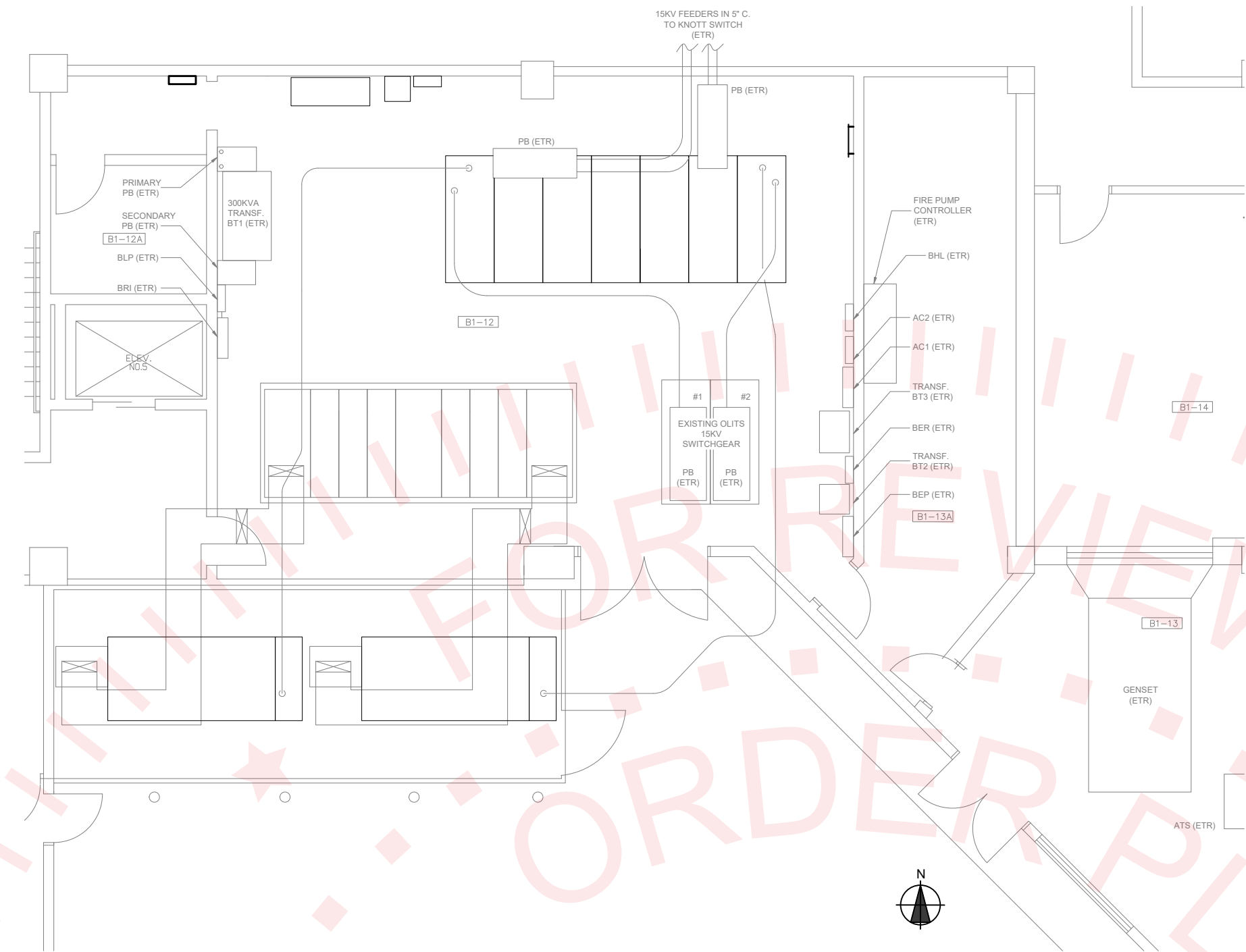
2311

PHASE 1 (INITIAL)



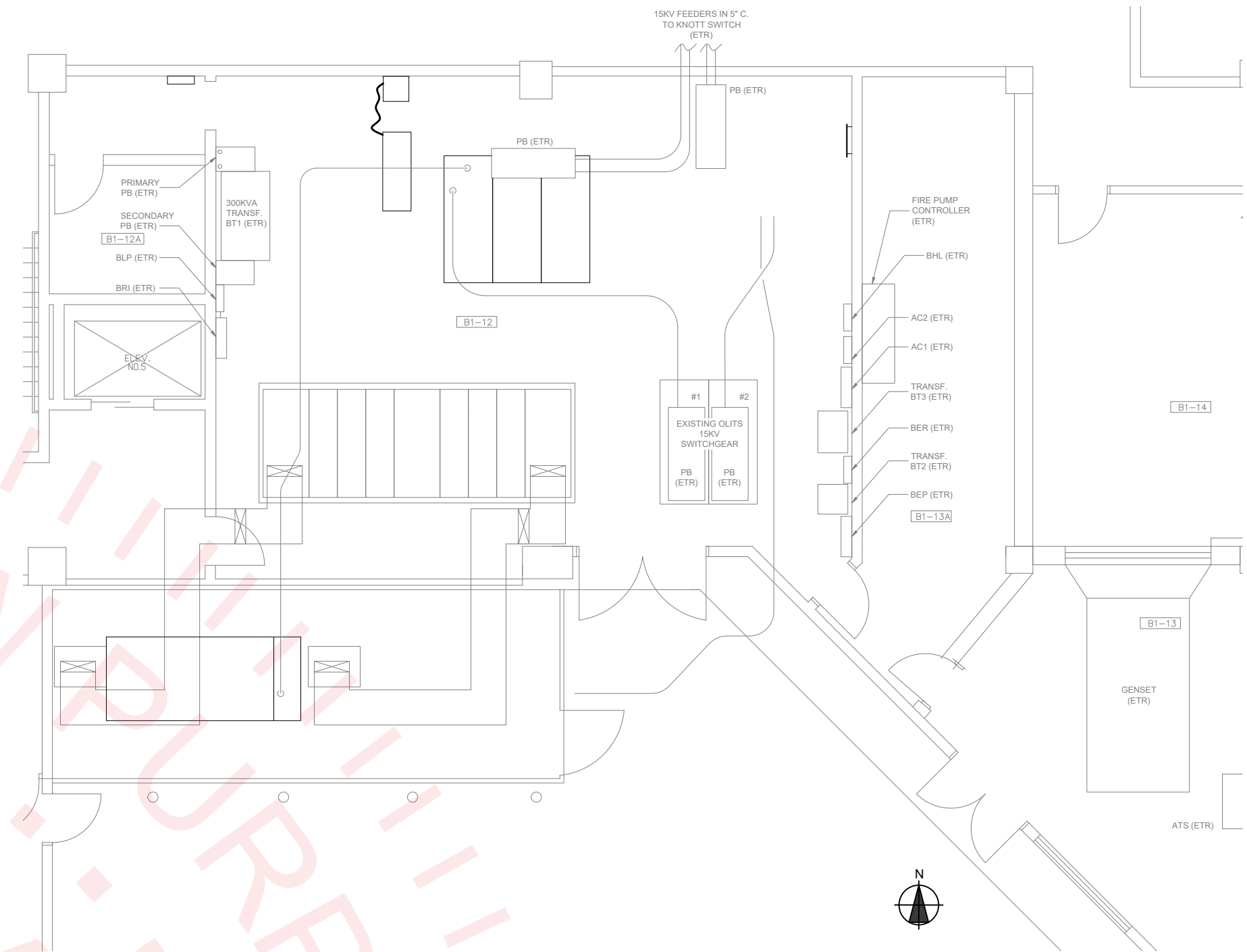
1. CONTRACTOR SHALL VISIT EACH GEAR TO BE OPERATED AS PART OF PROJECT AND CONFIRM INITIAL CONDITIONS ARE AS NOTED. ANY DISCREPANCIES SHALL BE IMMEDIATELY BROUGHT TO THE ENGINEER'S ATTENTION AND SWITCHING PLAN SHALL BE MODIFIED BY CONTRACTOR AS NEEDED.
2. CONTRACTOR SHALL SPEND TIME ONSITE REVIEWING PLANS AND BECOMING FAMILIAR WITH EACH SCOPE OF WORK AND ALL EQUIPMENT AFFECTED BY SCOPE OF WORK.

PHASE 2 (OUTAGE #1)



- << 8 HOUR OUTAGE FOR PEPPER BUILDING >>**
1. OPEN PEPPER A FEEDER AND PEPPER B FEEDER AT KNOTT PAD-MOUNT 15KV SWITCH.
 2. BOTH OLITS AND LIFE SAFETY GENERATORS WILL START TO PICKUP RESPECTIVE LOADS.
 3. VERIFY 15KV AND 480V SWGR's ARE DE-ENERGIZED.
 4. COMPLETE DEMOLITION OF ABANDONED FEEDER LEAVING 480V SWGR AND ROUTED THROUGH GARAGE. REMOVE FEEDER AND RACEWAY FROM WITHIN 480V SWGR DURING OUTAGE. REMAINING DEMOLITION CAN OCCUR DURING NORMAL HOURS.
 5. COMPLETE DEMOLITION OF METERING WIRING AND CONDUIT AT BOTH 480V GEAR AND 15KV GEAR.
 6. PERFORM INVESTIGATION OF 15KV GEAR IN PREPARATION FOR DIVIDING GEAR IN PHASE 3.
 7. CLOSE PEPPER A FEEDER AND PEPPER B FEEDER SWITCHES AT KNOTT SWITCH.
 8. VERIFY OLITS AND LIFE SAFETY ATS's TRANSFER BACK TO NORMAL AND THAT GENERATORS SHUTDOWN.
- << POWER IS RESTORED >>**
9. COMPLETE REFUELING OF GENERATOR TANKS.
 10. INSTALL NEW GROUNDING ELECTRODE BUSBAR AND CONNECTION TO WATER MAIN.
 11. INSTALL NEW SCADA CABINET IN MAIN ELECTRICAL ROOM AND CONNECT TO BUILDING'S EXISTING NETWORK SWITCH.

PHASE 3 (OUTAGE #2)



- << 48 HOUR OUTAGE >>**
1. OPEN PEPPER A FEEDER AND PEPPER B FEEDER AT KNOTT PAD-MOUNT 15KV SWITCH.
 2. BOTH OLITS AND LIFE SAFETY GENERATORS WILL START TO PICKUP RESPECTIVE LOADS.
 3. VERIFY 15KV AND 480V SWGR's ARE DE-ENERGIZED.
 4. DIVIDE 15KV SWGR BETWEEN AUX. 'A' AND TIE SECTIONS AND PREPARE FOR FOUR SECTIONS BETWEEN TIE AND FEEDER B TO BE REMOVED IN THEIR ENTIRETY. SWGR SHALL BE SETUP TO ALLOW FOR MAIN 'A' AND FEEDERS 'A' TO BE RE-ENERGIZED AND OPERATIONAL WHEN READY. INCLUDES PHASE, NEUTRAL, GROUND BUS, CONTROL WIRING, AC CIRCUITS, DC CIRCUITS, ETC.
 5. OPEN AND LOCK OUT OLITS SWGR MAIN #2. TRANSFER OLITS 480V SWGR TO CIRCUIT #1 SIDE.
 6. REMOVE FEEDER CONDUCTORS BETWEEN 15KV SWGR OLITS FEEDER #2 AND OLITS SWGR MAIN #2. STORE FEEDER FOR REUSE. REMOVE PORTION OF CONDUIT BACK TO ALLOW FOR GEAR REMOVAL.
 7. REMOVE FEEDER CONDUCTORS BETWEEN 15KV SWGR FEEDER 'B' AND TX-B. REMOVE PORTION OF CONDUIT BACK TO ALLOW FOR GEAR REMOVAL.
 8. OPEN AND LOCK OUT 480V SWGR MAIN 'A' (SEE PLANS FOR INCORRECT NAMING CONVENTION) AND CLOSE TIE.
 9. PULL BACK INCOMING 15KV FEEDER 'B' TO ALLOW FOR PULLBOX TO BE DISCONNECTED FROM TOP OF SWGR FOR GEAR REMOVAL.
 10. DISCONNECT AND PULL APART SWGR SECTIONS TO SEPARATE TIE FROM AUX. 'A' SECTION AND PROVIDE TEMPORARY BARRIER ON SIDE OF AUX 'A' SECTION TO ALLOW FOR PROTECTION WHILE ENERGIZED.
 11. DISCONNECT CIRCUIT 'B' 4000A BUSDUCT. COMPLETE ELECTRICAL TESTING, AND RECONNECT BUSDUCT.
 12. SHIFT EXISTING DC BATTERY AWAY FROM WALL IN OPEN AREA TO ALLOW FOR NEW DC BATTERY TO BE INSTALLED AGAINST WALL. RELOCATE DC CHARGER TEMPORARILY TO MAKE SPACE FOR NEW DC BATTERY RACK. EXTEND DC CIRCUITS AS NEEDED TO ALLOW DC BATTERY TO SIT AWAY FROM WALL. USE CAUTION AS DC BATTERY WILL BE CHARGED.
 13. REMOVE EXISTING CPT AC PANEL FROM BACK WALL. PANEL ONLY SERVES LIGHTS AND RECEPTACLES THAT ARE TO BE REMOVED.
 14. CLOSE PEPPER A FEEDER AT KNOTT PAD-MOUNT 15KV SWITCH.
 15. VERIFY 15KV SWGR MAIN 'A' AND FEEDER 'A' SECTIONS ARE RE-ENERGIZED. VERIFY TX-A AND 480V SWGR ARE RE-ENERGIZED. VERIFY OLITS 15KV MAIN 'A' IS RE-ENERGIZED.
 16. VERIFY OLITS AND LIFE SAFETY ATS's TRANSFER BACK TO NORMAL AND THAT GENERATORS SHUTDOWN.
- << POWER RESTORED >>**
17. COMPLETE REFUELING OF GENERATOR TANKS.
 18. COMPLETE DEMOLITION OF 15KV SWGR SECTIONS DISCONNECTED FROM REMAINDER OF SYSTEM.
 19. COMPLETE DEMOLITION OF TX-B.

- SHEET NOTES:**
1. FLOORPLAN CONFIGURATIONS SHOWN FOR EACH PHASE INDICATE CONDITION FOLLOWING PHASE SCOPE OF WORK.
 2. PHASING DIAGRAMS AND NOTES DO NOT INCLUDE ALL DETAILS OF CONSTRUCTION.
 3. CONTRACTOR SHALL BE RESPONSIBLE FOR PERFORMING ALL WORK IN A SAFE MANNER AND COMPLETING ALL NECESSARY LOCK-OUT, TAG-OUT PROCEDURES.
 4. CONTRACTOR SHALL BE RESPONSIBLE FOR CONFIRMING ALL SEQUENCING, SWITCHING, AND REQUIRED ACTIONS FOR SAFELY PERFORMING ALL WORK NOTED.
 5. MULTIPLE TESTING AND START-UP SERVICES WILL BE A REQUIREMENT OF THIS PROJECT TO ALLOW FOR EQUIPMENT REPLACEMENT IN PHASES.
 6. SWITCHING OF DMS EQUIPMENT AS LISTED IN SEQUENCING NOTES SHALL BE COMPLETED BY ELECTRICAL CONTRACTOR.
 7. CONTRACTOR SHALL COORDINATE SWITCHING SCHEDULE WITH DMS.
 8. SEQUENCING NOTES WERE DEVELOPED AT TIME OF DESIGN WITHOUT ANY COORDINATION WITH CONTRACTOR. SHOULD CONTRACTOR HAVE IDEAS TO SIMPLIFY CONSTRUCTION, PROVIDE SAFER WORK ENVIRONMENT, OR TO REDUCE OUTAGE QUANTITIES, CONTRACTOR SHALL PRESENT PLANS TO OWNER AND ENGINEER FOR DISCUSSION AND CONSIDERATION.



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PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 1

MSFM-02203280

Tallahassee, Florida

DATE:
March 25, 2024

REVISED:

DESIGNED BY: BW	DRAWN BY: TB
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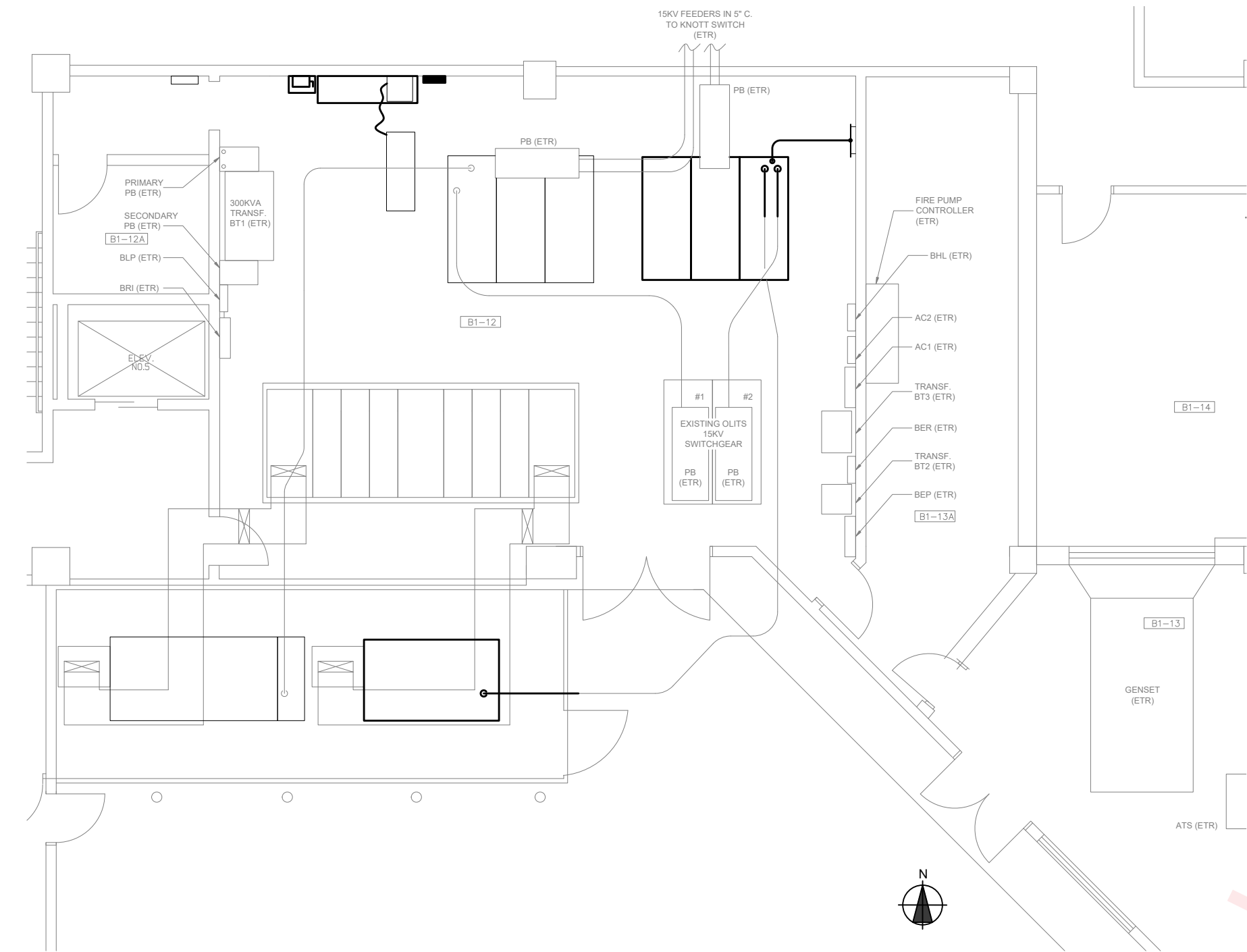
SUBMITTAL:
100% Construction Documents

SHEET TITLE:
SEQUENCING

SHEET:
E103

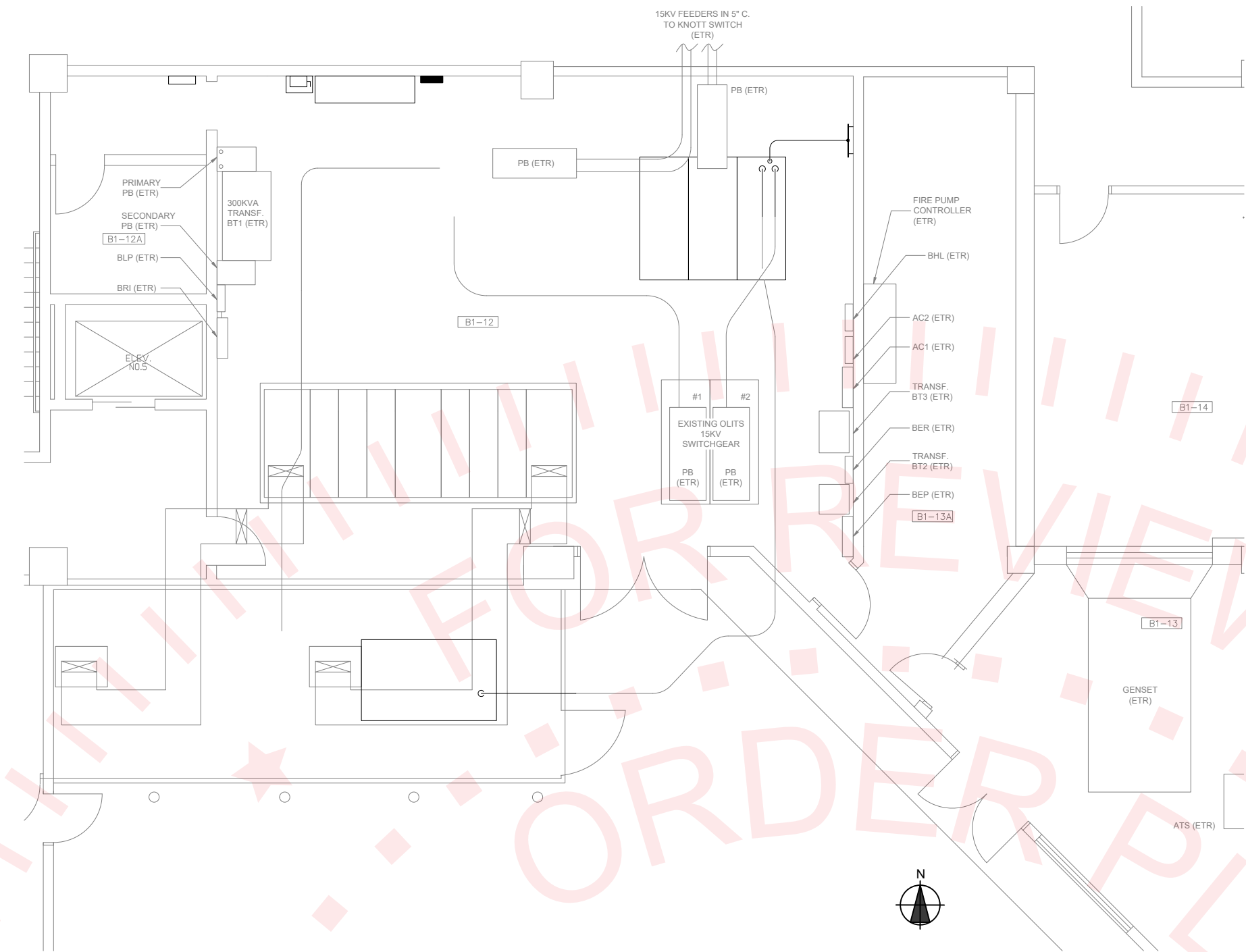
JOB NUMBER:
2311

PHASE 4



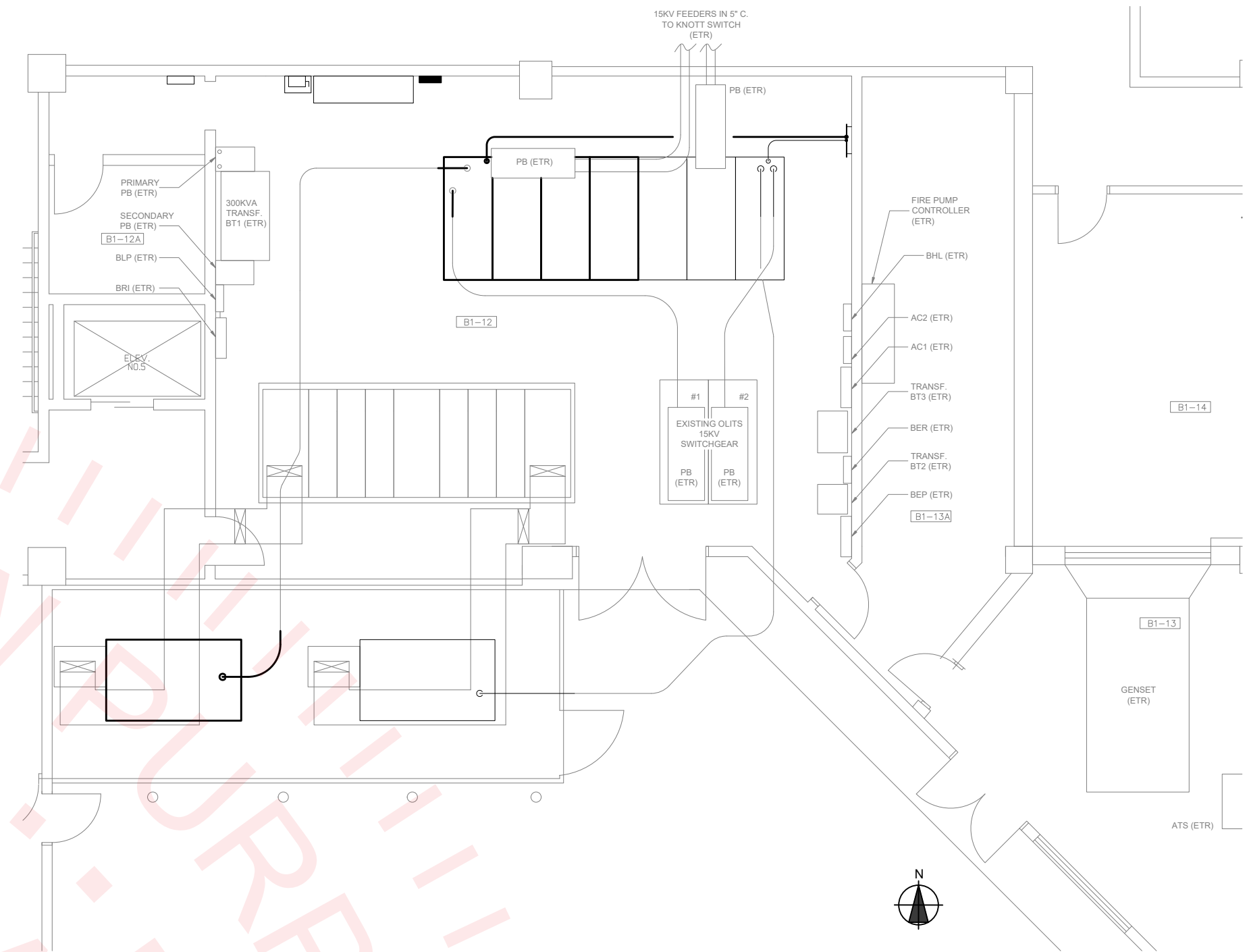
1. INSTALL NEW 15KV SWGR SECTIONS 5, 6, AND 7. PROVIDE TEMPORARY BARRIER ON END OF SECTION 5 TO ALLOW FOR PROTECTION WHILE ENERGIZING.
2. RECONNECT INCOMING FEEDER 'B' TO MAIN 'B' SECTION AND TERMINATE.
3. CONNECT TO NEW GROUNDING ELECTRODE SYSTEM BUSBAR.
4. EXTEND CONDUIT FROM OLITS SWGR TO NEW SWGR, REPULL EXISTING FEEDER, AND TERMINATE EACH END.
5. INSTALL NEW TX-B.
6. CONNECT TX-B SECONDARY TO EXISTING BUSDUCT TAP BOX.
7. EXTEND CONDUIT TO TX-B AND TO 15KV SWGR AND PREPARE FOR FEEDER INSTALL.
8. INSTALL NEW FEEDER FROM 15KV SWGR TX-B FEEDER TO TRANSFORMER AND TERMINATE EACH END.
9. INSTALL NEW DC BATTERY, CHARGER, SWITCH, AND DC PANEL.
10. INSTALL NEW DC CIRCUITS TO NEW SECTIONS 5, 6, AND 7.
11. TEMPORARY CONTROLS AND AC CIRCUITS WITHIN NEW 15KV SWGR SECTIONS 5, 6, AND 7 FOR ENERGIZING.
12. COMPLETE START-UP ON FIRST HALF OF 15KV SWGR. INPUT PROTECTIVE RELAY SETTINGS. PREPARE SECTIONS OF SWGR TO BE ENERGIZED.
13. REMOVE LIGHTS AND RECEPTACLES THAT WERE SERVED BY OPT AC PANEL PREVIOUSLY REMOVED.

PHASE 5 (OUTAGE #3)



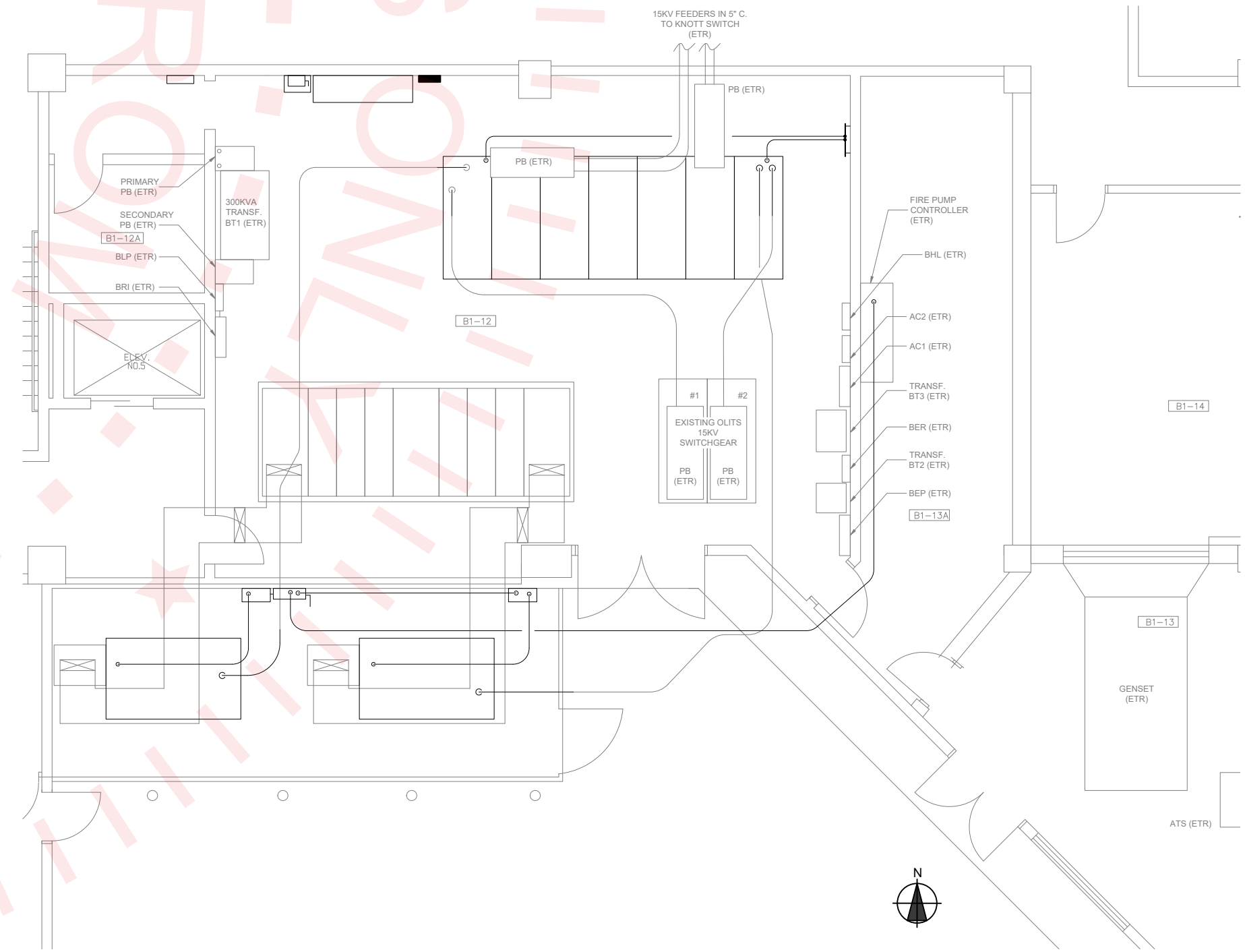
- << 48 HOUR OUTAGE FOR PEPPER BUILDING >>**
1. OPEN PEPPER A FEEDER AT KNOTT PAD-MOUNT 15KV SWITCH.
 2. BOTH OLITS AND LIFE SAFETY GENERATORS WILL START TO PICKUP RESPECTIVE LOADS.
 3. VERIFY 15KV AND 480V SWGR's ARE DE-ENERGIZED.
 4. DISCONNECT EXISTING AC AND DC CIRCUIT FROM OLITS SWGR.
 5. EXTEND NEW AC AND DC CIRCUITS TO EXISTING OLITS 15KV SWGR AND TERMINATE. DC CIRCUITS FROM DC PANEL AND AC CIRCUITS FROM NEW 15KV SWITCHGEAR LOAD CENTER.
 6. DISCONNECT CIRCUIT 'A' 4000A BUSDUCT, COMPLETE ELECTRICAL TESTING, AND RECONNECT BUSDUCT.
 7. SWAP OLITS 480V SWITCHGEAR OVER TO A-SIDE.
 8. OPEN 480V SWGR MAIN 'A' AND CLOSE MAIN 'B'.
 9. OPEN OLITS 15KV SWGR MAIN 1 AND CLOSE MAIN 2.
 10. CLOSE 15KV MAIN 'B' BREAKER, TX-B FEEDER BREAKER, AND OLITS 2 FEEDER BREAKER.
 11. CLOSE PEPPER B FEEDER AT KNOTT PAD-MOUNT 15KV SWITCH.
 12. VERIFY NEW SECTIONS OF 15KV SWGR ARE ENERGIZED, OLITS MAIN 2 IS ENERGIZED, AND 480V SWGR IS ENERGIZED VIA TX-B.
- << POWER IS RESTORED >>**
13. COMPLETE REFUELING OF GENERATOR TANKS.
 14. REMOVE EXISTING DC BATTERY IN ITS ENTIRETY, INCLUDING DC CHARGER AND CIRCUITS.
 15. REMOVE FEEDER CONDUCTORS BETWEEN 15KV SWGR OLITS FEEDER #1 AND OLITS SWGR MAIN #1. STORE FEEDER FOR REUSE. REMOVE PORTION OF CONDUIT BACK TO ALLOW FOR GEAR REMOVAL.
 16. REMOVE FEEDER CONDUCTORS BETWEEN 15KV SWGR FEEDER 'A' AND TX-A. REMOVE PORTION OF CONDUIT BACK TO ALLOW FOR GEAR REMOVAL.
 17. PULL BACK INCOMING 15KV FEEDER 'A' TO ALLOW FOR PULLBOX TO BE DISCONNECTED FROM TOP OF SWGR FOR GEAR REMOVAL.
 18. REMOVE REMAINING SECTIONS OF 15KV SWGR TO BE REMOVED.
 19. REMOVE TX-A.

PHASE 6



1. INSTALL NEW 15KV SWGR SECTIONS 1 - 4. INSTALL UP AGAINST SECTION 5 AND CAUTIOUSLY REMOVE BARRIER AT END OF SECTION 5.
2. RECONNECT INCOMING FEEDER 'A' TO MAIN 'A' SECTION AND TERMINATE.
3. EXTEND CONDUIT FROM OLITS SWGR MAIN 1 TO NEW SWGR OLITS 1 FEEDER, REPULL EXISTING FEEDER, AND TERMINATE EACH END.
4. INSTALL NEW TX-A.
5. CONNECT TX-A SECONDARY TO EXISTING BUSDUCT TAP BOX.
6. EXTEND CONDUIT TO TX-A AND TO 15KV SWGR TX-A FEEDER SECTION AND PREPARE FOR FEEDER INSTALL.
7. INSTALL NEW FEEDER FROM 15KV SWGR TX-A FEEDER TO TRANSFORMER AND TERMINATE EACH END.
8. INSTALL NEW DC CIRCUITS TO NEW SECTIONS 1 - 4.
9. INSTALL NEW AC CIRCUITS FROM SWGR LOAD CENTER TO NEW SECTIONS.
10. COMPLETE START-UP ON SECOND HALF OF 15KV SWGR. INPUT PROTECTIVE RELAY SETTINGS. PREPARE SECTIONS OF SWGR TO BE ENERGIZED.

PHASE 7 (OUTAGE #4)



- << 48 HOUR OUTAGE FOR PEPPER BUILDING >>**
1. OPEN PEPPER B FEEDER AT KNOTT PAD-MOUNT 15KV SWITCH. BOTH OLITS AND LIFE SAFETY GENERATORS WILL START TO PICKUP RESPECTIVE LOADS. VERIFY 15KV AND 480V SWGR's ARE DE-ENERGIZED.
 2. COMPLETE INTERCONNECTIONS OF ALL 15KV SWGR SECTIONS, INCLUDING AC CIRCUITS, PHASE, NEUTRAL, AND GROUND BUS, CONTROLS, ARC FLASH CABLES, ETC.
 3. INSTALL 15KV SWGR BREAKER TEST STATION.
 4. CLOSE 15KV SWGR MAIN A AND MAIN B BREAKERS, OLITS SWGR MAIN 1 AND MAIN 2 BREAKERS. 480V SWGR MAIN A AND MAIN B BREAKERS, OLITS 480V SWGR MAIN A AND MAIN B BREAKERS. BOTH 480V SWGR TIE BREAKERS SHOULD BE OPEN.
 5. CLOSE PEPPER A FEEDER AND PEPPER B FEEDER AT KNOTT PAD-MOUNT 15KV SWITCH.
 6. VERIFY NEW SECTIONS OF 15KV SWGR ARE ENERGIZED, OLITS SWGR IS ENERGIZED, AND 480V SWGR IS ENERGIZED.
- <POWER IS RESTORED>**
7. COMPLETE REFUELING OF GENERATOR TANKS.
 8. COMPLETE ALL SCADA CONNECTIONS AND IMPLEMENT SCADA CHANGES INTO PME.
 9. COMPLETE LIGHTING REPLACEMENT.
 10. COMPLETE ALL SIGNAGE, SITE CLEAN-UP, AND OWNER TRAINING.

- SHEET NOTES:**
1. FLOORPLAN CONFIGURATIONS SHOWN FOR EACH PHASE INDICATE CONDITION FOLLOWING PHASE SCOPE OF WORK.
 2. PHASING DIAGRAMS AND NOTES DO NOT INCLUDE ALL DETAILS OF CONSTRUCTION.
 3. CONTRACTOR SHALL BE RESPONSIBLE FOR PERFORMING ALL WORK IN A SAFE MANNER AND COMPLETING ALL NECESSARY LOCK-OUT, TAG-OUT PROCEDURES.
 4. CONTRACTOR SHALL BE RESPONSIBLE FOR CONFIRMING ALL SEQUENCING, SWITCHING, AND REQUIRED ACTIONS FOR SAFELY PERFORMING ALL WORK NOTED.
 5. MULTIPLE TESTING AND START-UP SERVICES WILL BE A REQUIREMENT OF THIS PROJECT TO ALLOW FOR EQUIPMENT REPLACEMENT IN PHASES.
 6. SWITCHING OF DMS EQUIPMENT AS LISTED IN SEQUENCING NOTES SHALL BE COMPLETED BY ELECTRICAL CONTRACTOR.
 7. CONTRACTOR SHALL COORDINATE SWITCHING SCHEDULE WITH DMS.
 8. SEQUENCING NOTES WERE DEVELOPED AT TIME OF DESIGN WITHOUT ANY COORDINATION WITH CONTRACTOR. SHOULD CONTRACTOR HAVE IDEAS TO SIMPLIFY CONSTRUCTION, PROVIDE SAFER WORK ENVIRONMENT, OR TO REDUCE OUTAGE QUANTITIES, CONTRACTOR SHALL PRESENT PLANS TO OWNER AND ENGINEER FOR DISCUSSION AND CONSIDERATION.



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PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 1

MSFM-02203280

Tallahassee, Florida

DATE:
March 25, 2024

REVISED:

DESIGNED BY:
BW

DRAWN BY:
TB

SUBMITTAL:
100% Construction Documents

SHEET TITLE:

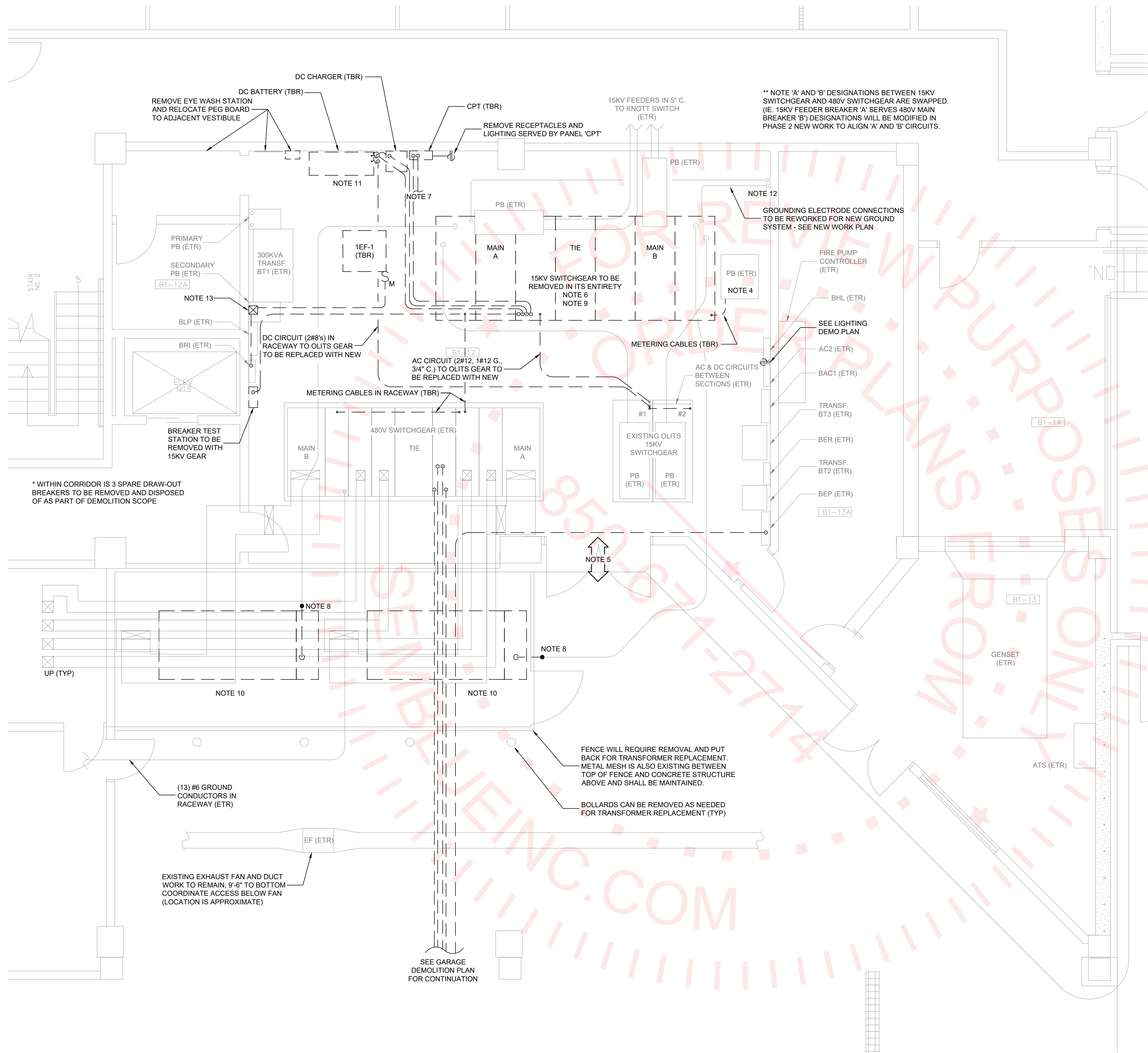
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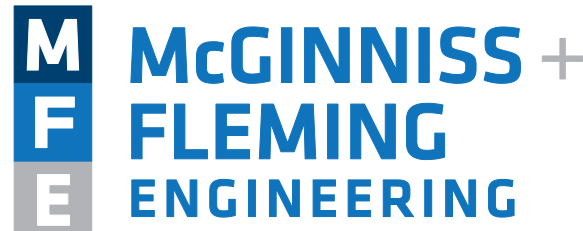
E104

JOB NUMBER:

2311



- SHEET NOTES:**
- REFER TO SEQUENCING NOTES FOR PHASING OF EQUIPMENT REPLACEMENT.
 - NO RACEWAY SHALL BE ABANDONED. ALL RACEWAY THAT IS PART OF EQUIPMENT BEING REMOVED SHALL BE REMOVED IN ITS ENTIRETY, UNLESS NOTED OTHERWISE.
 - NOTE THERE IS MUCH CONDUIT ROUTED OVERHEAD IN ELECTRICAL ROOM THAT IS NOT SERVED BY SWITCHGEAR BEING REPLACED AND IS NOT SHOWN HERE, BUT CONTRIBUTES TO CREATE A CONGESTED AREA ABOVE AND AROUND SWITCHGEAR. CONTRACTOR SHALL USE CAUTION WHEN WORKING AROUND FEEDERS / RACEWAY TO REMAIN.
 - EXISTING METERING SCADA CABLES (RS-485) ENTER DATA PULLBOX. ROUTING BEYOND PULLBOX IS UNKNOWN. REMOVE CABLES BACK TO PULLBOX, COIL, AND LABEL AS "ABANDONED - SOURCE UNKNOWN". IF THE OWNER IDENTIFIES POINT OF ORIGIN, CONTRACTOR SHALL REMOVE IN ITS ENTIRETY.
 - EXISTING DOUBLE DOORS SHALL BE USED FOR ACCESS INTO ELECTRICAL ROOM FOR EQUIPMENT REPLACEMENT. ELECTRICAL ROOM FLOOR ELEVATION IS 48" ABOVE GARAGE FLOOR. DOOR OPENING IS APPROX. 7'-10" WIDE AND HAS A CLEAR HEAD HEIGHT OF APPROX. 8'-5". DOOR HAS REMOVABLE CONCRETE THRESHOLD TO ALLOW FOR EQUIPMENT PASS-THROUGH. CONTRACTOR SHALL REMOVE AS NEEDED AND PUT BACK WHEN COMPLETED WITH MOVING EQUIPMENT THROUGH.
 - EXISTING 15KV SWITCHGEAR IS 94" DEEP, 92" HIGH AND HAS (7) 38" WIDE SECTIONS.
 - CIRCUIT TO LIGHTS AND RECEPTACLES TO BE REMOVED. REFER TO LIGHTING DEMOLITION PLAN.
 - CUT PRIMARY FEEDER RACEWAY AT TRANSFORMER WHERE SHOWN AND MAKE PROVISIONS TO REWORK TO NEW TRANSFORMER. IF COUPLING IN CLOSE PROXIMITY ALLOWS FOR DISCONNECTING RACEWAY TO REWORK, THAT IS PREFERRED TO CUTTING.
 - REMOVE 15KV SWITCHGEAR IN ITS ENTIRETY, INCLUDING GEAR STRUCTURE, FRAME, BREAKERS, CONTROL CIRCUITS, INSTRUMENT TRANSFORMERS, DC SYSTEM, RELAYS, METERING, ETC.
 - REMOVE DRY-TYPE TRANSFORMERS IN THEIR ENTIRETY. ADJOINING BUSDUCT CONNECTION BOXES SHALL REMAIN. BE DISCONNECTED FROM TRANSFORMER ENCLOSURE TO ALLOW REMOVAL, AND BE PROTECTED OR SUPPORTED TO ALLOW FOR RECONNECTING TO NEW TRANSFORMER ENCLOSURE. EXISTING BUSDUCT TAP BOX ENCLOSURE PANELS SHALL BE PROFESSIONALLY CLEANED AND PAINTED TO MATCH NEW TRANSFORMER FINISH. TRANSFORMER FINISH TO BE VERIFIED FROM SHOP DRAWINGS PRIOR TO PAINTING BUS ENCLOSURE. NOTE EXISTING TRANSFORMERS BOTH HAVE PRIMARY ON RIGHT SIDE AND SECONDARY ON LEFT SIDE WHEN VIEWED FROM FRONT.
 - REMOVE DC BATTERY SYSTEM IN ITS ENTIRETY. DC BATTERY REMOVAL SHALL BE COORDINATED CLOSELY AS IT PROVIDES DC POWER TO 15KV GEAR BEING REPLACED AS WELL AS OLITS 15KV GEAR TO REMAIN. PROPERLY DISPOSE OF BATTERIES PER GENERAL NOTES.
 - WHILE PERFORMING 15KV GEAR REPLACEMENT DURING AN OUTAGE, CONTRACTOR SHALL DISCONNECT GROUNDING ELECTRODE SYSTEM FROM GEAR AND TEST RESISTANCE OF GROUNDING SYSTEM. IF EXISTING GROUND SYSTEM RESISTANCE IS ABOVE 25 OHMS, CONTRACTOR SHALL PROVIDE NEW #3/0 GROUND CONDUCTORS IN RACEWAY TO NEW GROUNDING ELECTRODE RODS OUTSIDE OF BUILDING.
 - REMOVE 1EF-1 STARTER, CIRCUIT, RACEWAY, AND RACEWAY SUPPORTS BACK TO SERVING PANELBOARD. 15A, 2P BREAKER TO REMAIN FOR REUSE WITH NEW FAN.



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PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 1

MSFM-02203280

Tallahassee, Florida

DATE:
March 25, 2024

REVISED:

DESIGNED BY:
BW

DRAWN BY:
TB

SUBMITTAL:
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SHEET TITLE:

DEMOLITION
POWER PLAN

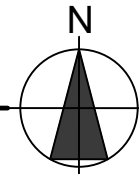
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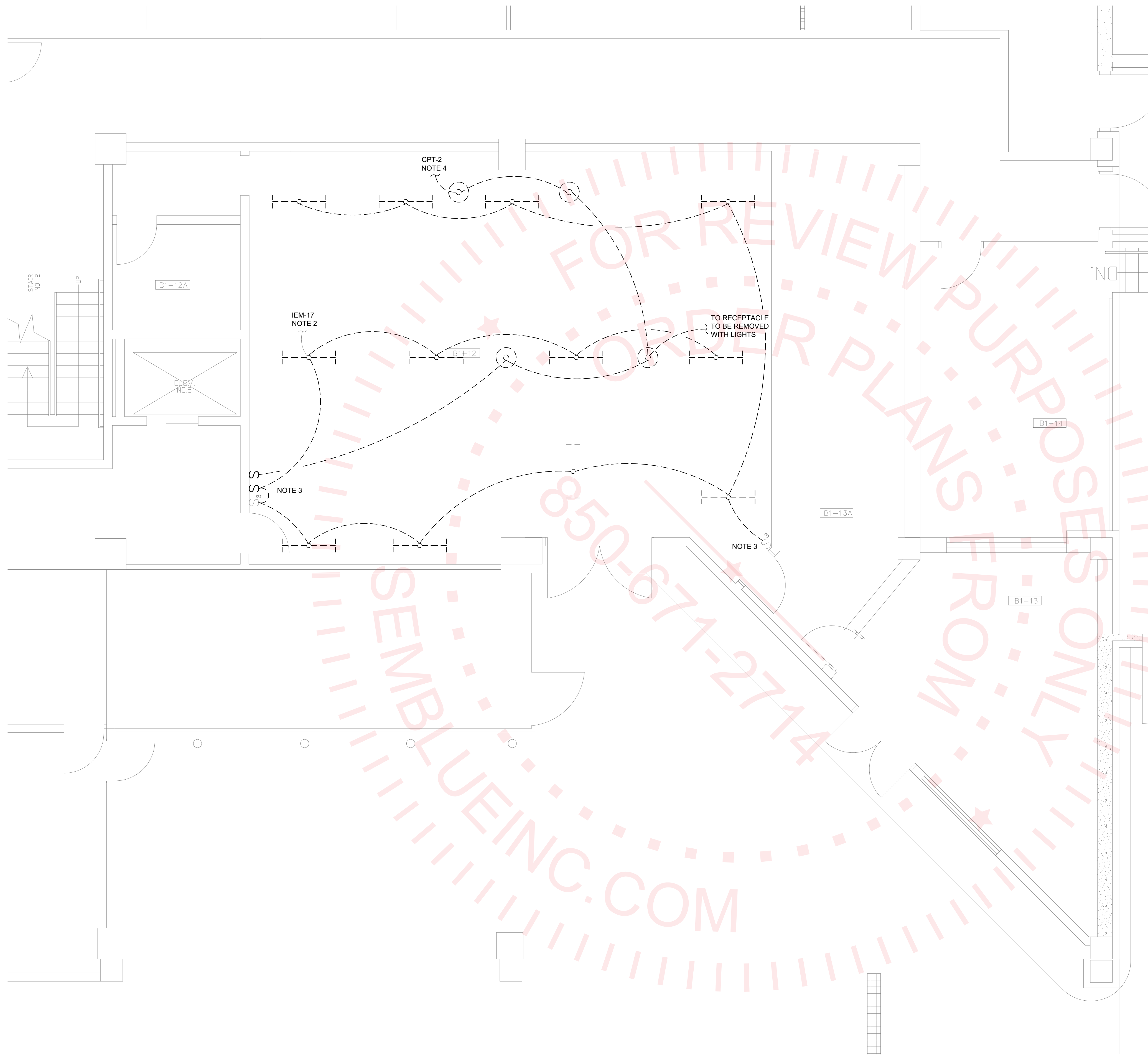
E200

JOB NUMBER:

2311

DEMOLITION POWER PLAN
SCALE: 1/4"=1'-0"





- SHEET NOTES:**
1. REMOVE ALL LIGHTING WITHIN MAIN ELECTRICAL ROOM. REWORK LIGHTING CIRCUITS AND CONTROLS AS NOTED BELOW. WHILE LIGHTING IS BEING REPLACED IN MAIN ELECTRICAL ROOM, CONTRACTOR SHALL PROVIDE TEMPORARY LIGHTING TO MAINTAIN NORMAL OPERATIONS WITHIN MAIN ELECTRICAL ROOM.
 2. DISCONNECT EMERGENCY LIGHTING CIRCUIT FOR LIGHT FIXTURE REMOVAL AND MAKE PROVISIONS TO RECONNECT EMERGENCY CIRCUIT TO SERVE NEW LIGHTS.
 3. REMOVE TWO SINGLE-POLE TOGGLE SWITCHES ALONG WITH LIGHTING. MAINTAIN TWO EXISTING 3-WAY SWITCHES FOR REUSE WITH NEW LIGHTING. 3-WAY SWITCHES WILL REQUIRE RECIRCUITING.
 4. REMOVE LIGHTING, CIRCUIT, AND RECEPTACLES SERVED BY 15KV SWITCHGEAR CONTROL PANEL. REMOVE DEVICES, CIRCUITS, RACEWAY, AND RACEWAY SUPPORTS IN THEIR ENTIRETY.



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PHASE 1

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Tallahassee, Florida

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SUBMITTAL:
100% Construction Documents

SHEET TITLE:
DEMOLITION
LIGHTING PLAN

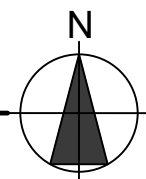
SHEET:

E201

JOB NUMBER:

2311

DEMOLITION LIGHTING PLAN
SCALE: 1/4"=1'-0"





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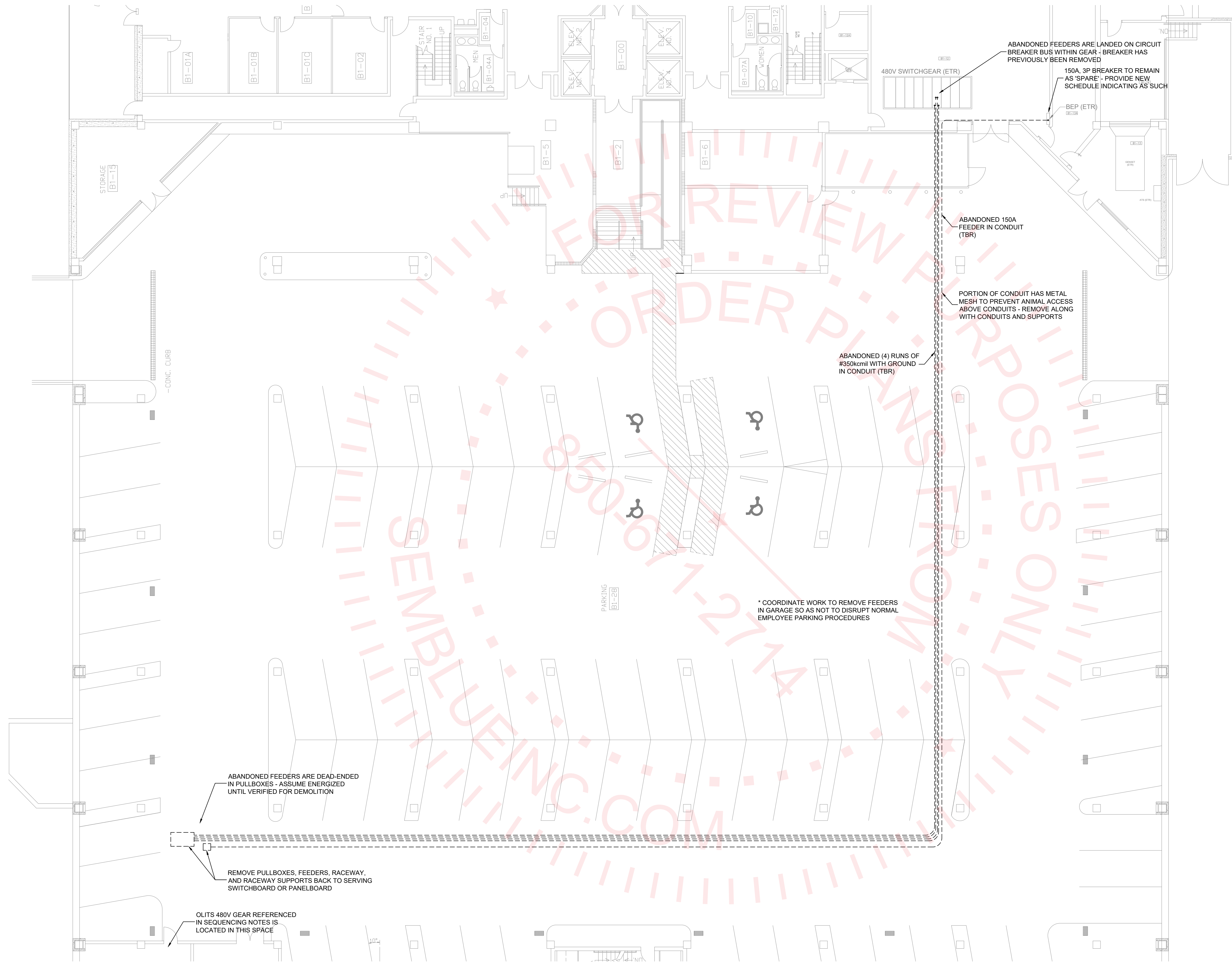
DEMOLITION
GARAGE PLAN

SHEET:

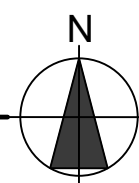
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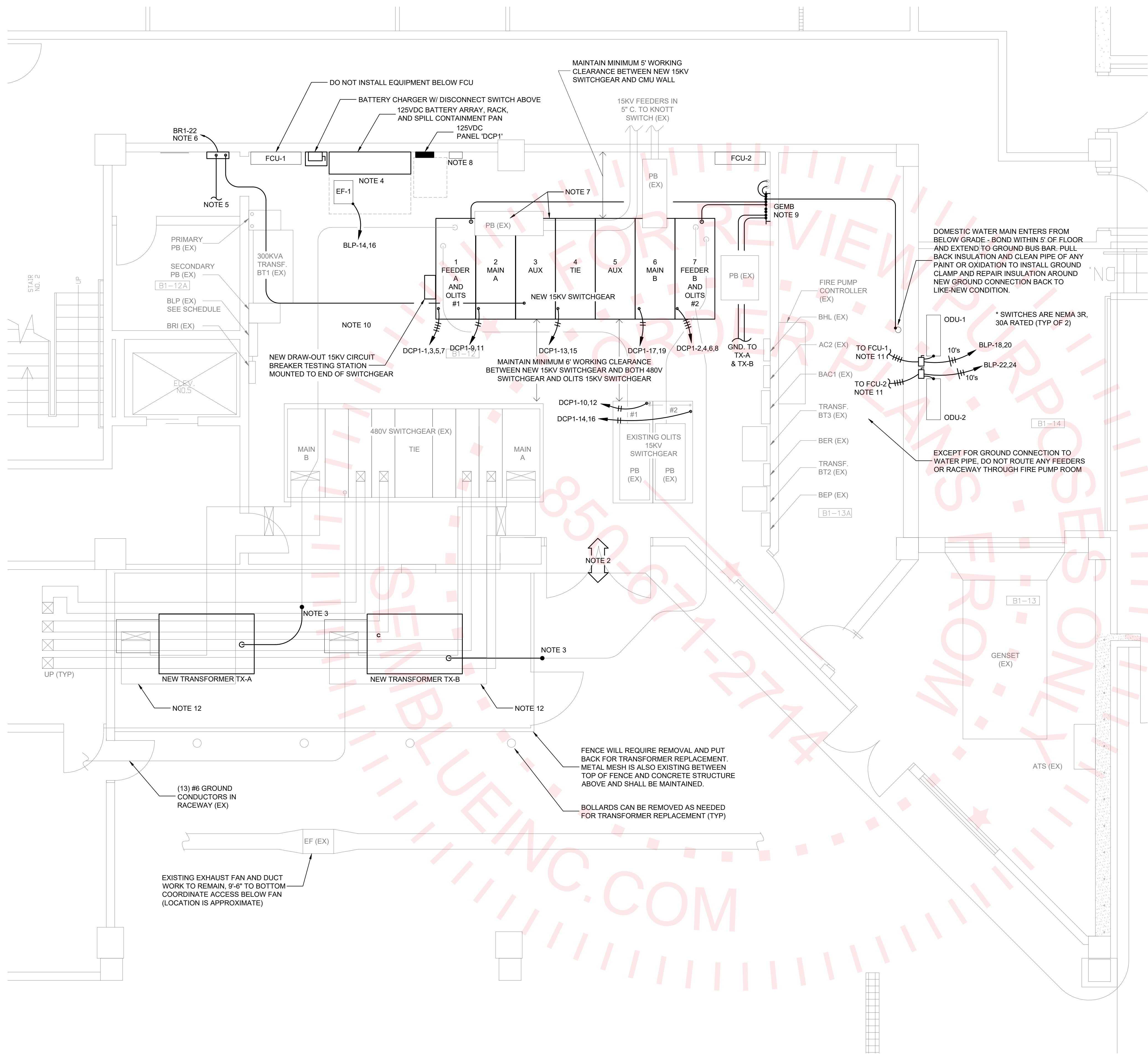
JOB NUMBER:

2311



DEMOLITION POWER GARAGE PLAN
SCALE: 3/32"=1'-0"





- SHEET NOTES:**
- REFER TO SEQUENCING NOTES FOR PHASING OF EQUIPMENT REPLACEMENT.
 - EXISTING DOUBLE DOORS SHALL BE USED FOR ACCESS INTO ELECTRICAL ROOM FOR EQUIPMENT REPLACEMENT. ELECTRICAL ROOM FLOOR ELEVATION IS 48" ABOVE GARAGE FLOOR. DOOR OPENING IS APPROX. 7'-10" WIDE AND HAS A CLEAR HEAD HEIGHT OF APPROX. 8'-5". DOOR HAS REMOVABLE CONCRETE THRESHOLD TO ALLOW FOR EQUIPMENT PASS-THROUGH. CONTRACTOR SHALL REMOVE AS NEEDED AND PUT BACK WHEN COMPLETED WITH MOVING EQUIPMENT THROUGH.
 - REWORK EXISTING CONDUIT TO EXTEND TO NEW TRANSFORMER LOCATION FOR NEW PRIMARY FEEDER.
 - BOND METALLIC BATTERY RACK TO DC SYSTEM GROUND BUS IN DC PANEL WITH #6 CU. IN RACEWAY.
 - ROUTE NEW FIBER WITHIN 1" CONDUIT FROM NETWORK ENCLOSURE TO EXISTING FIBER SWITCH LOCATION AT TELECOM ROOM. SEE SCADA PLAN SHEET FOR ADDITIONAL DETAILS.
 - NEW WALL-MOUNT NETWORK ENCLOSURE FOR SWITCHGEAR SCADA CONNECTIONS. REFER TO SCADA SHEET FOR DETAILS. CIRCUIT TO EXISTING SPARE 20A, 1P BREAKER IN PANEL NOTED.
 - TWO SECTIONS OF NEW 15KV SWITCHGEAR THAT CANNOT MEET WORKING CLEARANCE DUE TO REQUIREMENTS AT REAR PANELS DUE TO STRUCTURAL COLUMN SHALL HAVE TAMPER-PROOF HARDWARE FOR SECURING BACK PANELS. THESE PANELS WILL NOT BE PERMITTED TO BE OPENED WHILE ENERGIZED. PROVIDE SIGNAGE THAT READS "DO NOT OPEN WHILE ENERGIZED" AND SUPPLY OWNER WITH (2) SECURITY TIPS TO MATCH HARDWARE PROVIDED.
 - PROVIDE NEW WALL-MOUNT DOUBLE BOTTLE (32 Oz. EACH) SAFETY EYE WASH STATION ADJACENT TO DC EQUIPMENT WITH BUFFERED ISOTONIC EYESALINE® SOLUTION. (GUIDE: HONEYWELL EMERGENCY EYE/FACE WASH)
 - PROVIDE NEW GROUNDING ELECTRODE SYSTEM 24" x 4" x 1/4" THICK BARE COPPER BUSBAR. PROVIDE STAND-OFFS AND INSULATORS AND MOUNT TO WALL AT 24" AFF. BUSBAR SHALL BE PRE-DRILLED TO ACCEPT NEW CONDUCTORS AND LEAVE SPACE FOR FUTURE GROUND CONNECTIONS. ALL GROUNDING CONNECTIONS SHALL BE NON-REVERSIBLE COMPRESSION TYPE. PROVIDE NEW GROUND CONDUCTORS, SIZED AS SHOWN ON ONE LINE. RETERMINATE AND CONNECT EXISTING GROUND CONDUCTORS AS NOTED.
 - ONCE FINISHED WITH NEW SWITCHGEAR INSTALL, PAINT COMPLETE ELECTRICAL ROOM FLOOR, INCLUDING EQUIPMENT PADS, WITH TWO-PART EPOXY PAINT. (PAINT SHALL BE EQUIVALENT TO SHERWIN-WILLIAMS 6504-97928 570T08104 AS 8100 CLEAR/UD A, COLOR: OUTER SPACE GRAY). PROTECT NEW AND EXISTING ELECTRICAL EQUIPMENT DURING PAINTING.
 - ROUTE 4#12's IN 3/4" CONDUIT FROM OUTDOOR UNIT TO INDOOR UNIT. COORDINATE WITH MANUFACTURER'S REQUIREMENTS FOR FINAL WIRING QUANTITY AND CONFIGURATION. ROUTE ALONG WITH REFRIGERANT PIPING. INDOOR UNIT IS POWERED FROM OUTDOOR UNIT AND HAS INTEGRATED DISCONNECT SWITCH.
 - DURING SCHEDULED OUTAGE, DISCONNECT AND ISOLATE 4000A, 480V BUSWAY BETWEEN SWITCHGEAR AND TRANSFORMER SECONDARY, AND COMPLETE INSULATION RESISTANCE TESTING. CONDUCT AN INSULATION RESISTANCE TEST ON EACH BUSWAY WITH AN INSULATION RESISTANCE TESTER RATED 1000VDC TO ENSURE THAT THE SYSTEM IS FREE FROM SHORT CIRCUITS AND GROUNDS (PHASE-TO-GROUND, PHASE-TO-NEUTRAL, AND PHASE-TO-PHASE). A MINIMUM RESISTANCE RATING OF 100 MEGA-OHMS PER 1000 FEET OF BUSWAY SHOULD BE MET. TO ADJUST FOR ACTUAL BUSWAY LENGTH USE THE FOLLOWING: $R_{1000} = \text{MEASURED RESISTANCE} \times (\text{BUSWAY LENGTH} / 1000)$. IF RATINGS ARE FOUND BELOW THIS LEVEL, NOTIFY THE ENGINEER. PROVIDE WRITTEN REPORT WITH ALL TEST RESULTS, INCLUDING WEATHER AND HUMIDITY CONDITIONS AT TIME OF TESTING.



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MAJOR ELECTRICAL SYSTEMS
PHASE 1

MSFM-02203280

Tallahassee, Florida

DATE:
March 25, 2024

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DESIGNED BY:
BW

DRAWN BY:
TB

SUBMITTAL:
100% Construction Documents

SHEET TITLE:

NEW WORK
POWER PLAN

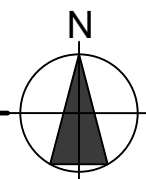
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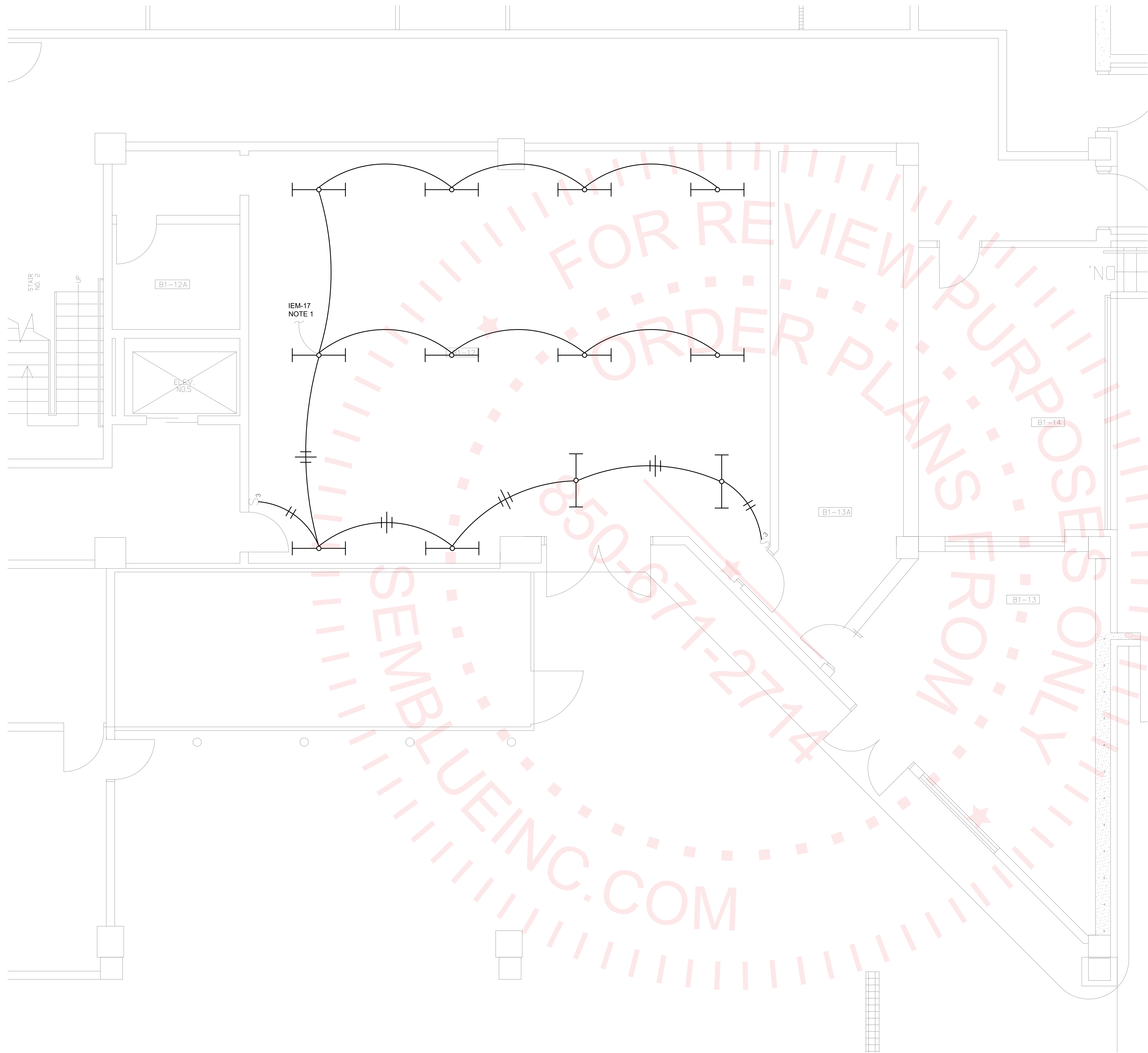
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JOB NUMBER:

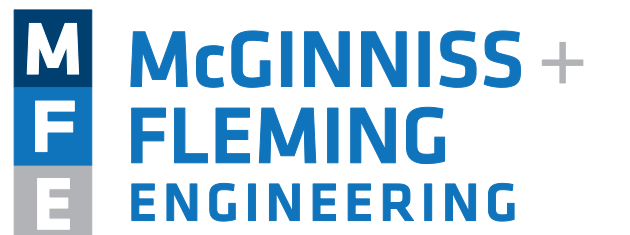
2311

NEW WORK POWER PLAN
SCALE: 1/4"=1'-0"





- SHEET NOTES:**
- CONNECT NEW LIGHTS TO EXISTING EMERGENCY CIRCUIT REMAINING FROM DEMOLITION. ALL NEW LIGHTS IN ELECTRICAL ROOM WILL BE EMERGENCY. REUSE EXISTING 3-WAY LIGHTING SWITCHES AT EACH ENTRY DOOR.
 - ALL LIGHTS ARE TYPE 'LS4'. REFER TO LIGHT FIXTURE SCHEDULE.



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PHASE 1

MSFM-02203280

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SHEET TITLE:

NEW WORK
LIGHTING PLAN

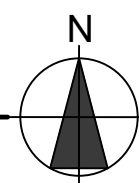
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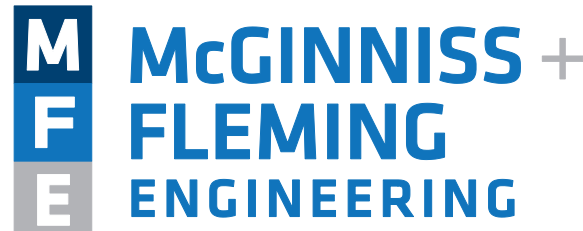
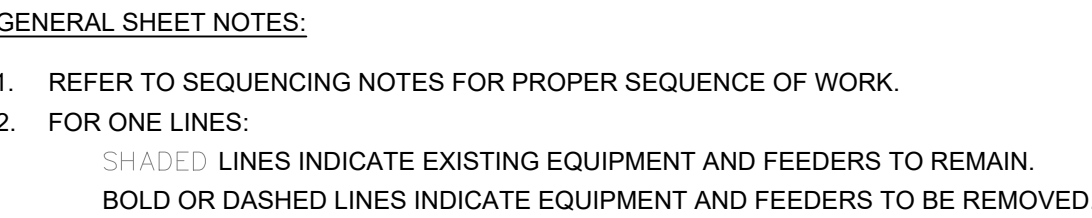
E301

JOB NUMBER:

2311

NEW WORK LIGHTING PLAN
SCALE: 1/4"=1'-0"





PEPPER BUILDING
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MAJOR ELECTRICAL SYSTEMS
PHASE 1
MSFM-02203280
Tallahassee, Florida

REVISÉ:

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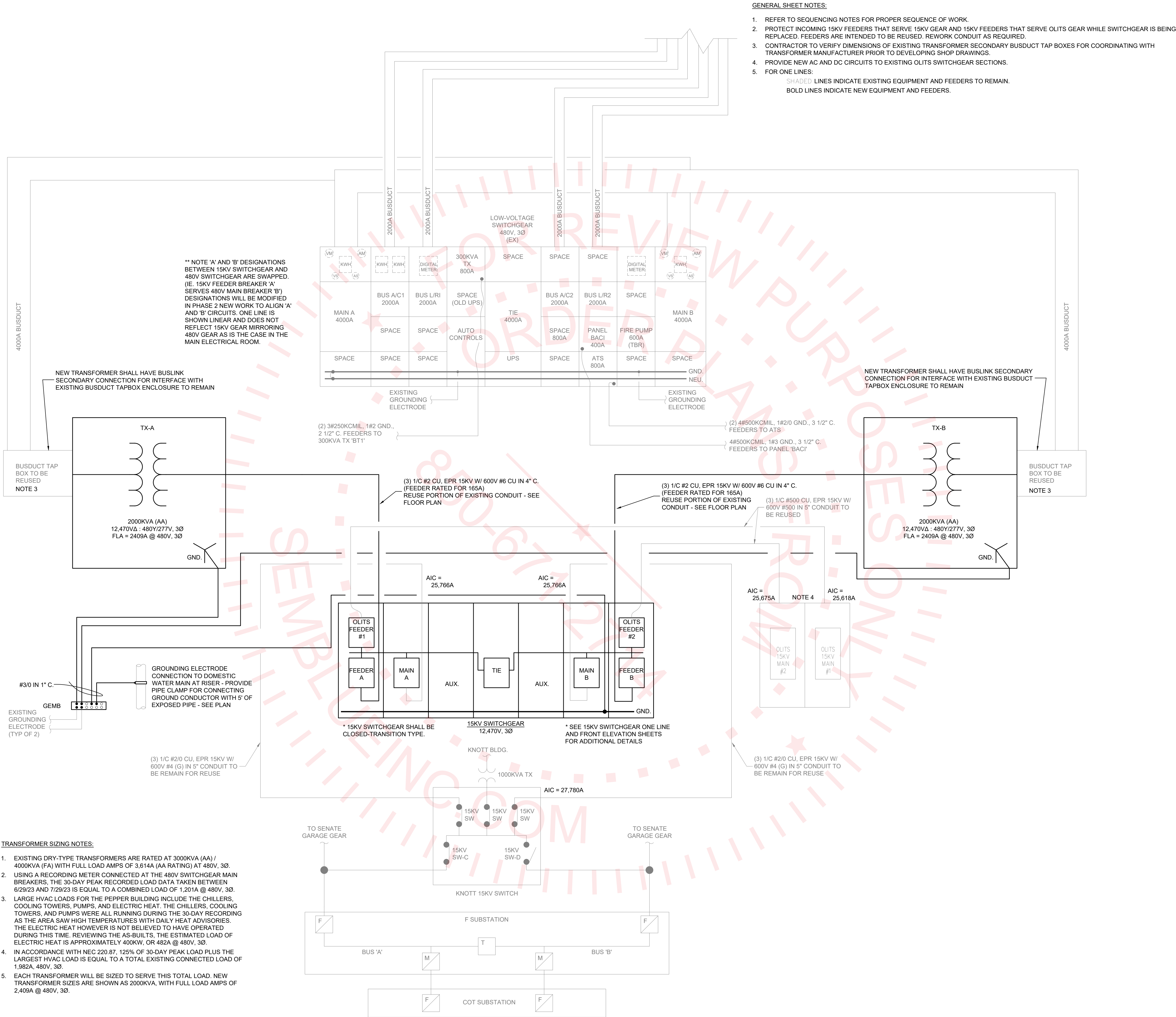
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EXISTING
ONE LINE DIAGRAM

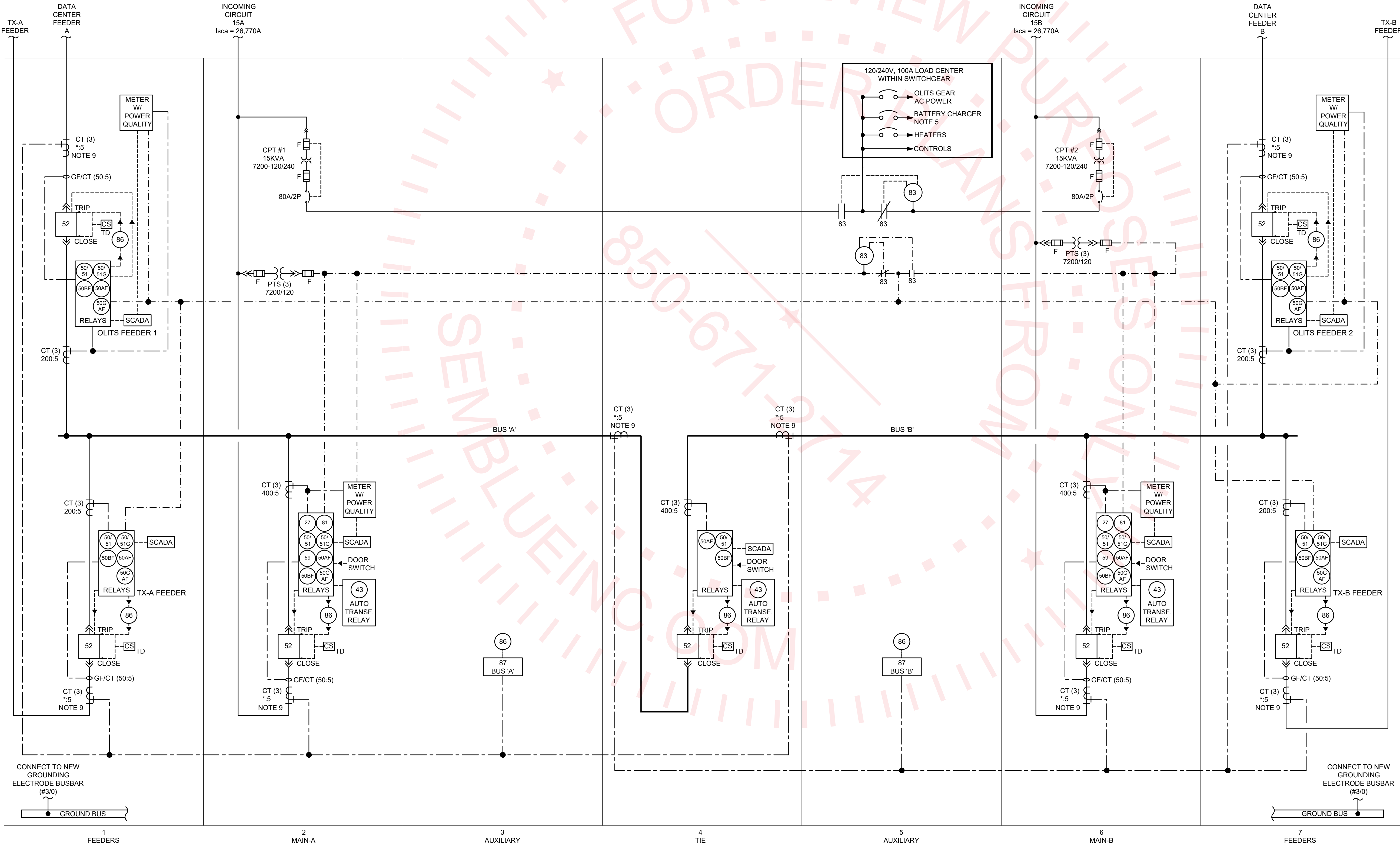
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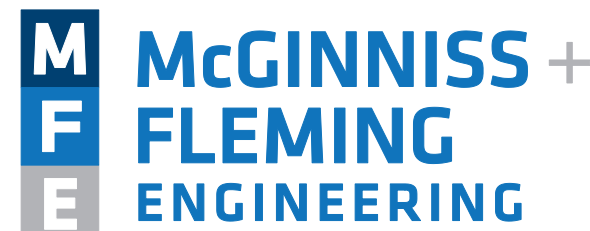


- SHEET NOTES:
- BREAKER CONTROL SWITCHES SHALL BE TIME DELAY TYPE AND BE INTEGRATED WITH RESPECTIVE RELAY TO PROVIDE A PROGRAMMABLE TEN (10) SECOND TIME DELAY FROM OPERATION OF CONTROL SWITCH TO CIRCUIT BREAKER CLOSING ACTION. THERE IS NO DELAY FOR OPENING BREAKER WITH CONTROL SWITCH.
 - ALL USER INTERFACE FUNCTIONS ON FRONT PANEL OF PROTECTIVE RELAYS SHALL BE PROGRAMMED TO OPERATE ACCORDINGLY. LED INDICATORS AND PUSH BUTTONS SHALL BE PROGRAMMED ACCORDING TO THEIR FUNCTION. NON-APPLICABLE LED INDICATORS OR PUSH BUTTONS SHALL BE DEACTIVATED. BREAKER OPEN / CLOSE STATUS ON FRONT OF RELAY SHALL BE OPERATIONAL AND ALIGN WITH SEPARATE BREAKER INDICATING LIGHTS.
 - PROVIDE NEW 120V, 250W HEATERS WITHIN EACH COMPARTMENT TO PREVENT CONDENSATION. CIRCUIT HEATERS FROM NEW AC CIRCUIT BREAKER IN LOAD CENTER WITHIN SWITCHGEAR. HEATERS SHALL HAVE THERMOSTAT CONTROL. THERMOSTAT SHALL BE SET AT 80° F.
 - ALL SWITCHGEAR COMPONENTS NECESSARY FOR THE SCADA FUNCTIONALITY NOTED ON SCADA PLAN SHALL BE PROVIDED, SETUP, CONFIGURED, AND TERMINATED TO NETWORK SWITCH WITHIN SWITCHGEAR.
 - COORDINATE WITH DC BATTERY CHARGER TO BE PROVIDED FOR FINAL CIRCUIT AND BREAKER SIZING.
 - RELAYS AND CIRCUIT BREAKER CONTROL POWER SHALL BE 125VDC, SUPPLIED BY DC BATTERY SYSTEM. MAIN BREAKER RELAYS SHALL BE CONFIGURED TO MONITOR AND ALARM FOR LOW VOLTAGE AND HIGH VOLTAGE OF DC SYSTEM IN SCADA NETWORK. PROTECTIVE RELAYS SHALL INCLUDE V-BATT CARD FOR MONITORING AND ALARMING ON HIGH AND LOW DC BATTERY VOLTAGE.
 - PROTECTIVE RELAYS SHALL INCLUDE ADAPTIVE PHASE OVERCURRENT (PEAK DETECTION) FEATURE TO ALLOW FOR RELIABLE PERFORMANCE IN THE EVENT OF CT SATURATION OR DC OFFSET.
 - ALL CT SECONDARY WIRING BETWEEN INSTRUMENT TRANSFORMER AND PROTECTIVE RELAY OR METER SHALL BE MINIMUM #12 AWG.
 - REFER TO SWITCHGEAR SPECIFICATIONS FOR BUS DIFFERENTIAL PROTECTION SYSTEM REQUIREMENTS BY MANUFACTURER.
 - DIGITAL METERS FOR TWO MAIN BREAKERS SHALL HAVE POWER QUALITY CAPABILITIES WITH A MINIMUM SAMPLING RATE OF 512 SAMPLES PER CYCLE. (EQUIVALENT TO ION9000 SERIES METER BY SCHNEIDER ELECTRIC)
 - DIGITAL METER FOR DATA CENTER FEEDER BREAKERS SHALL HAVE POWER QUALITY CAPABILITY WITH A MINIMUM SAMPLING RATE OF 256 SAMPLES PER CYCLE. (EQUIVALENT TO PM8000 SERIES METER BY SCHNEIDER ELECTRIC)
 - 15KV SWITCHGEAR SHALL BE CONFIGURED WITH MAIN-TIE-MAIN CLOSED TRANSITION. PROVIDE REQUIRED RELAYS AND COORDINATION SETTINGS BETWEEN INCOMING CIRCUITS A AND B AS REQUIRED FOR A PROPERLY FUNCTIONING SYSTEM.
 - LOW VOLTAGE LOAD CENTER BREAKERS SHALL BE SIZED PER NEC FOR LOAD SERVED AND CIRCUIT CONDUCTOR SIZE TO PROVIDE PROPER PROTECTION.



NEW 15KV SWITCHGEAR SINGLE LINE DIAGRAM

SCALE: N.T.S.



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PEPPER BUILDING
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MAJOR ELECTRICAL SYSTEMS
PHASE 1

MSFM-02203280

Tallahassee, Florida

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March 25, 2024

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TB

SUBMITTAL:
100% Construction Documents

SHEET TITLE:

NEW 15KV
SWITCHGEAR
ONE LINE DIAGRAM

SHEET:

E402

JOB NUMBER:

2311

1. PROVIDE AN INDUSTRIAL GRADE LIMIT SWITCH ON MAINS AND THE COMPARTMENT DOORS, AND WIRE SWITCH TO INPUT CONTACTS ON PROTECTIVE RELAY OR REMOTE I/O MODULE. RELAY SHALL BE PROGRAMMED SUCH THAT AN OPEN COMPARTMENT DOOR PROHIBITS THE AUTOMATIC CLOSING OF THE "RESPECTIVE" CIRCUIT BREAKER. AN OPEN DOOR OF ONE COMPARTMENT SHALL NOT PROHIBIT BREAKER IN SEPARATE COMPARTMENT FROM CLOSING. OPEN COMPARTMENT DOOR SHALL NOT PROHIBIT CIRCUIT BREAKER FROM TRIPPING OPEN. POSITION OF DOOR SHALL BE CONFIGURED WITHIN RELAY AND TRANSMITTED TO SCADA NETWORK. REFER TO SCADA DRAWING.
2. PROVIDE REMOTE RACKING COMPONENTS FOR EACH CIRCUIT BREAKER COMPARTMENT. REMOTE RACKING EQUIPMENT SHALL INCLUDE ALL NECESSARY COMPONENTS TO ALLOW FOR RACKING IN OR OUT THE CIRCUIT BREAKER FROM A DISTANCE OF 25' AWAY FROM THE BREAKER COMPARTMENT.
3. PROVIDE CONTINUOUS THERMAL MONITORING WITHIN SWITCHGEAR. THERMAL MONITORING POINTS SHALL BE PROVIDED AT ALL FEEDER TERMINATIONS. THERMAL MONITORING POINTS SHALL BE MONITORED BY SWITCHGEAR HMI TO DISPLAY TEMPERATURE AND ALARM ON HIGH TEMP. THERMAL MONITORING POINTS SHALL ALSO BE MONITORED WITHIN PMS SYSTEM TO SHOW CURRENT TEMP RATINGS AND ALERT OWNER IF HIGH TEMP OCCURS.
4. PROVIDE SIGNAGE ON FRONT OF BOTH MAIN SECTIONS FOR AVAILABLE FAULT CURRENT NOTIFICATION. SEE DETAIL.
5. 15KV SWITCHGEAR NORMAL CONFIGURATIONS WILL BE WITH BOTH MAINS CLOSED AND THE BREAKER OPENED.
6. PROVIDE CONTROL POWER TRANSFORMERS AT EACH INCOMING CIRCUIT WITH AUTO-TRANSFER SYSTEM TO MAINTAIN CONTROL POWER FROM EITHER POWER SOURCE. SECONDARY VOLTAGE SHALL BE AS NEEDED FOR CIRCUIT BREAKER CONTROL AND PLC CONTROL POWER.
7. EACH SECTION IS AN INDIVIDUAL SHIPPING SPLIT TO ENABLE ENTRY INTO BUILDING.
8. ENCLOSURE TYPE = NEMA 1.
9. SWITCHGEAR SHALL INCLUDE CAPABILITIES, RELAY SENSING, AND PROTECTION SCHEMES TO ALLOW FOR CLOSED TRANSITION SWITCHING OF MAINS AND TIE.
10. REFER TO AUTO-TRANSFER NOTES ON GENERAL NOTES SHEET FOR ADDITIONAL REQUIREMENTS.
11. PLC TOUCHSCREEN SHALL INCLUDE ONE LINE IMAGES WITH USER INTERFACE TO ALLOW FOR CONTROLLING ALL BREAKERS WITHIN SWITCHGEAR. PLC SHALL INCLUDE SAME 10-SECOND DELAY FOR CLOSING BREAKERS AS USED WITH CONTROL SWITCHES.
12. PROVIDE TEST SWITCHES IN DOOR PANEL OF EACH SECTION FOR ALL RELAYS, METERS, AND INSTRUMENTS.



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SHEET TITLE:

NEW 15KV
SWITCHGEAR
FRONT ELEVATION

SHEET:

E403

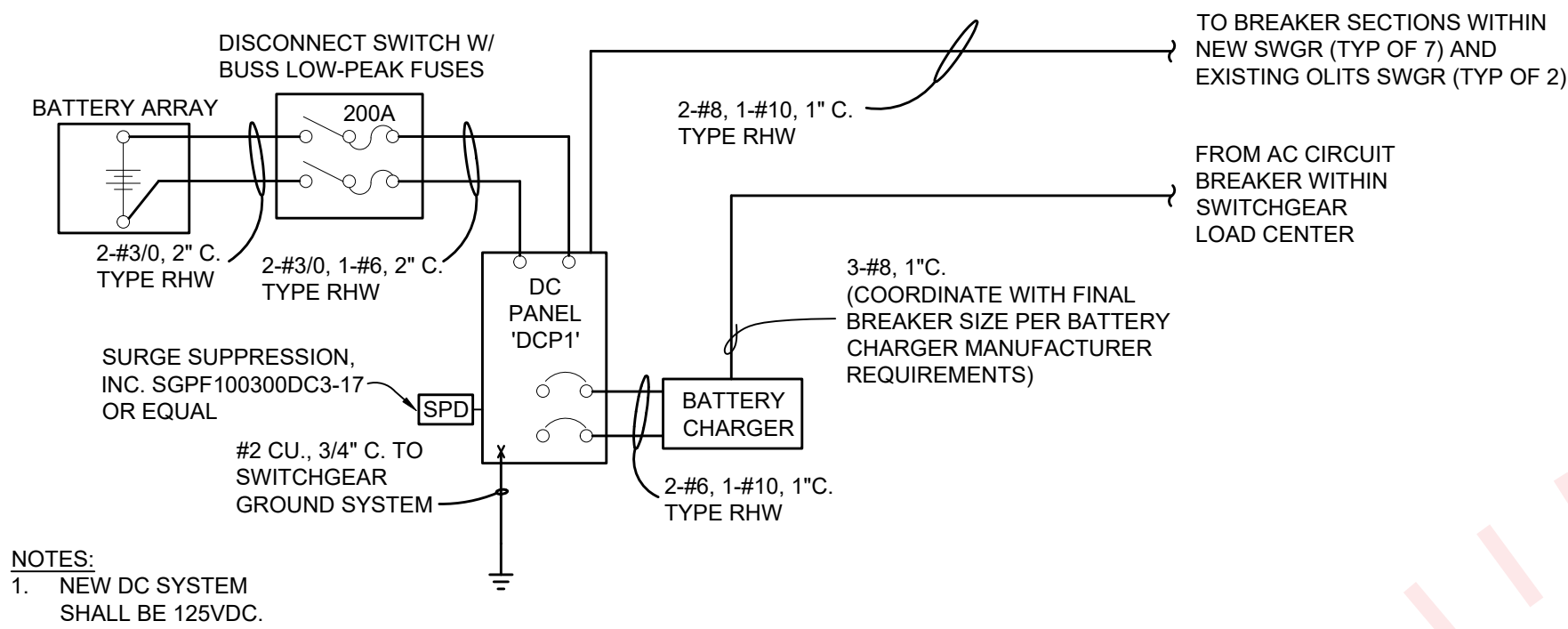
JOB NUMBER:

2311

NEW 15KV SWITCHGEAR FRONT ELEVATION

SCALE: N.T.S.

DC SYSTEM ONE LINE



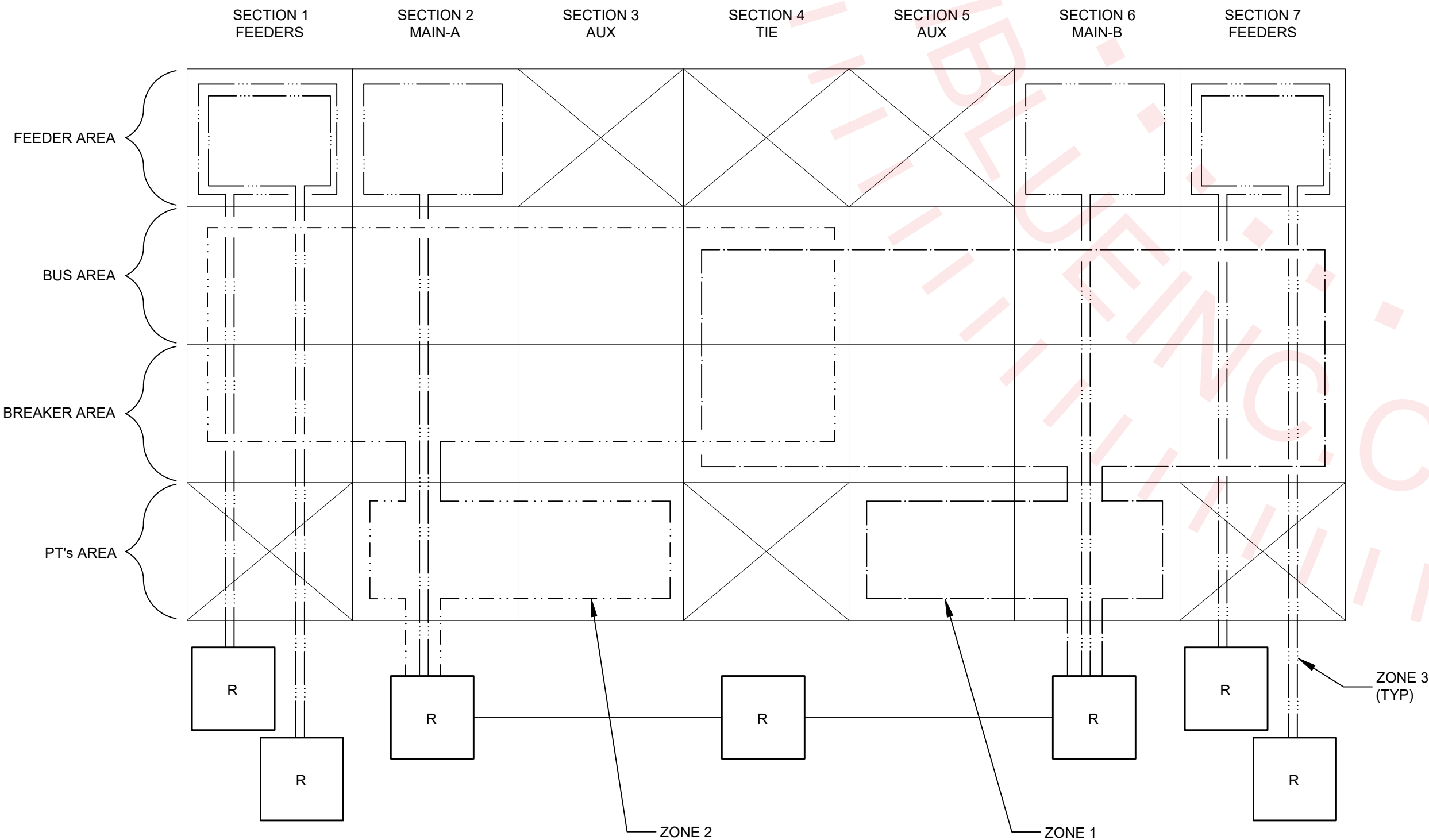
15KV SWITCHGEAR ARC FLASH PROTECTION NOTES AND FIBER LOOP DETAIL

- NOTES:
- THE PROTECTIVE RELAY MODULE SHALL INCLUDE INTEGRATED ARC FLASH RELAY PROTECTION WITH LIGHT SENSING FIBER CABLE INPUTS.
 - PROTECTIVE RELAYS SHALL MONITOR FIBER CABLE LOOPS AND OVERCURRENT TO PROVIDE HIGH-SPEED TRIPPING OF CIRCUIT BREAKER(S) WHEN BOTH LIGHT AND OVERCURRENT ARE PRESENT. PROTECTIVE RELAYS SHALL INCLUDE INTEGRATED HIGH-SPEED I/O (FAST HYBRID CARD) FOR BREAKER OPERATION.
 - INSTALL FIBER LOOPS AS NOTED BELOW. DETAIL IS INTENDED TO PROVIDE GENERAL GUIDANCE. ACTUAL CONDITIONS MAY VARY.
 - ARC FLASH PROTECTION SCHEME SHALL BE CONFIGURED SUCH THAT IN THE EVENT THAT BREAKERS ARE TRIPPED DUE TO AN ARC FLASH EVENT, THE AUTOMATIC TRANSFER SCHEME DOES NOT ALLOW A SECONDARY BREAKER, EITHER A TIE OR MAIN, TO CLOSE ONTO THE FAULTED BUS. (I.E. IN THE CASE WHERE MAIN 'A' IS CLOSED, TIE-A IS CLOSED, AND MAIN 'B' IS OPEN, SHOULD A FAULT OCCUR CAUSING MAIN 'A' AND TIE-A TO TRIP OPEN, AUTO-TRANSFER SHALL BE PROHIBITED FROM CLOSING MAIN 'B' ONTO THE FAULTED BUS.)
 - FIBER CABLES BETWEEN SECTIONS SHALL BE INSTALLED IN THE FIELD AFTER SWITCHGEAR SECTIONS ARE JOINED TOGETHER. FIBER CABLES SHALL BE INSTALLED WITH CAUTION SUCH THAT SWITCHGEAR MOVING PARTS OR BREAKERS BEING RACKED IN/OUT DO NOT DAMAGE FIBER CABLES.

ZONE 1: (---) MAIN 'B' ZONE OF PROTECTION. INSTALL FIBER LOOP THROUGH 'B' SIDE PT AREA, THROUGH 'B' SIDE BREAKER AREA, INCLUDING TIE, DOWN CENTER OF BREAKER CONNECTIONS ON EACH SIDE OF THE PROTECTIVE SHUTTER, AND THROUGH MIDDLE OF BUS AREA ON 'B' SIDE. AN INCIDENT IN THIS ZONE SHALL TRIP AND LOCK-OUT MAIN 'B' BREAKER AND TIE BREAKER.

ZONE 2: (---) MAIN 'A' ZONE OF PROTECTION. INSTALL FIBER LOOP THROUGH 'A' SIDE PT AREA, THROUGH 'A' SIDE BREAKER AREA, INCLUDING TIE, DOWN CENTER OF BREAKER CONNECTIONS ON EACH SIDE OF THE PROTECTIVE SHUTTER, AND THROUGH MIDDLE OF BUS AREA ON 'A' SIDE. AN INCIDENT IN THIS ZONE SHALL TRIP AND LOCK-OUT MAIN 'A' BREAKER AND TIE BREAKER.

ZONE 3: (---) FEEDER AREA ZONE OF PROTECTION(S). INSTALL FIBER LOOP UP AND DOWN SIDES AND ACROSS TOP OF FEEDER AREA IN COMPARTMENTS NOTED. AN INCIDENT IN THIS ZONE SHALL TRIP AND LOCK-OUT THE CORRESPONDING MAIN OR FEEDER BREAKER ASSOCIATED WITH THIS COMPARTMENT.



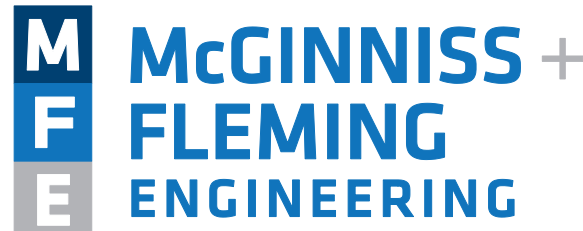
DC PANEL SCHEDULE

LOAD / REMARKS	CIR. NO.	BREAKERS		AMPS PER LEG			BREAKERS		CIR. NO.	LOAD / REMARKS	PANELBOARD SCHEDULE
		AMPS	POLES	L1	L2		AMPS	POLES			
SECTION 1 - TX-A FEEDER	1	20	2	15			20	2	2	SECTION 7 - TX-B FEEDER	Pepper Building Tallahassee, Florida
	3			15	15				4		
SECTION 1 - OLITS FEEDER 1	5	20	2	15			20	2	6	SECTION 7 - OLITS FEEDER 2	Panel <u>'DCP1'</u>
	7			15	15				8		Location <u>ELECTRICAL RM B1-12</u>
SECTION 2 - MAIN-A	9	20	2	15			20	2	10	OLITS GEAR MAIN 1	Service <u>2</u> Pole <u>3</u> Wire
	11			15	15				12	OLITS GEAR MAIN 2	☐ 208/120V ☒ 250VDC
SECTION 4 - TIE	13	20	2	15			20	2	14		☐ 240/120V
	15			15	15				16		Main Bkr. <u>-</u> <u>A</u> <u>-</u> <u>P</u>
SECTION 6 - MAIN-B	17	20	2	15					18	SPACE	Lugs Only <u>225</u> <u>A</u>
	19			15	15				20	SPACE	10,000 AIC Min.
SPACE	21	-	-	-	-				22	SPACE	<u>60</u> Hz.
SPACE	23	-	-	-	-				24	SPACE	☒ Surface mounted panel
SPACE	25	-	-	-	-				26	SPACE	Remarks
SPACE	27	-	-	-	-				28	SPACE	SQUARE D CLASS 2110 WITH
SPACE	29	-	-	-	-				30	SPACE	250VDC TYPE FA260 _AB
SPACE	31	-	-	-	-				32	SPACE	CIRCUIT BREAKERS. PROVIDE
SPACE	33	-	-	-	-				34	SPACE	GROUNDING BAR, COPPER
SPACE	35	-	-	-	-				36	TEST STATION	BUSSING, 63" BREAKER
SPACE	37	-	-	-	-		20	2	38		MOUNTING SPACE, NEMA 1
BATTERY CHARGER INPUT	39	35	2	15					40	SPD	ENCLOSURE.
	41			15			20	2	42		* COORDINATE FINAL BREAKER
TOTAL CONNECTED (AMPS)				150	150		TOTAL CONNECTED (AMPS)				SIZE AND CIRCUIT SIZE WITH

PANEL SCHEDULE - EXISTING

LOAD / REMARKS	CIR. NO.	BREAKERS		H.P. OR K.W.	LIGHTING			AMPS PER PHASE			SPECIAL			H.P. OR K.W.	BREAKERS		CIR. NO.	LOAD / REMARKS	PANELBOARD SCHEDULE
		AMPS	POLES		ØA	ØB	ØC	ØA	ØB	ØC	ØA	ØB	ØC		AMPS	POLES			
PANEL BR1	1	100	3	-	-	-	-	-	-	-	-	-	-	-	225	3	2	PANEL BR2	Pepper Building Tallahassee, Florida
	3			-	-	-	-	-	-	-	-	-	-	-			4		
	5			-	-	-	-	-	-	-	-	-	-	-			6		
PANEL BR3	7	225	3	-	-	-	-	-	-	-	-	-	-	-	225	3	8	PANEL BR4	
	9			-	-	-	-	-	-	-	-	-	-	-			10		
	11			-	-	-	-	-	-	-	-	-	-	-			12		
PANEL BR5	13	225	3	-	-	-	-	-	-	-	-	-	-	-			14	EF-1	
	15			-	-	-	-	-	-	-	-	-	-	-	1/4	20	2	16	
	17			-	-	-	-	-	-	-	-	-	-	-	4		18		
SPACE	19			-	-	-	-	-	-	-	-	-	-	-	19	-	25	2	18
SPACE	21			-	-	-	-	-	-	-	-	-	-	-			20		
SPACE	23			-	-	-	-	-	-	-	-	-	-	-	19	-	25	2	22
SPACE	25			-	-	-	-	-	-	-	-	-	-	-			24		
SPACE	27			-	-	-	-	-	-	-	-	-	-	-	19		26		
SPACE	29			-	-	-	-	-	-	-	-	-	-	-			28		
SPACE	31			-	-	-	-	-	-	-	-	-	-	-			30		
SPACE	33			-	-	-	-	-	-	-	-	-	-	-			32		
SPACE	35			-	-	-	-	-	-	-	-	-	-	-			34		
SPACE	37			-	-	-	-	-	-	-	-	-	-	-			36		
BRANCH MOUNTED	37	1000	3	-	-	-	-	-	-	-	-	-	-	-			38		
MAIN BREAKER	39			-	-	-	-	-	-	-	-	-	-	-			40		
	41			-	-	-	-	-	-	-	-	-	-	-			42		
ADDED CONNECTED LOAD (AMPS)				-	-	-	-	-	-	-	23	23	38	ADDED CONNECTED LOAD (AMPS)					

Panel	BLP
Location	RM. B1-12
Fed From	300KVA TRANSF.
Service	3 Phase 4 Wire
	208Y/120V
	480Y/277V
Main Bkr.	1000 A 3 P
Lugs Only	- A
22,000 AIC	
60 Hz.	
	Surface mounted panel
	Flush mounted panel
	200% Neutral Bar
Remarks	PANEL IS EXISTING TO REMAIN.
CONTRACTOR SHALL SAFELY MEASURE EXISTING LOAD ON PANEL 'BLP' PRIOR TO NEW WORK SCOPE AND NOTIFY ENGINEER IF EXISTING LOAD IS EXCEEDING 400A ON ANY PHASE. LOAD DATA FOR DOWNSTREAM PANELS IS NOT AVAILABLE AT TIME OF DESIGN.	



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PEPPER BUILDING
DESIGN REPLACEMENT OF
MAJOR ELECTRICAL SYSTEMS
PHASE 1
MSFM-02203280
Tallahassee, Florida

DATE:
March 25, 2024

REVISED:

DESIGNED BY:
BW

DRAWN BY:
TB

SUBMITTAL:
100% Construction Documents

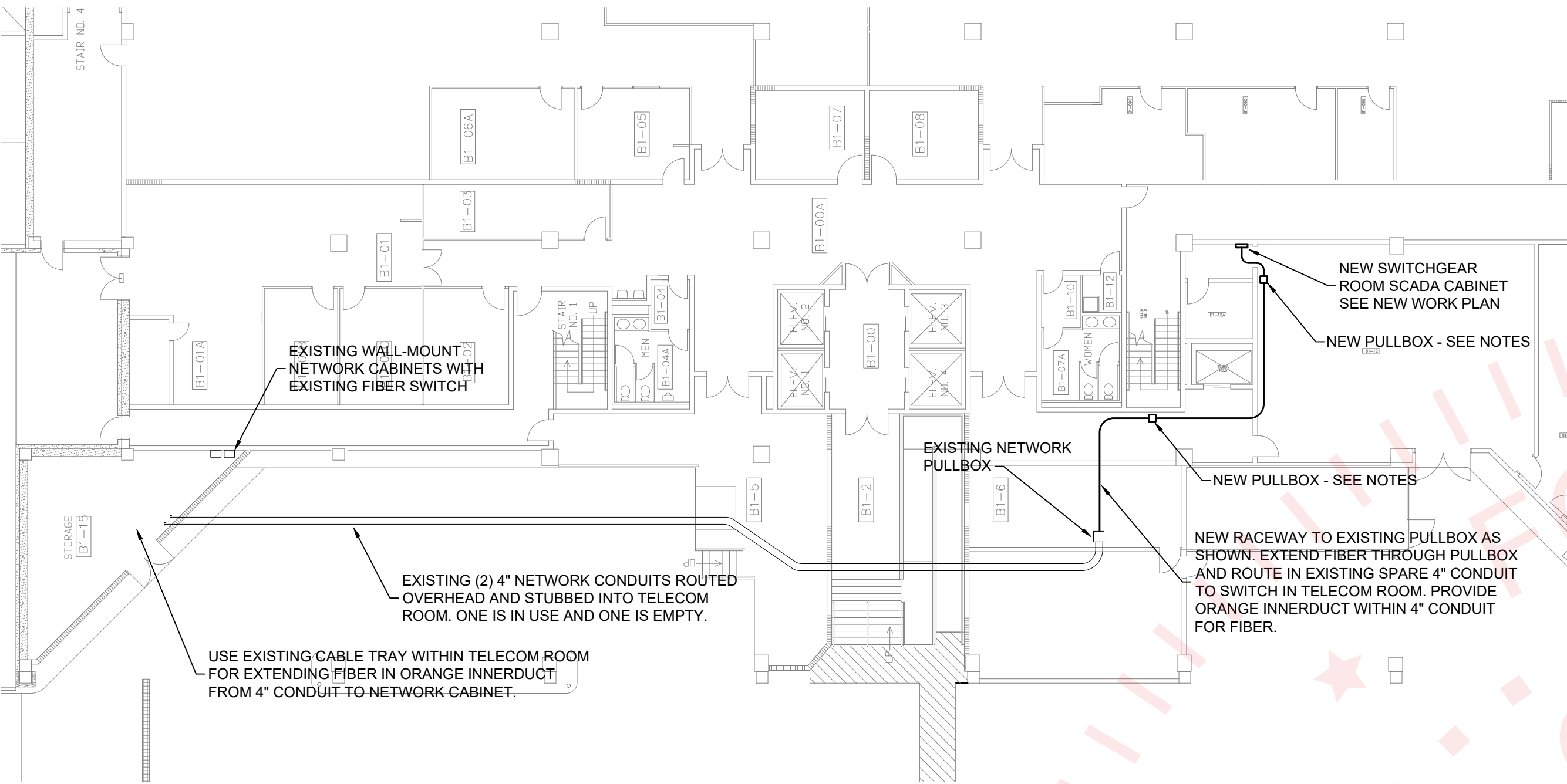
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LOW VOLTAGE
ONE LINES AND SCHEDULES

SHEET:

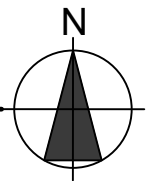
E404

JOB NUMBER:
2311

SCADA PATHWAY PLAN



SCADA PLAN
SCALE: 1/16\"/>



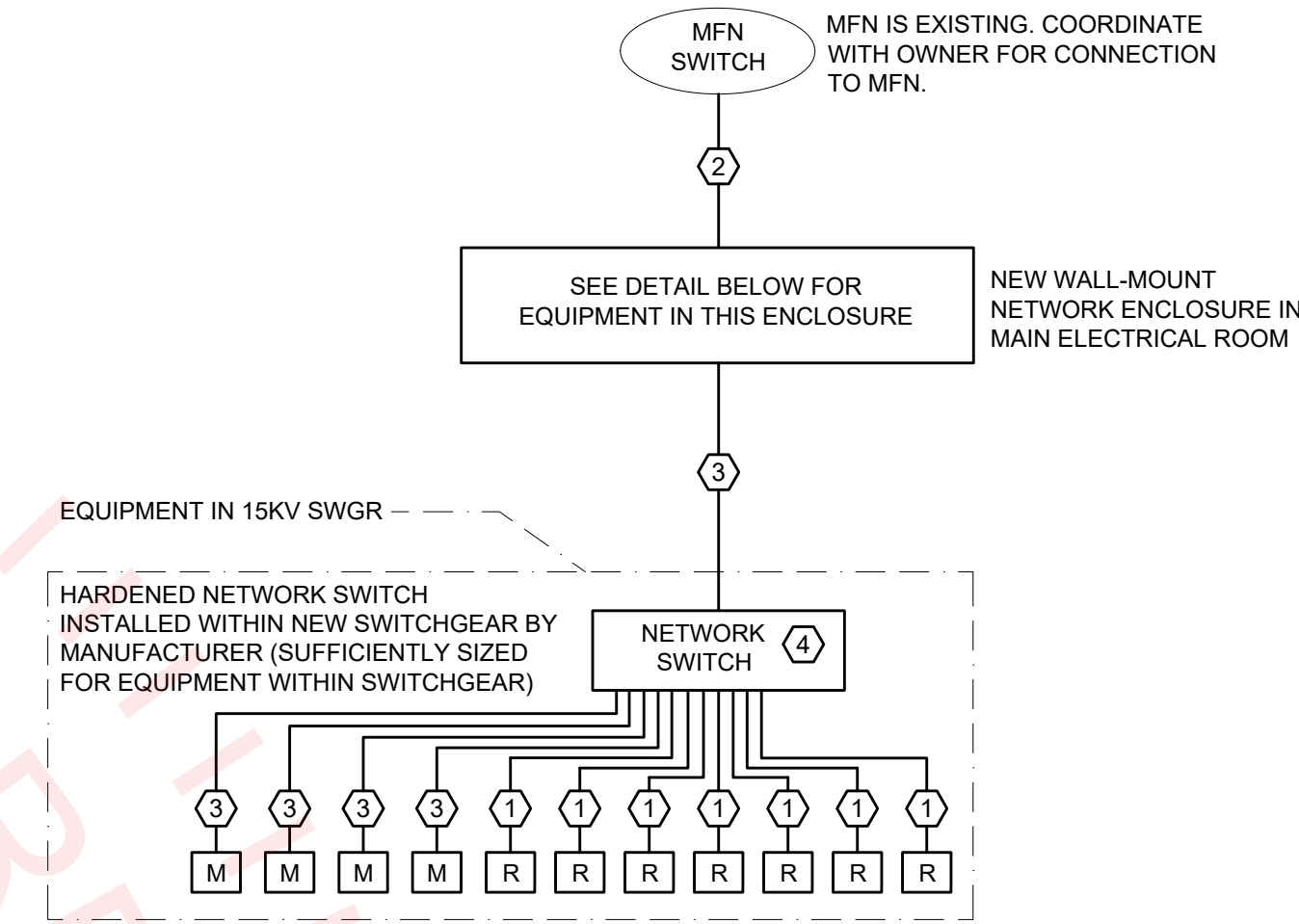
FOR HORIZONTAL FIBER RACEWAY ROUTING, THE FOLLOWING GUIDELINES SHALL BE USED:
3.1. UTILIZE LONG SWEEP 90's FOR ALL TURNS.
3.2. NO MORE THAN (2) 90 DEGREE BENDS BETWEEN PULLBOXES.
3.3. PULLBOXES SHALL BE 4\"/>

DATA (SCADA) NOTES

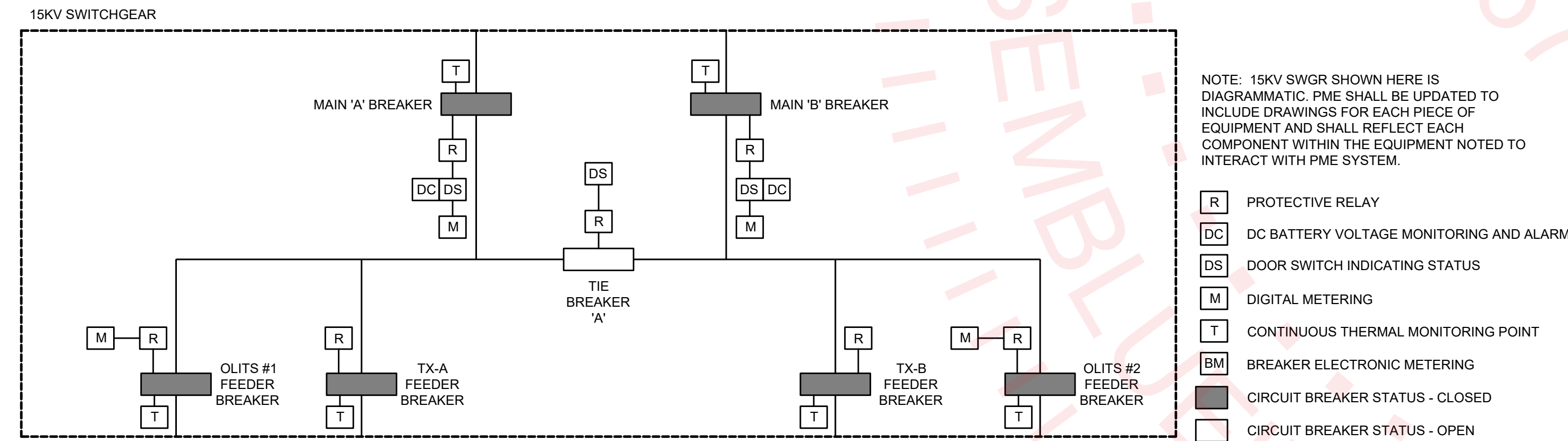
- PROGRAMMING CHANGES WITHIN THE SCADA NETWORK SHALL INCLUDE THE REMOVAL OF EXISTING 15KV SWITCHGEAR METERING DATA POINTS AND ASSOCIATED PROGRAMMING FROM THE SCADA SYSTEM.
- SCADA SHALL INTERACT WITH NEW EQUIPMENT COMPONENTS FOR THE ACTIONS NOTED.
- FOR 15KV SWGR, SCADA SHALL MONITOR ALL CIRCUIT BREAKERS AND INDICATE GRAPHICALLY THE STATUS OF EACH - OPEN OR CLOSED. DEMONSTRATE PROPER SIGNALING AT COMPLETION OF PROJECT.
- SCADA SHALL MONITOR 15KV SWITCHGEAR DOOR PANELS AND INDICATE STATUS OF EACH DOOR, OPEN OR CLOSED. FOR MAINS AND TIE. OPEN DOORS SHALL DISABLE ABILITY TO AUTOMATICALLY CLOSE CIRCUIT BREAKER VIA AUTO-TRANSFER SCHEME. DOOR SWITCHES SHALL NOT PROHIBIT TRIPPING OF CIRCUIT BREAKERS. AN INDUSTRIAL GRADE LIMIT SWITCH SHALL BE USED TO COMMUNICATE TO SCADA NETWORK THE DOOR STATUS VIA PROTECTIVE RELAY MODULE.
- SCADA SHALL MONITOR AND ALARM FOR LOW DC SYSTEM BATTERY VOLTAGE AND HIGH DC SYSTEM BATTERY VOLTAGE. DC BATTERY VOLTAGE SHALL BE MONITORED USING MAIN RELAYS IN 15KV SWGR. SYSTEM NOMINAL VOLTAGE: 125VDC. MINIMUM VOLTAGE ALARM THRESHOLD: 113VDC. MAXIMUM VOLTAGE ALARM THRESHOLD: 137VDC.
- SCADA SHALL MONITOR DEDICATED MAIN BREAKER POWER QUALITY DIGITAL METERING IN 15KV SWGR TO PROVIDE FULL METERING INFORMATION TO SCADA SOFTWARE.
- SCADA SHALL MONITOR DEDICATED RELAYS TO PROVIDE FULL METERING INFORMATION TO SCADA SOFTWARE.
- SCADA SYSTEM SHALL INCLUDE GRAPHIC OF 15KV SWGR SIMILAR TO GRAPHIC SHOWN BELOW WITH INTERACTIVE INTERFACE.
- ALL RELAYS, BREAKERS, AND METERING EQUIPMENT SHALL BE CONFIGURED AND PROGRAMMED ACCORDINGLY TO MEET NOTED REQUIREMENTS FOR PROPER SCADA CONFIGURATION.
- CONTINUOUS THERMAL MONITORING POINTS SHALL BE CONFIGURED AND PROGRAMMED WITHIN PME TO GIVE REAL TIME TEMPERATURE READINGS, TO TRACK HISTORICAL TEMPERATURE LEVELS, AND TO ALARM ON HIGH TEMP. SWITCHGEAR MANUFACTURER SHALL BE CONSULTED FOR RECOMMENDED HIGH TEMP SETPOINTS PRIOR TO PROGRAMMING.
- NO REMOTE CONTROL FEATURES OF SWITCHGEAR SHALL BE CONFIGURED WITH SCADA SYSTEM.
- EXISTING SCADA SYSTEM IS ECOSTRUXURE POWER MONITORING EXPERT (PME) BY SCHNEIDER ELECTRIC. CONTRACTOR SHALL SECURE THE SERVICES OF A CERTIFIED TECHNICIAN EXPERIENCED WITH PME SOFTWARE TO PERFORM CHANGES TO THE EXISTING SCADA SYSTEM TO ACCOMMODATE THE PROJECT REQUIREMENTS. CHANGES INCLUDE ALL NECESSARY SOFTWARE, UPDATES, LICENSES, HARDWARE COMPONENTS, PROGRAMMING OF NEW DEVICES, REMOVAL OF EXISTING DEVICES NOTED TO BE REMOVED FROM SYSTEM, GRAPHIC DEVELOPMENT, AND OWNER TRAINING.
- EQUIPMENT PROVIDED, REGARDLESS OF MANUFACTURER, SHALL BE CONFIGURED FOR A FULLY INTEGRATED SYSTEM AS DETAILED ABOVE. CONTRACTOR SHALL GUARANTEE THAT ALL EQUIPMENT PROVIDED ALLOWS FOR FULL INTEGRATION INTO THE OWNER'S PME SYSTEM AS NOTED.

SCADA NETWORK ONE LINE

- NETWORK EQUIPMENT SCHEDULE:
- 2-STRAND, 62.5/125, MM OPTICAL FIBER BY SWITCHGEAR MANUFACTURER
 - 6-STRAND, 62.5/125, MM OPTICAL FIBER IN 1\"/>



SCADA SYSTEM GRAPHICS

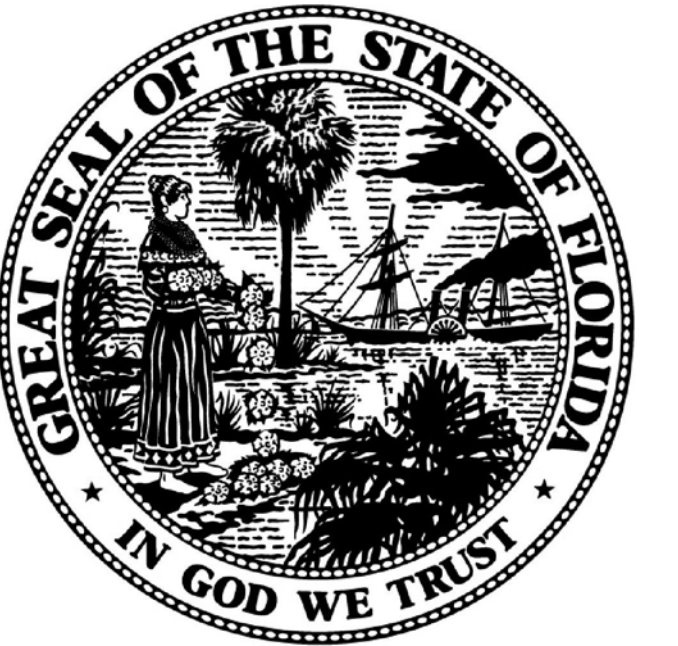
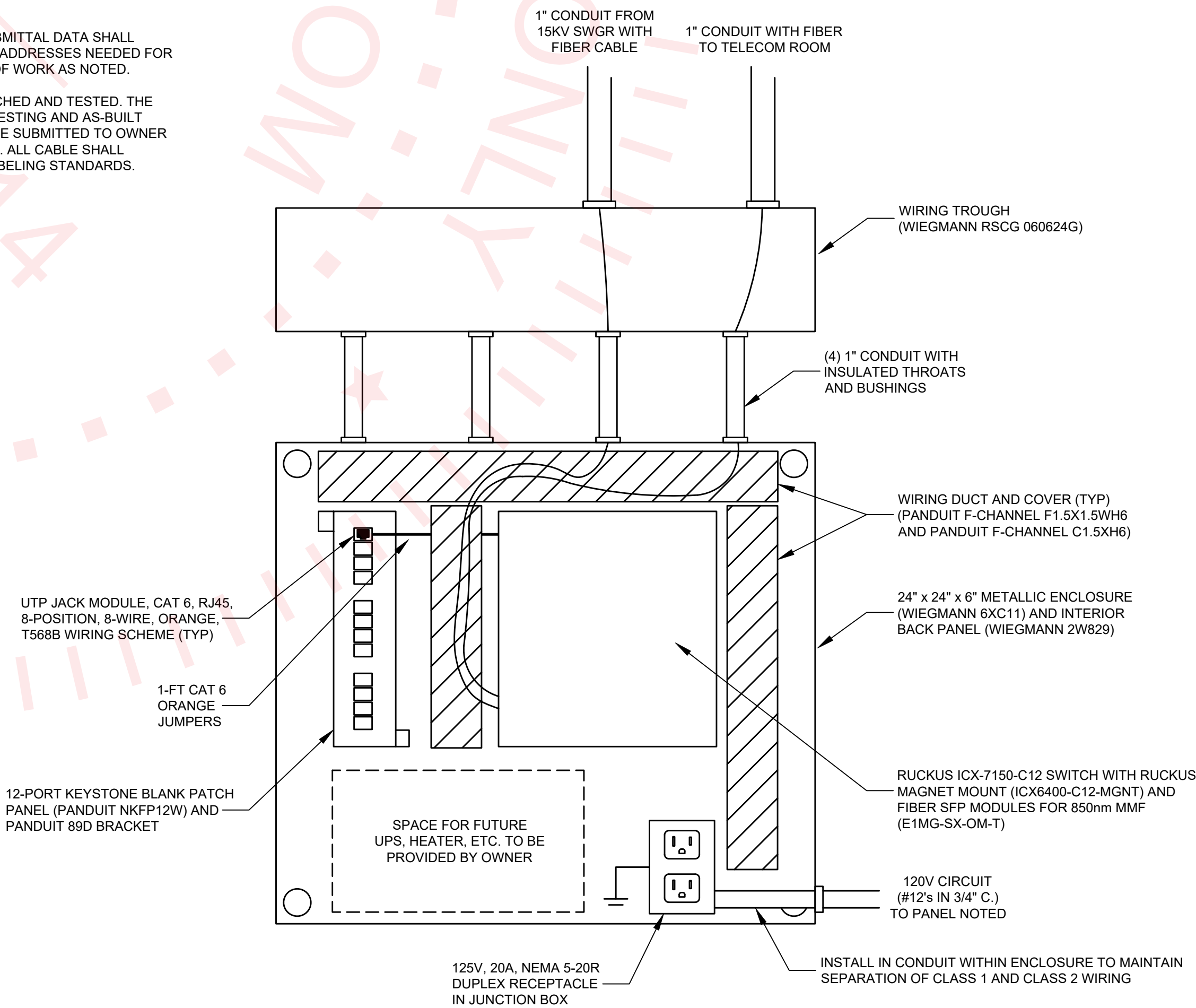


NETWORK ENCLOSURE DETAIL

* QUANTITIES OF FIBER AND ETHERNET CABLES ARE DIAGRAMMATIC. PROVIDE SUFFICIENT QUANTITIES FOR COMPLETING SCADA SCOPE OF WORK NOTED ABOVE.

SHOP DRAWINGS AND SUBMITTAL DATA SHALL INDICATE QUANTITY OF IP ADDRESSES NEEDED FOR ACCOMPLISHING SCOPE OF WORK AS NOTED.

ALL CABLE SHALL BE PATCHED AND TESTED. THE RESULTS OF THE CABLE TESTING AND AS-BUILT DOCUMENTATION SHALL BE SUBMITTED TO OWNER AT PROJECT COMPLETION. ALL CABLE SHALL ADHERE TO ANSI 606-A LABELING STANDARDS.



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DESIGN REPLACEMENT OF
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PHASE 1
MSFM-02203280
Tallahassee, Florida

DATE:
March 25, 2024

REVISED:

DESIGNED BY: BW	DRAWN BY: TB
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SUBMITTAL:
100% Construction Documents

SHEET TITLE:

SCADA PLAN
AND NOTES

SHEET:

E500

JOB NUMBER:

2311